

Chapter 8

Infinite Impulse Response (IIR) Filters

8.1 INTRODUCTION

There are several techniques available for the design of digital filters having an infinite duration unit impulse response. The analog filter design is well-developed and the techniques discussed in this chapter are all based on taking an analog filter and converting it into a digital filter. Thus, the design of an IIR filter involves design of a digital filter in the analog domain and transforming the design into the digital domain.

The system function describing an analog filter may be written as

$$H_a(S) = \frac{\sum_{k=0}^M b_k s^k}{\sum_{k=0}^N a_k s^k} \quad (8.1)$$

where $\{a_k\}$ and $\{b_k\}$ are the filter coefficients. The impulse response of these filter coefficients is related to $H_a(s)$ by the Laplace transform

$$H_a(s) = \int_{-\infty}^{\infty} h(t) e^{-st} dt \quad (8.2)$$

The analog filter having the rational system function $H(s)$ given in Eq. 8.1 can also be described by the linear constant-coefficient differential equation

$$\sum_{k=0}^N a_k \frac{d^k y(t)}{dt^k} = \sum_{k=0}^M b_k \frac{d^k x(t)}{dt^k} \quad (8.3)$$

where $x(t)$ is the input signal and $y(t)$ is the output of the filter.

The above three equivalent characterisation of an analog filter leads to three alternative methods for transforming the filter into the digital domain. The design techniques for IIR filters are presented with the

restriction that the filters be realisable and stable. Recall that an analog filter with system function $H(s)$ is stable if all its poles lie in the left-half of the s -plane. As a result, if the conversion techniques are to be effective, the technique should possess the following properties:

1. The $j\Omega$ axis in the s -plane should map onto the unit circle in the z -plane. This gives a direct relationship between the two frequency variables in the two domains.
2. The left-half plane of the s -plane should map into the inside of the unit circle in the z -plane to convert a stable analog filter into a stable digital filter.

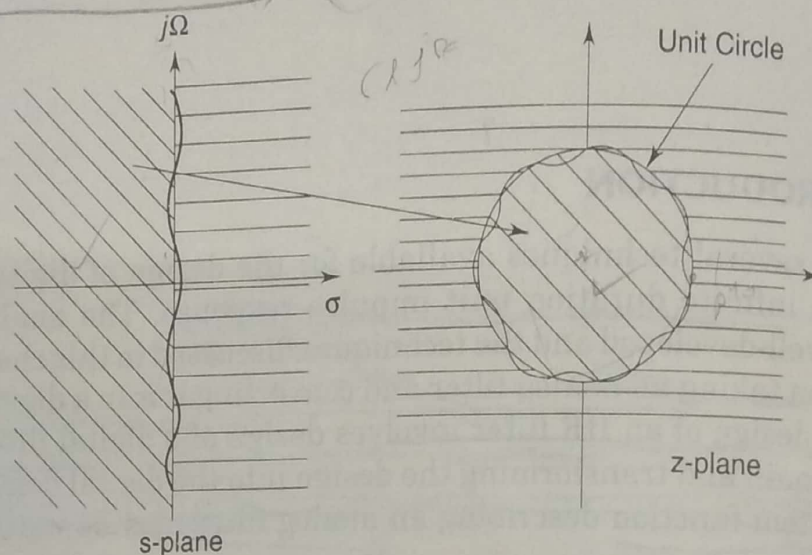


Fig. 8.1 Mapping from s -plane to z -plane

The physically realizable and stable IIR filter cannot have a linear phase. For a filter to have a linear phase, the condition is $h(n) = h(M-1-n)$ and the filter would have a mirror image pole outside the unit circle for every pole inside the unit circle. This results in an unstable filter. As a result, a causal and stable IIR filter cannot have a linear phase. In the design of IIR filters, only the desired magnitude response is specified and the phase response that is obtained from the design methodology is accepted.

8.2 IIR FILTER DESIGN BY APPROXIMATION OF DERIVATIVES

In this method an analog filter is converted into a digital filter by approximating the differential equation in Eq. 8.3 by an equivalent difference equation. The backward difference formula is substituted for the derivative $dy(t)/dt$ at time $t = nT$. Thus,

$$\begin{aligned} \left. \frac{dy(t)}{dt} \right|_{t=nT} &= \frac{y(nT) - y(nT-T)}{T} \\ &= \frac{y(n) - y(n-1)}{T} \end{aligned} \quad (8.4)$$

Where T is the sampling interval and $y(n) \equiv y(nT)$. The system function of an analog differentiator with an output dy/dt is $H(s) = s$, and the digital system that produces the output $[y(n) - y(n-1)]/T$ has the system function $H(z) = (1 - z^{-1})/T$. These two can be compared to get the frequency-domain equivalent for the relationship in Eq. 8.4 as

$$s = \frac{1 - z^{-1}}{T} \quad (8.5)$$

The second derivative $d^2y(t)/dt^2$ is replaced by the second backward difference,

$$\begin{aligned} \left. \frac{d^2y(t)}{dt^2} \right|_{t=nT} &= \frac{d}{dT} \left[\frac{dy(t)}{dt} \right]_{t=nT} \\ &= \frac{[y(nT) - y(nT - T)]/T - [y(nT - T) - y(nT - 2T)]/T}{T} \\ &= \frac{y(n) - 2y(n-1) + y(n-2)}{T^2} \end{aligned} \quad (8.6)$$

The equivalent to Eq. 8.6 in the frequency-domain is

$$s^2 = \frac{1 - 2z^{-1} + z^{-2}}{T^2} = \left(\frac{1 - z^{-1}}{T} \right)^2 \quad (8.7)$$

The i^{th} derivative of $y(t)$ results in the equivalent frequency-domain relationship

$$s^i = \left(\frac{1 - z^{-1}}{T} \right)^i \quad (8.8)$$

As a result, the digital filter's system function can be obtained by the method of approximation of the derivatives as,

$$H(z) = H_a(s) \Big|_{s=(1-z^{-1})/T} \quad (8.9)$$

where $H_a(s)$ is the system function of the analog filter characterised by the differential equation given in Eq. 8.3. The outcomes of the mapping of the z -plane from the s -plane are discussed below. Equation 8.5 can be, equivalently, written as

$$z = \frac{1}{1 - sT} \quad (8.10)$$

Substituting $s = j\Omega$ in the above equation,

$$\begin{aligned} z &= \frac{1}{1 - j\Omega T} \\ &= \frac{1}{1 + \Omega^2 T^2} + j \frac{\Omega T}{1 + \Omega^2 T^2} \end{aligned} \quad (8.11)$$

Varying Ω from $-\infty$ to ∞ , the corresponding locus of points in the z -plane is a circle with radius $\frac{1}{2}$ and with centre at $z = \frac{1}{2}$, as shown in Fig. 8.2.

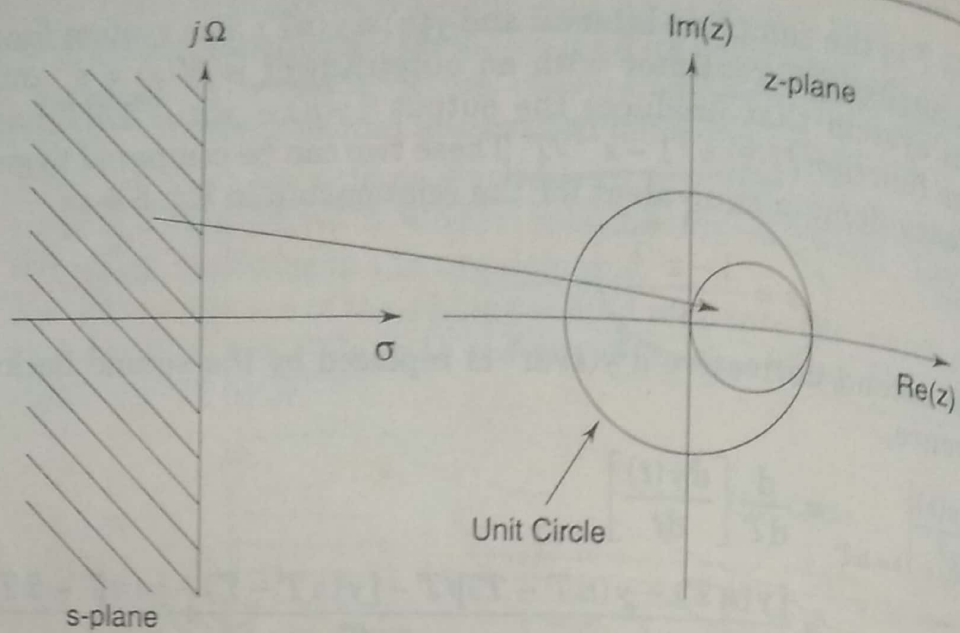


Fig. 8.2 The Mapping of Eq. 8.5 into the z-plane

It can be seen that the mapping of Eq. 8.5 takes the left-half plane of s-domain into the corresponding points inside the circle of radius 0.5 and centre at $z = 0.5$, and the right-half of the s-plane is mapped outside the unit circle. As a result, this mapping results in a stable analog filter transformed into a stable digital filter; however, as the locations of poles in the z-domain are confined to smaller frequencies, this design method can be used only for transforming analog low-pass filters and bandpass filters having smaller resonant frequencies. Neither a high-pass filter nor a band reject filter can be realised using this technique.

The forward difference can be substituted for the derivative instead of the backward difference. This gives,

$$\begin{aligned} \frac{dy(t)}{dt} &= \frac{y(nT + T) - y(nT)}{T} \\ &= \frac{y(n+1) - y(n)}{T} \end{aligned} \quad (8.12)$$

The transformation formula will be

$$s = \frac{z - 1}{T} \quad (8.13)$$

or,

$$z = 1 + sT \quad (8.14)$$

The mapping of Eq. 8.14 is shown in Fig. 8.3. This results in a worse situation than the backward difference substitution for the derivative. When $s = j\Omega$, the mapping of these points in the s-domain results in a straight line in the z-domain with coordinates $(z_{\text{real}}, z_{\text{imag}}) = (1, \Omega T)$. Consequently, stable analog filters do not always map into stable digital filters.

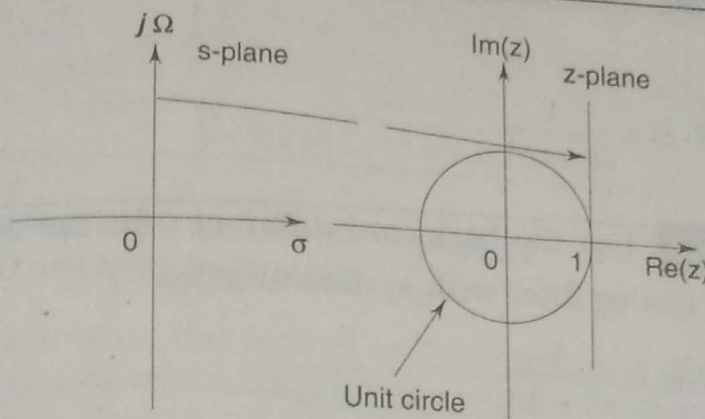


Fig. 8.3 Mapping of Eq. 8.14 into the z-plane

The limitations of the two mapping methods discussed above are overcome by using a more complex substitution for the derivatives. An N th order difference is proposed for the derivative, as shown.

$$\left. \frac{dy(t)}{dt} \right|_{t=nT} = \frac{1}{T} \sum_{k=1}^N a_k \frac{y(nT + kT) - y(nT - kT)}{T} \quad (8.15)$$

where $\{a_k\}$ are a set of parameters selected so as to optimise the approximation. The transformation from the s-plane to the z-plane is then,

$$s = \frac{1}{T} \sum_{k=1}^N a_k (z^k - z^{-k}) \quad (8.16)$$

By choosing proper values for $\{a_k\}$, the $j\Omega$ axis can be mapped into the unit circle and the left-half s-plane can be mapped into points inside the unit circle in the z-plane.

Example 8.1 Use the backward difference for the derivative to convert the analog low-pass filter with system function

$$H(s) = \frac{1}{s + 2}$$

Solution The mapping formula for the backward difference for the derivative is given in Eq. 8.5, i.e.

$$s = \frac{1 - z^{-1}}{T}$$

The system response of the digital filter is

$$\begin{aligned} H(z) &= H(s) \Big|_{s = \frac{1 - z^{-1}}{T}} = \frac{1}{\left(\frac{1 - z^{-1}}{T} \right) + 2} \\ &= \frac{T}{1 - z^{-1} + 2T} \end{aligned}$$

If $T = 1\text{s}$,

$$H(z) = \frac{1}{3 - z^{-1}}$$

Example 8.2 Use the backward difference for the derivative and convert the analog filter with system function

$$H(s) = \frac{1}{s^2 + 16}$$

Solution Using Eq. 8.5,

$$s = \frac{1 - z^{-1}}{T}$$

The system response of the digital filter is

$$H(z) = H(s) \Big|_{s = \frac{1 - z^{-1}}{T}} = \frac{1}{\left(\frac{1 - z^{-1}}{T}\right)^2 + 16}$$

$$H(z) = \frac{T^2}{1 - 2z^{-1} + z^{-2} + 16T^2}$$

If $T = 1\text{s}$,

$$H(z) = \frac{1}{z^{-2} - 2z^{-1} + 17}$$

Example 8.3 An analog filter has the following system function. Convert this filter into a digital filter using backward difference for the derivative.

$$H(s) = \frac{1}{(s + 0.1)^2 + 9}$$

Solution The system response of the digital filter is

$$H(z) = H(s) \Big|_{s = \frac{1 - z^{-1}}{T}} = \frac{1}{\left(\frac{1 - z^{-1}}{T} + 0.1\right)^2 + 9}$$

$$H(z) = \frac{T^2}{z^{-2} - 2(1 + 0.1T)z^{-1} + (1 + 0.2T + 9.01T^2)}$$

$$H(z) = \frac{T^2}{1 - 2\frac{(1 + 0.1T)}{(1 + 0.2T + 9.01T^2)}z^{-1} + \frac{z^{-2}}{(1 + 0.2T + 9.01T^2)}}$$

If $T = 1\text{s}$,

$$H(z) = \frac{0.0979}{1 - 0.2155 z^{-1} + 0.09792 z^{-2}}$$

8.3 IIR FILTER DESIGN BY IMPULSE INVARIANT METHOD

In this technique, the desired impulse response of the digital filter is obtained by uniformly sampling the impulse response of the equivalent analog filter. That is,

$$h(n) = h_a(nT) \quad (8.17)$$

where T is the sampling interval. The transformation technique can be well understood by first considering a simple distinct pole case for the analog filter's system function, as shown below.

$$H_a(s) = \sum_{i=1}^M \frac{A_i}{s - p_i} \quad (8.18)$$

The impulse response of the system specified by Eq. 8.18 can be obtained by taking the inverse Laplace transform and it will be of the form

$$h_a(t) = \sum_{i=1}^M A_i e^{p_i t} u_a(t) \quad (8.19)$$

where $u_a(t)$ is the unit step function in continuous time. The impulse response $h(n)$ of the equivalent digital filter is obtained by uniformly sampling $h_a(t)$, i.e. by applying Eq. 8.17

$$h(n) = h_a(nT) = \sum_{i=1}^M A_i e^{p_i nT} u_a(nT) \quad (8.20)$$

The system response of the digital system of Eq. 8.20, can be obtained by taking the z -transform, i.e.

$$H(z) = \sum_{n=0}^{\infty} h(n) z^{-n}$$

Using Eq. 8.20,

$$H(z) = \sum_{n=0}^{\infty} \left[\sum_{i=1}^M A_i e^{p_i nT} u_a(nT) \right] z^{-n} \quad (8.21)$$

Interchanging the order of summation,

$$H(z) = \sum_{i=1}^M \left[\sum_{n=0}^{\infty} A_i e^{p_i nT} u_a(nT) \right] z^{-n}$$

$$H(z) = \sum_{i=1}^M \frac{A_i}{1 - e^{p_i T} z^{-1}} \quad (8.22)$$

Now, by comparing Eqs. 8.18 and 8.22, the mapping formula for the impulse invariant transformation is given by

$$\frac{1}{s - p_i} \longrightarrow \frac{1}{1 - e^{p_i T} z^{-1}} \quad (8.23)$$

Equation 8.23 shows that the analog pole at $s = p_i$ is mapped into a digital pole at $z = e^{p_i T}$. Therefore, the analog poles and the digital poles are related by the relation

$$z = e^{sT} \quad (8.24)$$

The general characteristic of the mapping $z = e^{sT}$ can be obtained by substituting $s = \sigma + j\Omega$ and expressing the complex variable z in the polar form as $z = r e^{j\omega}$. With these substitutions, Eq. 8.24 becomes

$$r e^{j\omega} = e^{\sigma T} e^{j\Omega T}$$

Clearly,

$$\begin{aligned} r &= e^{\sigma T} \\ \omega &= \Omega T \end{aligned} \quad (8.25)$$

Consequently, $\sigma < 0$ implies that $0 < r < 1$ and $\sigma > 0$ implies that $r > 1$. Then $\sigma = 0$, we have $r = 1$. Therefore, the left-half of s -plane is mapped inside the unit circle in the z -plane and the right-half of s -plane is mapped into points that fall outside the unit circle in z . This is one of the desirable properties for stability. The $j\Omega$ -axis is mapped into the unit circle in z -plane. However, the mapping of the $j\Omega$ -axis is not one-to-one. The mapping $\omega = \Omega T$ implies that the interval $-\pi/T \leq \Omega \leq \pi/T$ maps into the corresponding values of $-\pi \leq \omega \leq \pi$. Further, the frequency interval $\pi/T \leq \Omega \leq 3\pi/T$ also maps into the interval $-\pi \leq \omega \leq \pi$ and, in general, any frequency interval $(2k - 1)\pi/T \leq \Omega \leq (2k + 1)\pi/T$, where k is an integer, will also map into the interval $-\pi \leq \omega \leq \pi$ in the z -plane. Thus the mapping from the analog frequency Ω to the frequency variable ω in the digital domain is many-to-one, which simply reflects the effects of aliasing due to sampling of the impulse response. Figure 8.4 illustrates the mapping from the s -plane to the z -plane.

Some of the properties of the impulse invariant transformation are given below.

Approximation
derivation
method

$$\frac{1}{(s + s_i)^m} \longrightarrow \frac{(-1)^{m-1}}{(m-1)!} \frac{d^{m-1}}{ds^{m-1}} \left[\frac{1}{1 - e^{-sT} z^{-1}} \right]; s \rightarrow s_i \quad (8.26)$$

Impulse
invariant
method

$$\frac{s + a}{(s + a)^2 + b^2} \longrightarrow \frac{1 - e^{-aT} (\cos bT) z^{-1}}{1 - 2e^{-aT} (\cos bT) z^{-1} + e^{-2aT} z^{-2}} \quad (8.27)$$

$$\frac{b}{(s + a)^2 + b^2} \longrightarrow \frac{e^{-aT} (\sin bT) z^{-1}}{1 - 2e^{-aT} (\cos bT) z^{-1} + e^{-2aT} z^{-2}} \quad (8.28)$$

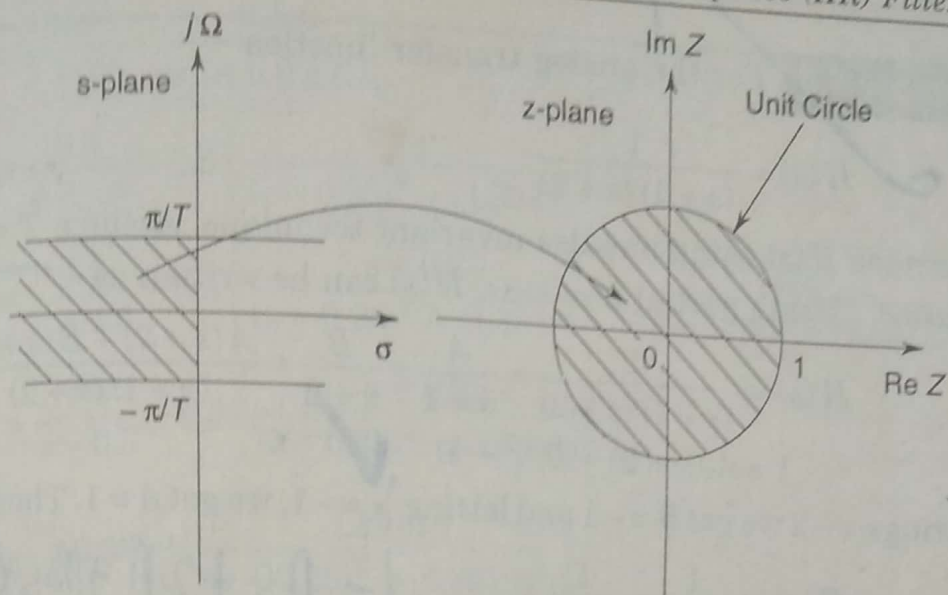


Fig. 8.4 The Mapping of $z = e^{sT}$

Example 8.4 Convert the analog filter into a digital filter whose system function is

$$H(s) = \frac{s + 0.2}{(s + 0.2)^2 + 9}$$

Use the impulse invariant technique. Assume $T = 1$ s.

Solution The system response of the analog filter is of the standard form

$$H(s) = \frac{s + a}{(s + a)^2 + b^2}$$

where $a = 0.2$ and $b = 3$. The system response of the digital filter can be obtained using Eq. 8.27.

$$\begin{aligned} H(z) &= \frac{1 - e^{-aT} (\cos bT) z^{-1}}{1 - 2e^{-aT} (\cos bT) z^{-1} + e^{-2aT} z^{-2}} \\ &= \frac{1 - e^{-0.2T} (\cos 3T) z^{-1}}{1 - 2e^{-0.2T} (\cos 3T) z^{-1} + e^{-0.4T} z^{-2}} \end{aligned}$$

Taking $T = 1$ s,

$$H(z) = \frac{1 - (0.8187)(-0.99) z^{-1}}{1 - 2(0.8187)(-0.99) z^{-1} + 0.6703 z^{-2}}$$

That is,

$$H(z) = \frac{1 + (0.8105) z^{-1}}{1 + 1.6210 z^{-1} + 0.6703 z^{-2}}$$

Example 8.5 For the analog transfer function

$$H(s) = \frac{1}{(s+1)(s+2)}$$

determine $H(z)$ using impulse invariant technique. Assume $T = 1s$.

Solution Using partial fractions, $H(s)$ can be written as

$$H(s) = \frac{1}{(s+1)(s+2)} = \frac{A}{s+1} + \frac{B}{s+2} = \frac{A(s+2) + B(s+1)}{(s+1)(s+2)}$$

$$1 = A(s+2) + B(s+1)$$

Letting $s = -2$ we get $B = -1$ and letting $s = -1$, we get $A = 1$. Therefore

$$H(s) = \frac{1}{s+1} - \frac{1}{s+2}$$

$$1 = As + 2A + Bs + B$$

The system function of the digital filter is obtained by using Eq. 8.23.

$$H(z) = \frac{1}{1 - e^{-T} z^{-1}} - \frac{1}{1 - e^{-2T} z^{-1}}$$

$$= \frac{z^{-1} [e^{-T} - e^{-2T}]}{1 - (e^{-T} + e^{-2T}) z^{-1} + e^{-3T} z^{-2}}$$

$$\begin{aligned} 2A + B &= 1 \\ 1 &= A + B \end{aligned}$$

Since $T = 1s$,

$$H(z) = \frac{0.2326 z^{-1}}{1 - 0.5032 z^{-1} + 0.0498 z^{-2}}$$

Example 8.6 Determine $H(z)$ using the impulse invariant technique for the analog system function

$$H(s) = \frac{1}{(s+0.5)(s^2+0.5s+2)}$$

Solution Using partial fractions, $H(s)$ can be written as

$$H(s) = \frac{1}{(s+0.5)(s^2+0.5s+2)} = \frac{A}{s+0.5} + \frac{Bs+C}{s^2+0.5s+2}$$

Therefore,

$$A(s^2+0.5s+2) + (Bs+C)(s+0.5) = 1$$

Comparing the coefficients of s^2 , s and the constants on either side of the above expression, we get

$$A + B = 0$$

$$0.5A + 0.5B + C = 0$$

$$2A + 0.5C = 1$$

Solving the above simultaneous equations, we get $A = 0.5$, $B = -0.5$ and $C = 0$. The system response can be written as,

$$\begin{aligned}
 H(s) &= \frac{0.5}{s+0.5} - \frac{0.5s}{s^2+0.5s+2} \\
 &= \frac{0.5}{s+0.5} - 0.5 \left(\frac{s}{(s+0.25)^2 + (1.3919)^2} \right) \\
 &= \frac{0.5}{s+0.5} - 0.5 \left(\frac{s+0.25}{(s+0.25)^2 + (1.3919)^2} - \frac{0.25}{(s+0.25)^2 + (1.3919)^2} \right) \\
 &= \frac{0.5}{s+0.5} - 0.5 \left(\frac{s+0.25}{(s+0.25)^2 + (1.3919)^2} \right) \\
 &\quad + 0.0898 \left(\frac{1.3919}{(s+0.25)^2 + (1.3919)^2} \right)
 \end{aligned}$$

Using Eqs 8.27 and 8.28,

$$\begin{aligned}
 H(z) &= \frac{0.5}{1-e^{-0.5T}z^{-1}} - 0.5 \left[\frac{1-e^{-0.25T}(\cos 1.3919T)z^{-1}}{1-2e^{-0.25T}(\cos 1.3919T)z^{-1}+e^{-0.5T}z^{-2}} \right] \\
 &\quad + 0.0898 \left[\frac{e^{-0.25T}(\sin 1.3919T)z^{-1}}{1-2e^{-0.25T}(\cos 1.3919T)z^{-1}+e^{-0.5T}z^{-2}} \right]
 \end{aligned}$$

Letting $T = 1s$,

$$\begin{aligned}
 H(z) &= \frac{0.5}{1-0.6065z^{-1}} - 0.5 \left(\frac{1-0.1385z^{-1}}{(1+0.277z^{-1}+0.606z^{-2})} \right) \\
 &\quad + 0.0898 \left(\frac{0.7663z^{-1}}{(1-0.277z^{-1}+0.606z^{-2})} \right)
 \end{aligned}$$

On simplifying we get,

$$H(z) = \frac{0.30325z^2 + 0.2194z}{z^3 - 0.8835z^2 + 0.774z - 0.3675}$$

8.4 IIR FILTER DESIGN BY THE BILINEAR TRANSFORMATION

The IIR filter design using (i) approximation of derivatives method and (ii) the impulse invariant method are appropriate for the design of low-pass filters and bandpass filters whose resonant frequencies are low. These techniques are not suitable for high-pass or band-reject filters. This limitation is overcome in the mapping technique called the **bilinear transformation**. This transformation is a one-to-one mapping from the s -domain to the z -domain. That is, the bilinear transformation is a conformal mapping that transforms the $j\Omega$ -axis into the unit circle in the z -plane only once, thus avoiding aliasing of frequency components. Also, the transformation of a stable analog filter

results in a stable digital filter as all the poles in the left half of the s-plane are mapped onto points inside the unit circle of the z-domain. The bilinear transformation is obtained by using the trapezoidal formula for numerical integration. Let the system function of the analog filter be

$$H(s) = \frac{b}{s+a} \quad (8.29)$$

The differential equation describing the analog filter can be obtained from Eq. 8.29 as shown below.

$$H(s) = \frac{Y(s)}{X(s)} = \frac{b}{s+a}$$

$$sY(s) + aY(s) = bX(s) \quad (8.30)$$

Taking inverse Laplace transform,

$$\frac{dy(t)}{dt} + ay(t) = bx(t) \quad (8.31)$$

Eq. 8.31 is integrated between the limits $(nT - T)$ and nT

$$\int_{nT-T}^{nT} \frac{dy(t)}{dt} dt + a \int_{nT-T}^{nT} y(t) dt = b \int_{nT-T}^{nT} x(t) dt \quad (8.32)$$

The trapezoidal rule for numeric integration is given by

$$\int_{nT-T}^{nT} a(t) dt = \frac{T}{2} [a(nT) + a(nT - T)] \quad (8.33)$$

Applying Eq. 8.33 in Eq. 8.32, we get

$$y(nT) - y(nT - T) + \frac{aT}{2} y(nT) + \frac{aT}{2} y(nT - T) = \frac{bT}{2} x(nT) + \frac{bT}{2} x(nT - T)$$

Taking z-transform, the system function of the digital filter is,

$$H(z) = \frac{Y(z)}{X(z)} = \frac{b}{\frac{2}{T} \left(\frac{1-z^{-1}}{1+z^{-1}} \right) + a} \quad (8.34)$$

Comparing Eqs. 8.29 and 8.34 we get,

$$s = \frac{2}{T} \left(\frac{1-z^{-1}}{1+z^{-1}} \right) = \frac{2}{T} \left(\frac{z-1}{z+1} \right) \quad (8.35)$$

The general characteristic of the mapping $z = e^{sT}$ can be obtained by substituting $s = \sigma + j\Omega$ and expressing the complex variable z in the polar form as $z = re^{j\omega}$ in Eq. 8.35.

$$s = \frac{2}{T} \left(\frac{z-1}{z+1} \right) = \frac{2}{T} \left(\frac{re^{j\omega} - 1}{re^{j\omega} + 1} \right)$$

Therefore,

$$\frac{1 + e^{j\omega T}}{1 + r^2 + 2r \cos \omega} = \frac{2}{T} \left(\frac{r^2 - 1}{1 + r^2 + 2r \cos \omega} + j \frac{2r \sin \omega}{1 + r^2 + 2r \cos \omega} \right) \quad (8.36)$$

$$\sigma = \frac{2}{T} \left(\frac{r^2 - 1}{1 + r^2 + 2r \cos \omega} \right) \quad (8.37)$$

$$\Omega = \frac{2}{T} \left(\frac{2r \sin \omega}{1 + r^2 + 2r \cos \omega} \right) \quad (8.38)$$

From Eq. 8.37, it can be noted that if $r < 1$, then $\sigma < 0$, and if $r > 1$, then $\sigma > 0$. Thus, the left-half of the s -plane maps onto the points inside the unit circle in the z -plane and the transformation results in a stable digital system. Consider Eq. 8.38, for unity magnitude ($r = 1$), σ is zero. In this case,

$$\begin{aligned} \Omega &= \frac{2}{T} \left(\frac{\sin \omega}{1 + \cos \omega} \right) \\ &= \frac{2}{T} \left(\frac{2 \sin \omega/2 \cos \omega/2}{\cos^2 \omega/2 + \sin^2 \omega/2 + \cos^2 \omega/2 - \sin^2 \omega/2} \right) \\ \Omega &= \frac{2}{T} \tan \frac{\omega}{2} \end{aligned} \quad (8.39)$$

or equivalently,

$$\omega = 2 \tan^{-1} \frac{\Omega T}{2} \quad (8.40)$$

Equation 8.40 gives the relationship between the frequencies in the two domains and this is shown in Fig. 8.5. It can be noted that the entire range in Ω is mapped only once into the range $-\pi \leq \omega \leq \pi$. However, as seen in Fig. 8.5, the mapping is non-linear and the lower frequencies in analog domain are expanded in the digital domain, whereas the higher frequencies are compressed. This is due to the non-linearity of the arc tangent function and usually called as **frequency warping**.

Example 8.7 Convert the analog filter with system function

$$H(s) = \frac{s + 0.1}{(s + 0.1)^2 + 9}$$

into a digital IIR filter using bilinear transformation. The digital filter should have a resonant frequency of $\omega_r = \frac{\pi}{4}$.

Solution From the system function, we note that $\Omega_c = 3$. The sampling period T can be determined using Eq. 8.39, i.e.

$$\Omega_c = \frac{2}{T} \tan \frac{\omega_r}{2}$$

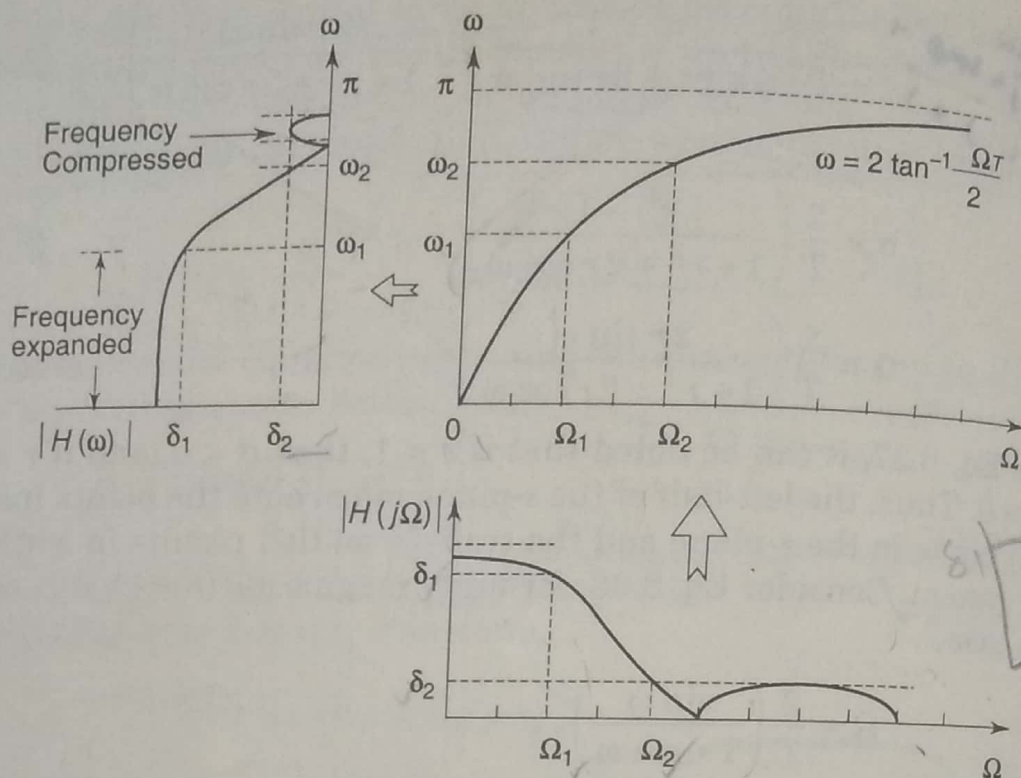


Fig. 8.5 Relationship Between ω and Ω as Given in Eq. 8.40

The sampling period is obtained from the above equation using

$$T = \frac{2}{\Omega_c} \tan \frac{\omega_r}{2} = \frac{2}{3} \tan \frac{\pi}{8} = 0.276 \text{ s}$$

Using bilinear transformation,

$$H(z) = H(s) \Big|_{s = \frac{2}{T} \frac{(z-1)}{(z+1)}}$$

$$H(z) = \frac{\frac{2}{T} \frac{(z-1)}{(z+1)} + 0.1}{\left[\frac{2}{T} \frac{(z-1)}{(z+1)} + 0.1 \right]^2 + 9}$$

$$= \frac{(2/T)(z-1)(z+1) + 0.1(z+1)^2}{[(2/T)(z-1) + 0.1(z+1)]^2 + 9(z+1)^2}$$

Substituting $T = 0.276 \text{ s}$,

$$H(z) = \frac{1 + 0.027 z^{-1} - 0.973 z^{-2}}{8.572 - 11.84 z^{-1} + 8.177 z^{-2}}$$

Example 8.8

Apply bilinear transformation to

$$H(s) = \frac{2}{(s+1)(s+3)}$$

with

$$T = 0.1 \text{ s.}$$

Solution For bilinear transformation,

$$H(z) = H(s) \Big|_{s = \frac{2}{T} \frac{(z-1)}{(z+1)}}$$

$$= \frac{2}{\left(\frac{2}{T} \frac{(z-1)}{(z+1)} + 1 \right) \left(\frac{2}{T} \frac{(z-1)}{(z+1)} + 3 \right)}$$

Using $T = 0.1 \text{ s}$,

$$H(z) = \frac{2}{\left(20 \frac{(z-1)}{(z+1)} + 1 \right) \left(20 \frac{(z-1)}{(z+1)} + 3 \right)}$$

$$= \frac{2(z+1)^2}{(21z-19)(23z-17)}$$

Simplifying further,

$$H(z) = \frac{0.0041(1+z^{-1})^2}{1 - 1.644z^{-1} + 0.668z^{-2}}$$

Example 8. A digital filter with a 3 dB bandwidth of 0.25π is to be designed from the analog filter whose system response is

$$H(s) = \frac{\Omega_c}{s + \Omega_c}$$

Use bilinear transformation and obtain $H(z)$.

Solution Using Eq. 8.39,

$$\Omega_c = \frac{2}{T} \tan \frac{\omega_r}{2} = \frac{2}{T} \tan 0.125 \pi = 0.828/T$$

The system response of the digital filter is given by

$$H(z) = H(s) \Big|_{s = \frac{2}{T} \frac{(z-1)}{(z+1)}}$$

$$= \frac{\Omega_c}{\frac{2}{T} \frac{(z-1)}{(z+1)} + \Omega_c} = \frac{\frac{0.828}{T}}{\frac{2}{T} \frac{(z-1)}{(z+1)} + \frac{0.828}{T}}$$

$$= \frac{0.828(z+1)}{2(z-1) + 0.828(z+1)}$$

Simplifying we get further,

$$H(z) = \frac{1+z^{-1}}{3.414 - 1.414z^{-1}}$$

Example 8.10 Using bilinear transformation obtain $H(z)$ if

$$H(s) = \frac{1}{(s+1)^2}$$

and $T = 0.1\text{s}$.

Solution For the bilinear transformation,

$$\begin{aligned} H(z) &= H(s) \Big|_{s = \frac{2}{T} \frac{(z-1)}{(z+1)}} \\ &= \frac{1}{\left(\frac{2}{T} \frac{(z-1)}{(z+1)} + 1 \right)^2} \end{aligned}$$

Substituting $T = 0.1\text{s}$,

$$H(z) = \frac{1}{\left(20 \frac{(z-1)}{(z+1)} + 1 \right)^2} = \frac{(z+1)^2}{(21z-19)^2}$$

Further simplifying,

$$H(z) = \frac{0.0476 (1+z^{-1})^2}{(1-0.9048 z^{-1})^2}$$

8.5 BUTTERWORTH FILTERS

The Butterworth low-pass filter has a magnitude response given by

$$|H(j\Omega)| = \frac{A}{[1 + (\Omega/\Omega_c)^{2N}]^{0.5}} \quad (8.41)$$

where A is the filter gain and Ω_c is the 3 dB cut-off frequency and N is the order of the filter. The magnitude response of the Butterworth filter is shown in Fig. 8.6. The magnitude response has a maximally flat passband and stopband. It can be seen that by increasing the filter order

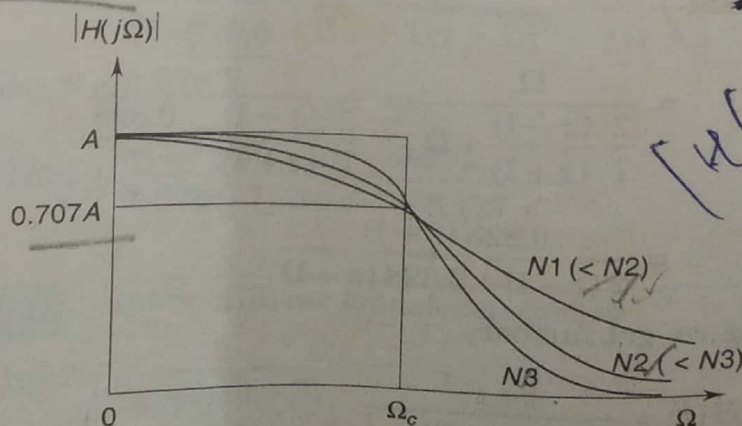


Fig. 8.6 Magnitude Response of a Butterworth Low-pass Filter

N , the Butterworth response approximates the ideal response. However, the phase response of the Butterworth filter becomes more non-linear with increasing N .

The design parameters of the Butterworth filter are obtained by considering the low-pass filter with the desired specifications as given below.

$$\delta_1 \leq |H(e^{j\omega})| \leq 1, 0 \leq \omega \leq \omega_1 \quad (8.42a)$$

$$|H(e^{j\omega})| \leq \delta_2, \omega_2 \leq \omega \leq \pi \quad (8.42b)$$

The corresponding analog magnitude response is to be obtained in the design process. Using Eq. 8.41 in Eq. 8.42 and if $A = 1$, we get

$$\delta_1^2 \leq \frac{1}{1 + (\Omega_1/\Omega_c)^{2N}} \leq 1 \quad (8.43a)$$

$$\frac{1}{1 + (\Omega_2/\Omega_c)^{2N}} \leq \delta_2^2 \quad (8.43b)$$

Equation 8.43 can be written in the form

$$(\Omega_1/\Omega_c)^{2N} \leq \frac{1}{\delta_1^2} - 1 \quad (8.44a)$$

$$(\Omega_2/\Omega_c)^{2N} \geq \frac{1}{\delta_2^2} - 1 \quad (8.44b)$$

Equality is assumed in Eq. 8.44 in order to obtain the filter order N and the 3 dB cut-off frequency Ω_c . Dividing Eq. 8.44b by Eq. 8.44a

$$(\Omega_2/\Omega_1)^{2N} = (1/\delta_2^2 - 1)/(1/\delta_1^2 - 1) \quad (8.45)$$

From Eq. 8.45, the order of the filter N is given by

$$N = \frac{1}{2} \frac{\log \{[(1/\delta_2^2 - 1)/(1/\delta_1^2 - 1)]\}}{\log (\Omega_2/\Omega_1)} \quad (8.46)$$

The value of N is chosen to be next nearest integer to the value of N as given by Eq. 8.46. Using Eq. 8.44a, we get

$$\Omega_c = \frac{\Omega_1}{[(1/\delta_1^2) - 1]^{1/2N}} \quad (8.47)$$

The values of Ω_1 and Ω_2 are obtained using the bilinear transformation or impulse invariant transformation techniques, $\Omega = \frac{2}{T} \tan\left(\frac{\omega}{2}\right)$ for bilinear transformation or $\Omega = \frac{\omega}{T}$ for the impulse invariant transformation.

The transfer function of the Butterworth filter is usually written in the factored form as given below.

$$H(s) = \prod_{k=1}^{N/2} \frac{B_k \Omega_c^2}{s^2 + b_k \Omega_c s + c_k \Omega_c^2} \quad N = 2, 4, 6, \dots \quad (8.48)$$

or

$$H(s) = \frac{B_0 \Omega_c}{s + c_0 \Omega_c} \prod_{k=1}^{(N-1)/2} \frac{B_k \Omega_c^2}{s^2 + b_k \Omega_c s + c_k \Omega_c^2} \quad N = 3, 5, 7, \dots \quad (8.49)$$

The coefficients b_k and c_k are given by

$$b_k = 2 \sin [(2k-1) \pi / 2N] \text{ and } c_k = 1 \quad (8.50)$$

The parameter B_k can be obtained from

$$A = \prod_{k=1}^{N/2} B_k, \text{ for even } N$$

and

$$A = \prod_{k=1}^{(N-1)/2} B_k, \text{ for odd } N \quad (8.51)$$

The system function of the equivalent digital filter is obtained from $H(s)$ (Eq. 8.48 or Eq. 8.49) using the specified transformation technique, viz. impulse invariant technique or bilinear transformation.

For bilinear transformation, $s = \frac{2}{T} \left(\frac{1-z^{-1}}{1+z^{-1}} \right) = \frac{2}{T} \left(\frac{z-1}{z+1} \right)$, and for

impulse invariant transformation, $\frac{1}{s-p_i} \rightarrow \frac{1}{1-e^{p_i T} z^{-1}}$ or Eq. 8.26, 8.27 or 8.28 may be used.

Poles of a normalised Butterworth filter

The Butterworth low-pass filter has a magnitude squared response given by

$$|H(j\Omega)|^2 = \frac{1}{1 + (\Omega/\Omega_c)^{2N}}$$

For a normalised filter, $\Omega_c = 1$. Thus,

$$|H(j\Omega)|^2 = \frac{1}{1 + \Omega^{2N}}$$

The normalised poles in the s -domain can be obtained by substituting $\Omega = s/j$ and equating the denominator polynomial to zero, i.e.

$$1 + \left(\frac{s}{j} \right)^{2N} = 0, \text{ or } 1 + (-s^2)^N = 0$$

The solution to the above expression gives us the poles of the filter. The above expression can be written as

$$(-1)^N s^{2N} = -1$$

Expressing -1 in the polar form,

$$(-1)^N s^{2N} = e^{j(2n-1)\pi}, \quad n = 1, 2, \dots, N$$

The poles in the left-half of the s -plane are given by,

$$s_n = \sigma_n + j\Omega_n = e^{j(2n+N-1)\pi/2N} = je^{j(2n-1)\pi/2N}$$

Using the polar to rectangular conversion, $e^{j\theta} = \cos \theta + j \sin \theta$, the normalised poles are obtained as

$$s_n = -\sin\left(\frac{2n-1}{2N}\pi\right) + j \cos\left(\frac{2n-1}{2N}\pi\right)$$

where

$$n = \begin{cases} 1, 2, \dots, (N+1)/2, & \text{for } N \text{ odd} \\ 1, 2, \dots, N/2, & \text{for } N \text{ even} \end{cases}$$

The unnormalised poles, s'_n , can also be obtained from the normalised poles as shown below,

$$s'_n = s_n (\Omega_c)^{-1/N}$$

The normalised poles lie on the unit circle spaced π/N apart.

Example 8.11 Determine $H(z)$ for a Butterworth filter satisfying the following constraints

$$\begin{aligned} \sqrt{0.5} \leq |H(e^{j\omega})| &\leq 1 & 0 \leq \omega \leq \pi/2 \\ |H(e^{j\omega})| &\leq 0.2 & 3\pi/4 \leq \omega \leq \pi \end{aligned}$$

with $T = 1$ s. Apply impulse invariant transformation.

Solution Given $\delta_1 = \sqrt{0.5} = 0.707$, $\delta_2 = 0.2$, $\omega_1 = \pi/2$ and $\omega_2 = 3\pi/4$.

Step (i) Determination of the analog filter's edge frequencies.

From Eq. 8.25,

$$\Omega_1 = \frac{\omega_1}{T} = \frac{\pi}{2} \quad \text{and} \quad \Omega_2 = \frac{\omega_2}{T} = \frac{3\pi}{4}$$

Therefore, $\Omega_2/\Omega_1 = 1.5$

Step (ii) Determination of the order of the filter.

From Eq. 8.46,

$$\begin{aligned} N &\geq \frac{1}{2} \frac{\log \{[(1/\delta_2^2) - 1]/[(1/\delta_1^2) - 1]\}}{\log(\Omega_2/\Omega_1)} \\ &= \frac{1}{2} \frac{\log \{24/1\}}{\log(1.5)} = 3.91 \end{aligned}$$

Let $N = 4$.

Step (iii) Determination of -3 dB cut-off frequency.

From Eq. 8.47,

$$\Omega_c = \frac{\Omega_1}{[(1/\delta_1^2) - 1]^{1/2N}} = \frac{\pi/2}{[(1/0.707^2) - 1]^{1/8}} = \frac{\pi}{2}$$

Step (iv) Determination of $H_a(s)$.

From Eq. 8.48,

$$H(s) = \prod_{k=1}^{N/2} \frac{B_k \Omega_c^2}{s^2 + b_k \Omega_c s + c_k \Omega_c^2}$$

$$= \left(\frac{B_1 \Omega_c^2}{s^2 + b_1 \Omega_c s + c_1 \Omega_c^2} \right) \left(\frac{B_2 \Omega_c^2}{s^2 + b_2 \Omega_c s + c_2 \Omega_c^2} \right)$$

From Eq. 8.50,

$$b_1 = 2 \sin \frac{\pi}{8} = 0.76536, \quad c_1 = 1$$

$$b_2 = 2 \sin \frac{3\pi}{8} = 1.84776, \quad c_2 = 1$$

$$B_1 B_2 = 1. \text{ Therefore } B_1 = B_2 = 1.$$

Therefore,

$$H(s) = \left(\frac{2.467}{s^2 + 1.2022s + 2.467} \right) \left(\frac{2.467}{s^2 + 2.9025s + 2.467} \right)$$

Using partial fractions,

$$H(s) = \left(\frac{As + B}{s^2 + 1.2022s + 2.467} \right) + \left(\frac{Cs + D}{s^2 + 2.9025s + 2.467} \right)$$

Comparing the above two expressions, we get

$$6.086 = (s^2 + 2.9025s + 2.467)(As + B) + (s^2 + 1.2022s + 2.467)(Cs + D)$$

Comparing the coefficients of s^3 , s^2 , s and the constants, we get a set of simultaneous equations.

$$A + C = 0$$

$$2.9025A + B + 1.2022C + D = 0$$

$$2.467A + 2.9025B + 2.467C + 1.2022D = 0$$

$$B + D = 2.467$$

Solving, we get $A = -1.4509$, $B = -1.7443$, $C = 1.4509$ and $D = 4.2113$.
Therefore,

$$H(s) = - \left(\frac{1.4509s + 1.7443}{s^2 + 1.2022s + 2.467} \right) + \left(\frac{1.4509s + 4.2113}{s^2 + 2.9025s + 2.467} \right)$$

Let $H(s) = H_1(s) + H_2(s)$, where

$$H_1(s) = - \left(\frac{1.4509s + 1.7443}{s^2 + 1.2022s + 2.467} \right) \text{ and } H_2(s) = \left(\frac{1.4509s + 4.2113}{s^2 + 2.9025s + 2.467} \right)$$

Rearranging $H_1(s)$ into the standard form,

$$H_1(s) = - \left(\frac{1.4509s + 1.7443}{s^2 + 1.2022s + 2.467} \right)$$

$$\begin{aligned}
 &= -1.4509 \left(\frac{s + 1.2022}{(s + 0.601)^2 + 1.451^2} \right) \\
 &= (-1.4509) \left[\frac{s + 0.601}{(s + 0.601)^2 + 1.451^2} + \frac{0.601}{(s + 0.601)^2 + 1.451^2} \right] \\
 &= (-1.4509) \left(\frac{s + 0.601}{(s + 0.601)^2 + 1.451^2} \right) - (0.601) \left(\frac{1.451}{(s + 0.601)^2 + 1.451^2} \right)
 \end{aligned}$$

Similarly, $H_2(s)$ can be written as

$$H_2(s) = (1.4509) \left(\frac{s + 1.45}{(s + 1.45)^2 + 0.604^2} \right) + (3.4903) \left(\frac{0.604}{(s + 1.45)^2 + 0.604^2} \right)$$

Step (v) Determination of $H(z)$. Using Eqs. 8.27 and 8.28,

$$\begin{aligned}
 H_1(z) &= (-1.4509) \frac{1 - e^{-0.601T} (\cos 1.451T) z^{-1}}{1 - 2e^{-0.601T} (\cos 1.451T) z^{-1} + e^{-1.202T} z^{-2}} \\
 &\quad - (0.601) \frac{e^{-0.601T} (\sin 1.451T) z^{-1}}{1 - 2e^{-0.601T} (\cos 1.451T) z^{-1} + e^{-1.202T} z^{-2}}
 \end{aligned}$$

and

$$\begin{aligned}
 H_2(z) &= (1.4509) \frac{1 - e^{-1.45T} (\cos 0.604T) z^{-1}}{1 - 2e^{-1.45T} (\cos 0.604T) z^{-1} + e^{-2.9T} z^{-2}} \\
 &\quad + (3.4903) \frac{e^{-1.45T} (\sin 0.604T) z^{-1}}{1 - 2e^{-1.45T} (\cos 0.604T) z^{-1} + e^{-2.9T} z^{-2}}
 \end{aligned}$$

where $H(z) = H_1(z) + H_2(z)$.

Upon simplifying we get,

$$H(z) = \frac{-1.4509 - 0.2321z^{-1}}{1 - 0.1310z^{-1} + 0.3006z^{-2}} + \frac{1.4509 + 0.1848z^{-1}}{1 - 0.3862z^{-1} + 0.055z^{-2}}$$

Example 8.12 Design a digital Butterworth filter that satisfies the following constraint using bilinear transformation. Assume $T = 1$ s.

$$\begin{aligned}
 0.9 \leq |H(e^{j\omega})| &\leq 1 & 0 \leq \omega \leq \pi/2 \\
 |H(e^{j\omega})| &\leq 0.2 & 3\pi/4 \leq \omega \leq \pi
 \end{aligned}$$

Solution Given $\delta_1 = 0.9$, $\delta_2 = 0.2$, $\omega_1 = \pi/2$ and $\omega_2 = 3\pi/4$.

Step (i) Determination of the analog filter's edge frequencies.

From Eq. 8.39,

$$\Omega_1 = \frac{2}{T} \tan \frac{\omega_1}{2} = 2 \tan \frac{\pi}{4} = 2 \text{ and } \Omega_2 = \frac{2}{T} \tan \frac{\omega_2}{2} = 2 \tan \frac{3\pi}{8} = 4.828$$

Therefore, $\Omega_2/\Omega_1 = 2.414$

Step (ii) Determination of the order of the filter.

From Eq. 8.46,

$$N \geq \frac{1}{2} \frac{\log \left\{ [(1/\delta_2^2 - 1)/(1/\delta_1^2 - 1)] \right\}}{\log (\Omega_2/\Omega_1)}$$

$$= \frac{1}{2} \frac{\log \{24/0.2346\}}{\log (2.414)} = 2.626$$

Let $N = 3$.

Step (iii) Determination of -3 dB cut-off frequency.

From Eq. 8.47,

$$\Omega_c = \frac{\Omega_1}{[(1/\delta_1^2) - 1]^{1/2N}} = \frac{2}{[(1/0.9^2) - 1]^{1/6}} = 2.5467$$

Step (iv) Determination of $H_a(s)$.

From Eq. 8.49,

$$H(s) = \frac{B_0 \Omega_c}{s + c_0 \Omega_c} \prod_{k=1}^{(N-1)/2} \frac{B_k \Omega_c^2}{s^2 + b_k \Omega_c s + c_k \Omega_c^2}$$

$$= \left(\frac{B_0 \Omega_c}{s + c_0 \Omega_c} \right) \left(\frac{B_1 \Omega_c^2}{s^2 + b_1 \Omega_c s + c_1 \Omega_c^2} \right)$$

From Eq. 8.50,

$$b_1 = 2 \sin \frac{\pi}{6} = 1, \quad c_0 = 1 \quad c_1 = 1$$

$$B_0 B_1 = 1. \text{ Therefore } B_0 = B_1 = 1.$$

Therefore,

$$H(s) = \left(\frac{2.5467}{s + 2.5467} \right) \left(\frac{6.4857}{s^2 + 2.5467s + 6.4857} \right)$$

Step (v) Determination of $H(z)$.

$$H(z) = H(s) \Big|_{s = \frac{2}{T} \frac{(z-1)}{(z+1)}}$$

That is,

$$H(z) = \left(\frac{2.5467}{2 \frac{(z-1)}{(z+1)} + 2.5467} \right) \left(\frac{6.4857}{\left[2 \frac{(z-1)}{(z+1)} \right]^2 + 2.5467s + 6.4857} \right)$$

Simplifying we get,

$$H(z) = \frac{16.5171(z+1)^3}{70.83z^3 + 31.1205z^2 + 27.2351z + 2.948}$$

or

$$H(z) = \frac{0.2332 (1 + z^{-1})^3}{1 + 0.4394 z^{-1} + 0.3845 z^{-2} + 0.0416 z^{-3}}$$

8.6 CHEBYSHEV FILTERS

The Chebyshev low-pass filter has a magnitude response given by

$$|H(j\Omega)| = \frac{A}{[1 + \epsilon^2 C_N^2(\Omega/\Omega_c)]^{0.5}} \quad (8.52)$$

where A is the filter gain, ϵ is a constant and Ω_c is the 3 dB cut-off frequency. The Chebyshev polynomial of the I kind of N th order, $C_N(x)$ is given by

$$C_N(x) = \begin{cases} \cos(N \cos^{-1} x), & \text{for } |x| \leq 1 \\ \cosh(N \cosh^{-1} x), & \text{for } |x| \geq 1 \end{cases} \quad (8.53)$$

The magnitude response of the Chebyshev filter is shown in Fig. 8.7. The magnitude response has equiripple passband and maximally flat stopband. It can be seen that by increasing the filter order N , the Chebyshev response approximates the ideal response. The phase response of the Chebyshev filter is more non-linear than the Butterworth filter for a given filter length N .

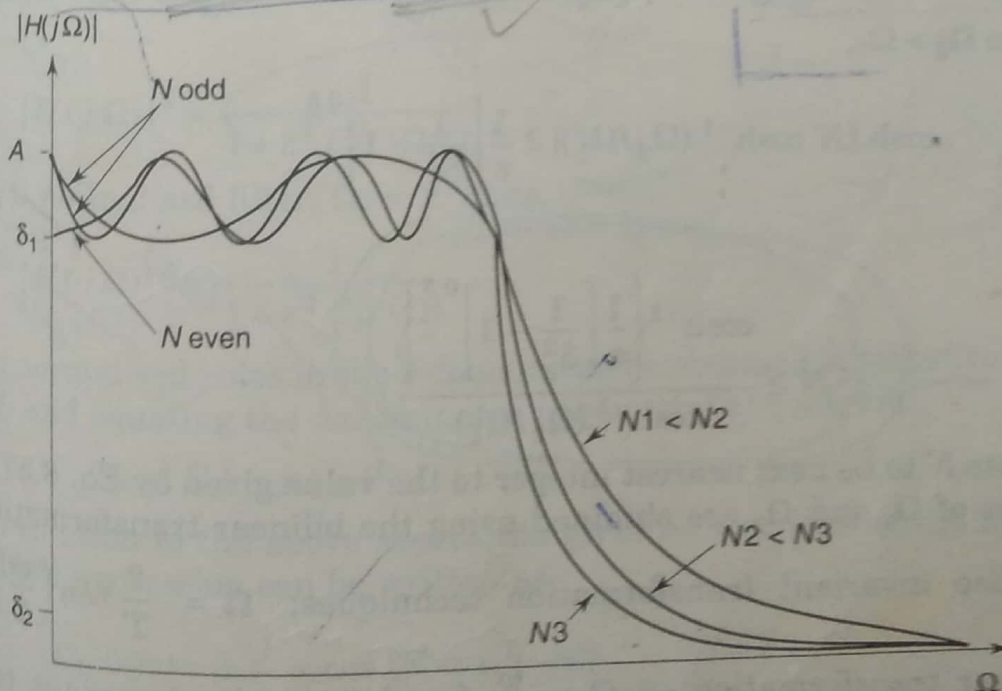


Fig. 8.7 Magnitude Response of a Low-pass Chebyshev Filter

The design parameters of the Chebyshev filter are obtained by considering the low-pass filter with the desired specifications as given below.

$$\delta_1 \leq |H(e^{j\omega})| \leq 1 \quad 0 \leq \omega \leq \omega_1 \quad (8.54a)$$

$$|H(e^{j\omega})| \leq \delta_2 \quad \omega_2 \leq \omega \leq \pi \quad (8.54b)$$

The corresponding analog magnitude response is to be obtained in the design process. Using Eq. 8.52 in 8.54 and if $A = 1$, we get

$$\delta_1^2 \leq \frac{1}{1 + \epsilon^2 C_N^2(\Omega_1/\Omega_c)} \leq 1 \quad (8.55a)$$

$$= \frac{1}{1 + \epsilon^2 C_N^2(\Omega_2/\Omega_c)} \leq \delta_2^2 \quad (8.55b)$$

Assuming $\Omega_c = \Omega_1$, we will have $C_N(\Omega_c/\Omega_c) = C_N(1) = 1$. Therefore, Eq. 8.55a can be written as

$$\delta_1^2 \leq \frac{1}{1 + \epsilon^2}$$

Assuming equality in the above equation, the expression for ϵ is

$$\epsilon = \left[\frac{1}{\delta_1^2} - 1 \right]^{0.5} \quad (8.56)$$

The order of the analog filter, N can be determined from Eq. 8.55b. Assuming $\Omega_c = \Omega_1$,

$$C_N(\Omega_2/\Omega_1) \geq \frac{1}{\epsilon} \left[\frac{1}{\delta_2^2} - 1 \right]^{0.5}$$

Since $\Omega_2 > \Omega_1$,

$$\cosh [N \cosh^{-1}(\Omega_2/\Omega_1)] \geq \frac{1}{\epsilon} \left[\frac{1}{\delta_2^2} - 1 \right]^{0.5}$$

or

$$N \geq \frac{\cosh^{-1} \left\{ \frac{1}{\epsilon} \left[\frac{1}{\delta_2^2} - 1 \right]^{0.5} \right\}}{\cosh^{-1}(\Omega_2/\Omega_1)} \quad (8.57)$$

Choose N to be next nearest integer to the value given by Eq. 8.57. The values of Ω_1 and Ω_2 are obtained using the bilinear transformation or impulse invariant transformation techniques; $\Omega = \frac{2}{T} \tan\left(\frac{\omega}{2}\right)$ for

bilinear transformation or $\Omega = \frac{\omega}{T}$ for the impulse invariant transformation.

The transfer function of Chebyshev filters are usually written in the factored form as given in Eqs. 8.48 or 8.49. The coefficients b_k and c_k are given by

$$b_k = 2y_N \sin [(2k-1)\pi/2N]$$

$$c_k = y_N^2 + \cos^2 \frac{(2k-1)\pi}{2N} \quad (8.58)$$

$$c_0 = y_N$$

The parameter y_N is given by

$$y_N = \frac{1}{2} \left\{ \left[\left(\frac{1}{\epsilon^2} + 1 \right)^{0.5} + \frac{1}{\epsilon} \right]^{\frac{1}{N}} - \left[\left(\frac{1}{\epsilon^2} + 1 \right)^{0.5} + \frac{1}{\epsilon} \right]^{-\frac{1}{N}} \right\} \quad (8.59)$$

The parameter B_k can be obtained from

$$\frac{A}{(1 + \epsilon^2)^{0.5}} = \prod_{k=1}^{N/2} \frac{B_k}{c_k}, \text{ for } N \text{ even}$$

and

$$A = \prod_{k=0}^{\frac{N-1}{2}} \frac{B_k}{c_k} \text{ for } N \text{ odd.}$$

The system function of the equivalent digital filter is obtained from $H(s)$ (Eq. 8.48 or Eq. 8.49) using the specified transformation technique, viz. impulse invariant technique or Bilinear transformation.

Poles of a Normalised Chebyshev Filter

The Chebyshev low-pass filter has a magnitude squared response given by

$$|H(j\Omega)|^2 = \frac{1}{1 + \epsilon^2 C_N^2(\Omega/\Omega_c)}$$

For a normalised filter, $\Omega_c = 1$. Thus,

$$|H(j\Omega)|^2 = \frac{1}{1 + \epsilon^2 C_N^2(\Omega)}$$

The normalised poles in the s -domain can be obtained by substituting $\Omega = s/j$ and equating the denominator polynomial to zero, i.e.

$$1 + \epsilon^2 C_N^2(-js) = 0$$

The solution to the above expression gives us the poles of the filter. The above expression can be written as

$$C_N(-js) = \pm \frac{j}{\epsilon} = \cos [N \cos^{-1}(-js)]$$

The cosine term in the above equation has a complex argument. Using trigonometric identities for the imaginary terms and with minor manipulations, the poles of the normalised low-pass analog Chebyshev filter is given by

$$s_n = -\sin x \sinh y + j \cos x \cosh y = \sigma_n + j \Omega_n$$

where

$$n = \begin{cases} 1, 2, \dots, (N+1)/2 & \text{for } N \text{ odd} \\ 1, 2, \dots, N/2 & \text{for } N \text{ even} \end{cases}$$

and

$$x = (2n-1) \frac{\pi}{2N} \quad n = 1, 2, \dots, N$$

$$y = \pm \frac{1}{N} \sinh^{-1} \left(\frac{1}{\varepsilon} \right)$$

The unnormalised poles, s'_n , can also be obtained from the normalised poles as shown below,

$$s'_n = s_n \Omega_c$$

The normalised poles lie on an ellipse in the s -plane and the equation of the ellipse is given by

$$\frac{\sigma_n^2}{\sinh^2 y} + \frac{\Omega_n^2}{\cosh^2 y} = 1$$

Example 8.13 Design a digital Chebyshev filter to satisfy the constraints

$$\begin{aligned} 0.707 \leq |H(e^{j\omega})| &\leq 1, & 0 \leq \omega \leq 0.2\pi \\ |H(e^{j\omega})| &\leq 0.1, & 0.5\pi \leq \omega \leq \pi \end{aligned}$$

using bilinear transformation and assuming $T = 1$ s.

Solution Given $\delta_1 = 0.707$, $\delta_2 = 0.1$, $\omega_1 = 0.2\pi$ and $\omega_2 = 0.5\pi$.

Step (i) Determination of the analog filter's edge frequencies.

From Eq. 8.39,

$$\Omega_c = \Omega_1 = \frac{2}{T} \tan \frac{\omega_1}{2} = 2 \tan 0.1\pi = 0.6498$$

$$\Omega_2 = \frac{2}{T} \tan \frac{\omega_2}{2} = 2 \tan 0.25\pi = 2$$

Therefore, $\Omega_2/\Omega_1 = 3.0779$

Step (ii) Determination of the order of the filter.

From Eq. 8.56,

$$\varepsilon = \left[\frac{1}{\delta_1^2} - 1 \right]^{0.5} = \left[\frac{1}{0.707^2} - 1 \right]^{0.5} = 1$$

From Eq. 8.57,

$$N \geq \frac{\cosh^{-1} \left\{ \frac{1}{\varepsilon} \left[\frac{1}{\delta_2^2} - 1 \right]^{0.5} \right\}}{\cosh^{-1} (\Omega_2/\Omega_1)} = \frac{\cosh^{-1} \left\{ 1 \left[\frac{1}{0.1^2} - 1 \right]^{0.5} \right\}}{\cosh^{-1} (3.0779)} = 1.669$$

Let $N=2$.

Step (iii) Determination of $H(s)$.

From Eq. 8.48,

$$H(s) = \prod_{k=1}^{N/2} \frac{B_k \Omega_c^2}{s^2 + b_k \Omega_c s + c_k \Omega_c^2} = \frac{B_1 \Omega_c^2}{s^2 + b_1 \Omega_c s + c_1 \Omega_c^2}$$

From Eqs. 8.58, 8.59 and 8.60,

$$y_N = \frac{1}{2} \left\{ \left[\left(\frac{1}{\epsilon^2} + 1 \right)^{0.5} + \frac{1}{\epsilon} \right]^{\frac{1}{N}} - \left[\left(\frac{1}{\epsilon^2} + 1 \right)^{0.5} + \frac{1}{\epsilon} \right]^{-\frac{1}{N}} \right\}$$

$$= \frac{1}{2} \left\{ [2.414]^{\frac{1}{2}} - [2.414]^{-\frac{1}{2}} \right\} = 0.455$$

$$b_1 = 2y_2 \sin [(2k-1)\pi/2N] = 0.6435$$

$$c_1 = y_2^2 + \cos^2 \frac{(2k-1)\pi}{2N} = 0.707$$

For N even,

$$\prod_{k=1}^{N/2} \frac{B_k}{c_k} = \frac{A}{(1+\epsilon^2)^{0.5}} = 0.707$$

That is,

$$\frac{B_1}{c_1} = 0.707, \text{ and hence } B_1 = 0.5$$

The system function is

$$H(s) = \frac{0.5 (0.6498)^2}{s^2 + (0.6435) (0.6498) s + (0.707) (0.6498)^2}$$

On simplifying we get,

$$H(s) = \frac{0.2111}{s^2 + 0.4181s + 0.2985}$$

Step (iv) Determination of $H(z)$. Using bilinear transformation,

$$H(z) = H(s) \Big|_{s = \frac{2}{T} \frac{(z-1)}{(z+1)}}$$

That is,

$$\begin{aligned} H(z) &= \frac{0.2111}{\left(2 \frac{(z-1)}{(z+1)} \right)^2 + 0.4181 \left(2 \frac{(z-1)}{(z+1)} \right) + 0.2985} \\ &= \frac{0.2111 (z+1)^2}{5.1347 z^2 - 7.403 z - 3.4623} \end{aligned}$$

Rearranging,

$$H(z) = \frac{0.041(1+z^{-1})^2}{1-1.4418z^{-1}+0.6743z^{-2}}$$

8.7 INVERSE CHEBYSHEV FILTERS

Inverse Chebyshev filters are also called Type-II Chebyshev filters. A low-pass inverse Chebyshev filter has a magnitude response given by

$$|H(j\Omega)| = \frac{\epsilon C_N(\Omega_2/\Omega)}{[1 + \epsilon^2 C_N^2(\Omega_2/\Omega)]^{0.5}} \quad (8.61)$$

where ϵ is a constant and Ω_c is the 3 dB cut-off frequency. The Chebyshev polynomial, $C_N(x)$ is given by Eq. 8.53.

The magnitude response of the inverse Chebyshev filter is shown in Fig. 8.8. The magnitude response has maximally flat passband and equiripple stopband, just the opposite of the Chebyshev filter's response. That is why the Type-II Chebyshev filters are called the inverse Chebyshev filters.

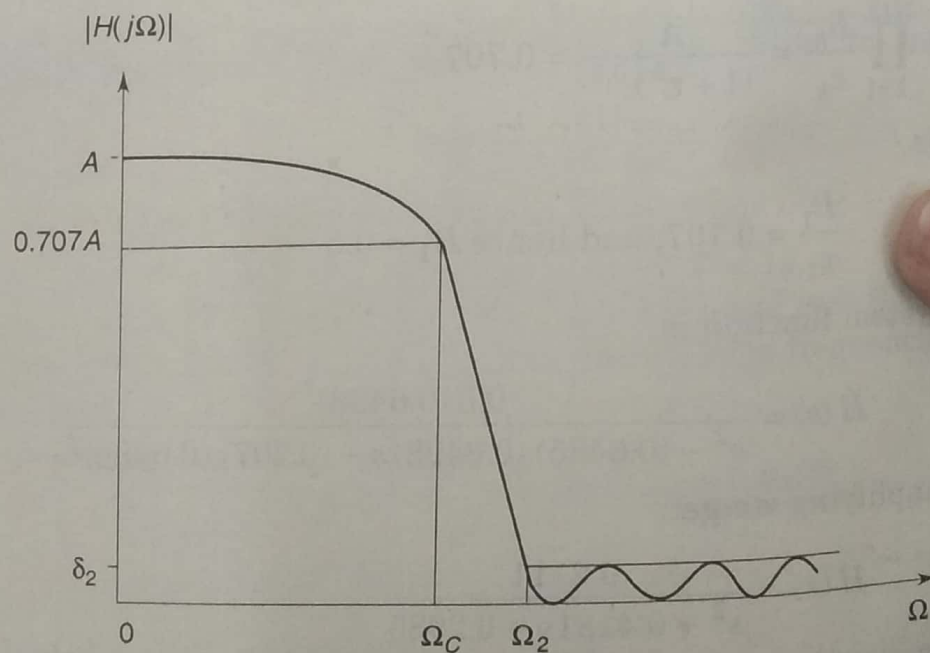


Fig. 8.8 Magnitude Response of the Low-pass Inverse Chebyshev Filter

The parameters of the inverse Chebyshev filter are obtained by considering the low-pass filter with the desired specifications as given below.

$$0.707 \leq |H(j\Omega)| \leq 1, \quad 0 \leq \Omega \leq \Omega_c \quad (8.62a)$$

$$|H(j\Omega)| \leq \delta_2, \quad \Omega \geq \Omega_2 \quad (8.62b)$$

Using Eq. 8.61 in Eq. 8.62, we get,

$$0.707^2 \leq \frac{\epsilon^2 C_N^2(\Omega_2/\Omega)}{1 + \epsilon^2 C_N^2(\Omega_2/\Omega)} \leq 1, \quad 0 \leq \Omega \leq \Omega_c \quad (8.63a)$$

$$\frac{\epsilon^2 C_N^2(\Omega_2/\Omega)}{1 + \epsilon^2 C_N^2(\Omega_2/\Omega)} \leq \delta_2^2, \quad \Omega \geq \Omega_2 \quad (8.63b)$$

When $\Omega = \Omega_2$, Eq. 8.63b becomes

$$\delta_2^2 = \frac{\epsilon^2}{1 + \epsilon^2}$$

Rearranging,

$$\epsilon = \frac{\delta_2}{(1 - \delta_2^2)^{0.5}} \quad (8.64)$$

When $\Omega = \Omega_c$, Eq. 8.63a becomes,

$$0.5 = \frac{\epsilon^2 C_N^2(\Omega_2/\Omega_c)}{1 + \epsilon^2 C_N^2(\Omega_2/\Omega_c)}$$

$$0.5 + 0.5 \epsilon^2 C_N^2(\Omega_2/\Omega_c) = \epsilon^2 C_N^2(\Omega_2/\Omega_c)$$

or, simplifying

$$C_N(\Omega_2/\Omega_c) = \frac{1}{\epsilon} \quad (8.65)$$

Using Eq. 8.53,

$$\cosh [N \cosh^{-1}(\Omega_2/\Omega_c)] = \frac{1}{\epsilon} \quad (8.66)$$

From Eqs. 8.66 and 8.64 we can get the order of the filter, N .

$$N = \frac{\cosh^{-1}(1/\epsilon)}{\cosh^{-1}(\Omega_2/\Omega_c)} = \frac{\cosh^{-1} \left[\frac{1}{\delta_2^2} - 1 \right]^{0.5}}{\cosh^{-1}(\Omega_2/\Omega_c)} \quad (8.67)$$

The value of N is chosen to be the nearest integer greater than the value given by Eq. 8.67.

8.8 ELLIPTIC FILTERS

The elliptic filter is sometimes called the Caier filter. This filter has equiripple passband and stopband. Among the filter types discussed so far, for a given filter order, passband and stopband deviations, elliptic filters have the minimum transition bandwidth. The magnitude response of an odd ordered elliptic filter is shown in Fig. 8.9. The magnitude squared response is given by

$$|H(j\Omega)|^2 = \frac{1}{1 + \epsilon^2 U_N(\Omega/\Omega_c)} \quad (8.68)$$

where $U_N(x)$ is the Jacobian elliptic function of order N and ϵ is a constant related to the passband ripple.

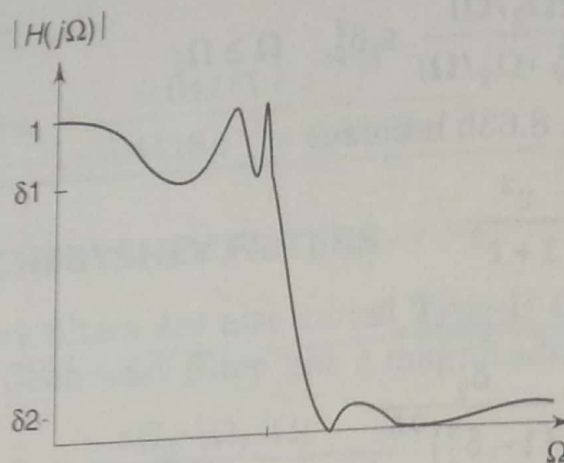


Fig. 8.9 Magnitude Response of a Low-pass Elliptic Filter

8.9 FREQUENCY TRANSFORMATION

There are basically four types of frequency selective filters, viz., low-pass, highpass, bandpass and bandstop. In the design techniques discussed so far we have considered only lowpass filters. This low-pass filter can be considered as a prototype filter and its system function can be obtained. Then, if a highpass or bandpass or bandstop filter is to be designed, it can be easily obtained by using frequency transformation. Frequency transformation can be accomplished in two ways. In the analog frequency transformation, the analog system function $H_p(s)$ of the prototype filter is converted into another analog system function $H(s)$ of the desired filter. Then using any of the mapping techniques, it is converted into the digital filter having a system function $H(z)$. In the digital frequency transformation, the analog prototype filter is first transformed to the digital domain, to have a system function $H_p(z)$. Then using frequency transformation, it can be converted into the desired digital filter.

8.9.1 Analog Frequency Transformation

The frequency transformation formulae used to convert a prototype lowpass filter into a lowpass (with a different cut-off frequency), highpass, bandpass or bandstop are given below.

(i) Low-pass with cut-off frequency Ω_c to low-pass with a new cut-off frequency Ω_c^*

$$s \rightarrow \frac{\Omega_c}{\Omega_c^*} s \quad (8.69)$$

Thus, if the system response of the prototype filter is $H_p(s)$, the system response of the new low-pass filter will be

$$H(s) = H_p\left(\frac{\Omega_c}{\Omega_c^*} s\right) \quad (8.70)$$

Low-pass with cut-off frequency Ω_c to high-pass with cut-off frequency Ω_c^* :

$$s \rightarrow \frac{\Omega_c \Omega_c^*}{s} \quad (8.71)$$

The system function of the high-pass filter is then,

$$H(s) = H_p \left(\frac{\Omega_c \Omega_c^*}{s} \right) \quad (8.72)$$

(iii) Low-pass with cut-off frequency Ω_c to band-pass with lower cut-off frequency Ω_1 and higher cut-off frequency Ω_2 :

$$s \rightarrow \Omega_c \frac{s^2 + \Omega_1 \Omega_2}{s(\Omega_2 - \Omega_1)} \quad (8.73)$$

The system function of the high-pass filter is then

$$H(s) = H_p \left(\Omega_c \frac{s^2 + \Omega_1 \Omega_2}{s(\Omega_2 - \Omega_1)} \right) \quad (8.74)$$

(iv) Low-pass with cut-off frequency Ω_c to bandstop with lower cut-off frequency Ω_1 and higher cut-off frequency Ω_2 :

$$s \rightarrow \Omega_c \frac{s(\Omega_2 - \Omega_1)}{s^2 + \Omega_1 \Omega_2} \quad (8.75)$$

The system function of the band stop filter is then,

$$H(s) = H_p \left(\Omega_c \frac{s(\Omega_2 - \Omega_1)}{s^2 + \Omega_1 \Omega_2} \right) \quad (8.76)$$

Table 8.1 gives the analog frequency transformations formulae.

Table 8.1 Analog frequency transformation

Type	Transformation
Low-pass	$s \rightarrow \frac{\Omega_c}{\Omega_c^*} s$
High-pass	$s \rightarrow \frac{\Omega_c \Omega_c^*}{s}$
Bandpass	$s \rightarrow \Omega_c \frac{s^2 + \Omega_1 \Omega_2}{s(\Omega_2 - \Omega_1)}$
Bandstop	$s \rightarrow \Omega_c \frac{s(\Omega_2 - \Omega_1)}{s^2 + \Omega_1 \Omega_2}$

Example 8.14 A prototype low-pass filter has the system response $H(s) = \frac{1}{s^2 + 2s + 1}$. Obtain a bandpass filter with $\Omega_0 = 2$ rad/s and

$$Q = 10. \quad \Omega_0^2 = \Omega_1 \cdot \Omega_2 \text{ and } Q = \frac{\Omega_0}{\Omega_2 - \Omega_1}$$

Solution From Table 8.1, the required transformation is

$$s \rightarrow \Omega_c \frac{s^2 + \Omega_1 \Omega_2}{s(\Omega_2 - \Omega_1)}, \text{ i.e.}$$

$$s = \Omega_c \frac{s^2 + \Omega_0^2}{s(\Omega_0/Q)} = \Omega_c \frac{s^2 + 2^2}{s(2/10)} = 5\Omega_c \left(\frac{s^2 + 4}{s} \right)$$

Therefore,

$$H(s) = H(s) \Big|_{s=5\Omega_c \left(\frac{s^2+4}{s} \right)}$$

$$H(s) = \frac{0.04 s^2}{\Omega_c^2 s^4 + 0.4 \Omega_c s^3 + (8 \Omega_c^2 + 0.01) s^2 + 1.6 \Omega_c s + 16 \Omega_c^2}$$

Example 8.15 Transform the prototype low-pass filter with system function

$$H(s) = \frac{\Omega_c}{s + \Omega_c}$$

into a high-pass filter with cutoff frequency Ω_c^* .

Solution From the Table 8.1, the desired transformation is,

$$s \rightarrow \frac{\Omega_c \Omega_c^*}{s}$$

Thus we have,

$$\begin{aligned} H_{hpf}(s) &= \frac{\Omega_c}{\left(\frac{\Omega_c \Omega_c^*}{s} \right) + \Omega_c} \\ &= \frac{s}{s + \Omega_c^*} \end{aligned}$$

8.9.2 DIGITAL FREQUENCY TRANSFORMATION

As in the analog domain, frequency transformation is possible in the digital domain also. The frequency transformation is done in the digital domain by replacing the variable z^{-1} by a function of z^{-1} , i.e. $f(z^{-1})$. This mapping must take into account the stability criterion. All the poles lying within the unit circle must map onto itself and the unit circle must

also map onto itself. For the unit circle to map onto itself, the implication is that for $r = 1$,

$$e^{-j\omega} = f(e^{-j\omega})$$

$$= |f(e^{-j\omega})| e^{j \arg[f(e^{-j\omega})]}$$

Hence, we must have $|f(e^{-j\omega})| = 1$ for all frequencies. So, the mapping is that of an all pass filter and of the form

$$f(z^{-1}) = \pm \prod_{k=1}^n \frac{z^{-1} - a_k}{1 - a_k^* z^{-1}} \quad (8.77)$$

To get a stable filter from the stable prototype filter, we must have $|a_k| < 1$. The transformation formulae can be obtained from Eq. 8.77 for converting the prototype low-pass digital filter into a digital low-pass, high-pass, bandpass or bandstop filter. Table 8.2 gives these transformations.

Table 8.2 Digital frequency transformation

Type	Transformation	Design Parameter
Low-pass	$z^{-1} \rightarrow \frac{z^{-1} - a}{1 - a z^{-1}}$	$a = \frac{\sin [(\omega_c - \omega_c^*)/2]}{\sin [(\omega_c + \omega_c^*)/2]}$
High-pass	$z^{-1} \rightarrow -\frac{z^{-1} + a}{1 + a z^{-1}}$	$a = -\frac{\cos [(\omega_c - \omega_c^*)/2]}{\cos [(\omega_c + \omega_c^*)/2]}$
Bandpass	$z^{-1} \rightarrow -\frac{z^{-2} - a_1 z^{-1} + a_2}{a_2 z^{-2} - a_1 z^{-1} + 1}$	$a_1 = -2\alpha K/(K + 1)$ $a_2 = (K - 1)/(K + 1)$ $\alpha = \frac{\cos [(\omega_2 + \omega_1)/2]}{\cos [(\omega_2 - \omega_1)/2]}$ $K = \cot\left(\frac{\omega_2 - \omega_1}{2}\right) \tan\left(\frac{\omega_c}{2}\right)$
Bandstop	$z^{-1} \rightarrow \frac{z^{-2} - a_1 z^{-1} + a_2}{a_2 z^{-2} - a_1 z^{-1} + 1}$	$a_1 = -2\alpha/(K + 1)$ $a_2 = (1 - K)/(1 + K)$ $\alpha = \frac{\cos [(\omega_2 + \omega_1)/2]}{\cos [(\omega_2 - \omega_1)/2]}$ $K = \tan\left(\frac{\omega_2 - \omega_1}{2}\right) \tan\left(\frac{\omega_c}{2}\right)$

The frequency transformation may be accomplished in any of the available two techniques, however, caution must be taken to which technique to use. For example, the impulse invariant transformation is not suitable for high-pass or bandpass filters whose resonant frequencies are higher. In such a case, suppose a low-pass prototype filter is converted into a high-pass filter using *analog frequency*

transformation and transformed later to a digital filter using impulse invariant technique. This will result in aliasing problems. However, if the same prototype low-pass filter is first transformed into a digital filter using impulse-invariant technique and later converted into a high-pass filter using **digital frequency transformation** will not have any aliasing problem. Whenever the bilinear transformation is used, it is of no significance whether analog frequency transformation is used or digital frequency transformation. In this case, both analog and digital frequency transformation techniques will give the same result.



REVIEW QUESTIONS

- 8.1 What is an IIR filter ? Compare its characteristics with an FIR filter.
- 8.2 What are the requirements for converting a stable analog filter into a stable digital filter?
- 8.3 What are the different design techniques available for IIR filters?
- 8.4 Obtain the mapping formula for the approximation of derivatives method using backward difference.
- 8.5 Comment on the stability of backward difference approximation for the derivative method of transformation.
- 8.6 Why is the forward difference formula for the approximation of derivatives not used ?
- 8.7 Use the backward difference for the derivative to convert the analog low-pass filter with the following system function, using impulse invariant transformation.

$$H(s) = \frac{1}{(s + 2)}$$

- 8.8 Transform the analog filter with the transfer function shown below into a digital filter, using backward difference for the derivative.

$$H(s) = \frac{1}{(s + 2)(s + 3)}$$

- 8.9 What is the impulse invariant technique ?
- 8.10 Discuss the stability of the impulse invariant mapping technique.
- 8.11 Obtain the mapping formula for the impulse invariant transformation.

Realisation of Digital Linear Systems

9.1 INTRODUCTION

For the design of digital filters, the system function $H(z)$ or the impulse response $h(n)$ must be specified. Then the digital filter structure can be implemented or synthesised in hardware/software form by its difference equation obtained directly from $H(z)$ or $h(n)$. Each difference equation or computational algorithm can be implemented by using a digital computer or special purpose digital hardware or special programmable integrated circuit.

In order to implement the specified difference equation of the system, the required basic operations are addition, delay and multiplication by a constant. A chosen structure determines a computational algorithm. Generally, different structures give different results. The most common methods for realising digital linear systems are direct, cascade and parallel forms, and state variable realisations.

In this chapter, direct-forms, cascade and parallel forms, ladder form and state variable realisations of IIR filters, direct-form and cascade realisations of FIR filters, and realisations of linear phase FIR systems are discussed.

9.2 BASIC REALISATION BLOCK DIAGRAM AND THE SIGNAL-FLOW GRAPH

The output of a finite order linear time invariant system at time n can be expressed as a linear combination of the inputs and outputs,

$$y(n) = \sum_{k=1}^N a_k y(n-k) + \sum_{k=0}^M b_k x(n-k) \quad (9.1)$$

where a_k and b_k are constants with $a_0 \neq 0$ and $M \leq N$.

$$\sum_{k=1}^N a_k y(n-k) + \sum_{k=0}^M b_k x(n-k)$$

The current output $y(n)$ is equal to the sum of past outputs, from $y(n-1)$ to $y(n-N)$, which are scaled by the **delay-dependent feedback coefficient** a_k , plus the sum of future, present and past inputs, which are scaled by the **delay-dependent feed forward coefficient** b_k .

Taking the z -transform of the output sequence $y(n)$ given in Eq. 9.1, we get

$$\begin{aligned} Y(z) &= \sum_{n=-\infty}^{\infty} y(n) z^{-n} \\ &= \sum_{n=-\infty}^{\infty} \left(\sum_{k=1}^N a_k y(n-k) + \sum_{k=0}^M b_k x(n-k) \right) z^{-n} \end{aligned}$$

Changing the order of summation

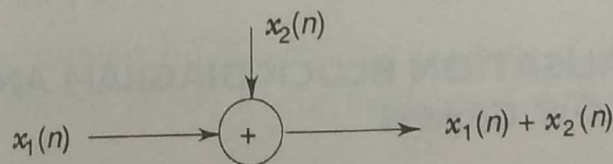
$$\begin{aligned} Y(z) &= \sum_{k=1}^N a_k \left\{ \sum_{n=-\infty}^{\infty} y(n-k) z^{-n} \right\} + \sum_{k=0}^M b_k \left\{ \sum_{n=-\infty}^{\infty} x(n-k) z^{-n} \right\} \\ &= \sum_{k=1}^N a_k z^{-k} Y(z) + \sum_{k=0}^M b_k z^{-k} X(z) \end{aligned}$$

$$Y(z) \left\{ 1 - \sum_{k=1}^N a_k z^{-k} \right\} = \sum_{k=0}^M b_k z^{-k} X(z)$$

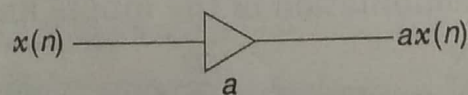
Therefore,

$$H(z) = \frac{Y(z)}{X(z)} = \frac{\sum_{k=0}^M b_k z^{-k}}{1 - \sum_{k=1}^N a_k z^{-k}} \quad (9.2)$$

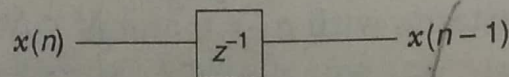
The computational algorithm of an LTI digital filter can be conveniently represented as a block diagram using basic building blocks representing the unit delay (z^{-1}) or storage element, the multiplier and the adder. These basic building blocks are shown in Fig. 9.1.



(a) Addition of two sequences



(b) Multiplier



(c) Unit delay

Fig. 9.1 Basic Building Blocks

For example, the difference equation of the first-order digital system may be written as

$$y(n] = a_1 y(n - 1) + x(n) + b_1 x(n - 1)$$

The basic realisation block diagram for this equation and the corresponding structure of the signal flow graph are shown in Figs 9.2 (a) and (b). Here, it is clear that there is direct correspondence between branches in the digital realisation structure and branches in the signal flow graph. But in the signal flow graph, the nodes represent both branch points and adders in the digital realisation block diagram.

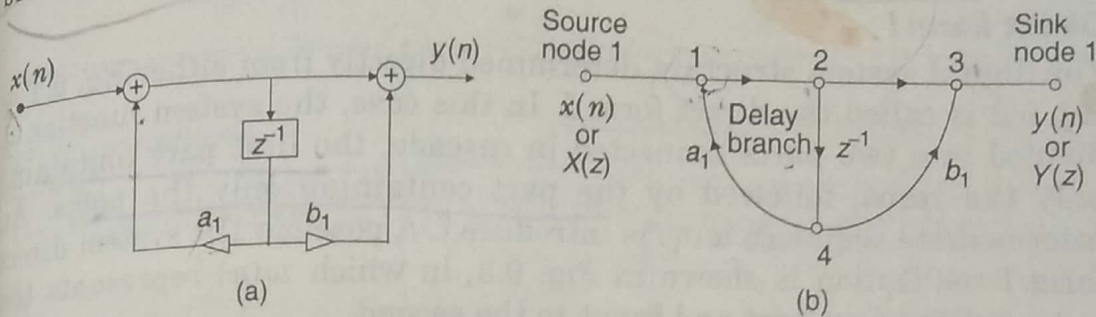


Fig. 9.2 (a) Basic Realisation Block Diagram Representing a First-order Digital System and (b) Its Corresponding Signal Flow Graph.

Advantages of representing the digital system in block diagram form

- Just by inspection, the computation algorithm can be easily written.
- The hardware requirements can be easily determined.
- A variety of equivalent block diagram representations can be easily developed from the transfer function.
- The relationship between the output and the input can be determined.

9.2.1 Canonic and Non-Canonic Structures

If the number of delays in the realisation block diagram is equal to the order of the difference equation or the order of the transfer function of a digital filter, then the realisation structure is called **canonic**. Otherwise, it is a **non-canonic** structure.

9.3 BASIC STRUCTURES FOR IIR SYSTEMS

Causal IIR systems are characterised by the constant coefficient difference equation of Eq. 9.1 or equivalently, by the real rational transfer function of Eq. 9.2. From these equations, it can be seen that the realisation of infinite duration impulse response (IIR) systems involves a recursive computational algorithm. In this section, the most important filter structures namely direct Forms I and II, cascade and parallel realisations for IIR systems are discussed.

9.3.1 Direct Form Realisation of IIR Systems

Equation 9.2 is the standard form of the system transfer function. By inspection of this equation, the block diagram representation can be drawn directly for the **direct form realisation**. The multipliers in the feed forward paths are the numerator coefficients and the multipliers in the feedback paths are the negatives of the denominator coefficients. Since the multiplier coefficients in the structures are exactly the coefficients of the transfer function, they are called direct form structures.

Direct Form I

The digital system structure determined directly from either Eq. 9.1 or Eq. 9.2 is called the direct form I. In this case, the system function is divided into two parts connected in cascade, the first part containing only the zeros, followed by the part containing only the poles. An intermediate sequence $w(n)$ is introduced. A possible IIR system direct form I realisation is shown in Fig. 9.3, in which $w(n)$ represents the output of the first part and input to the second.

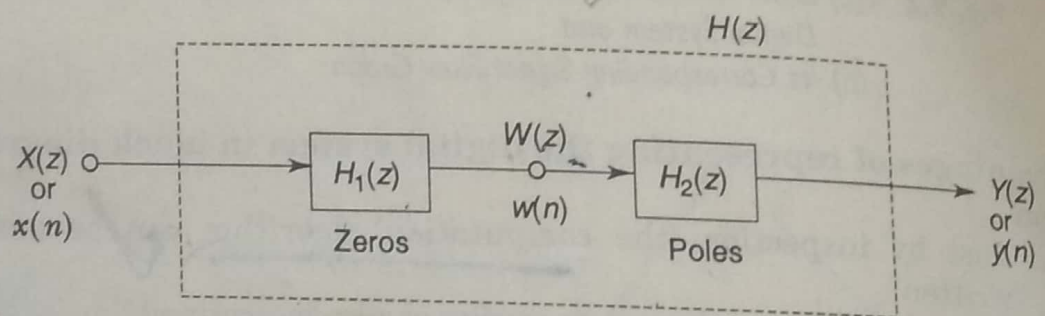


Fig. 9.3 A Possible IIR System direct-form I Realisation

$$w(n) = \sum_{k=0}^M b_k x(n-k)$$

and

$$y(n) = \sum_{k=1}^N a_k y(n-k) + w(n)$$

Taking z-transform, we get

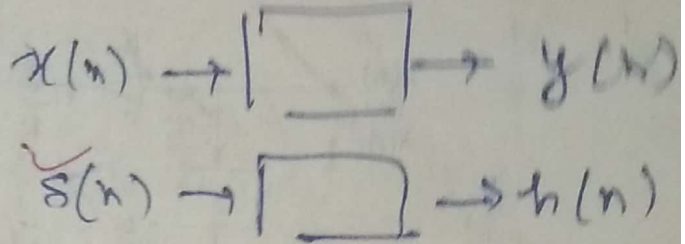
$$W(z) = X(z) \sum_{k=0}^M b_k z^{-k}$$

and

$$Y(z) = Y(z) \sum_{k=1}^N a_k z^{-k} + W(z)$$

Therefore,

$$Y(z) = \frac{W(z)}{1 - \sum_{k=1}^N a_k z^{-k}}$$



The direct form I realisation is shown in Fig. 9.4. The direct form I realisation requires $M + N$ storage elements. For our convenience, we have assumed that $M = N$, for drawing the network.

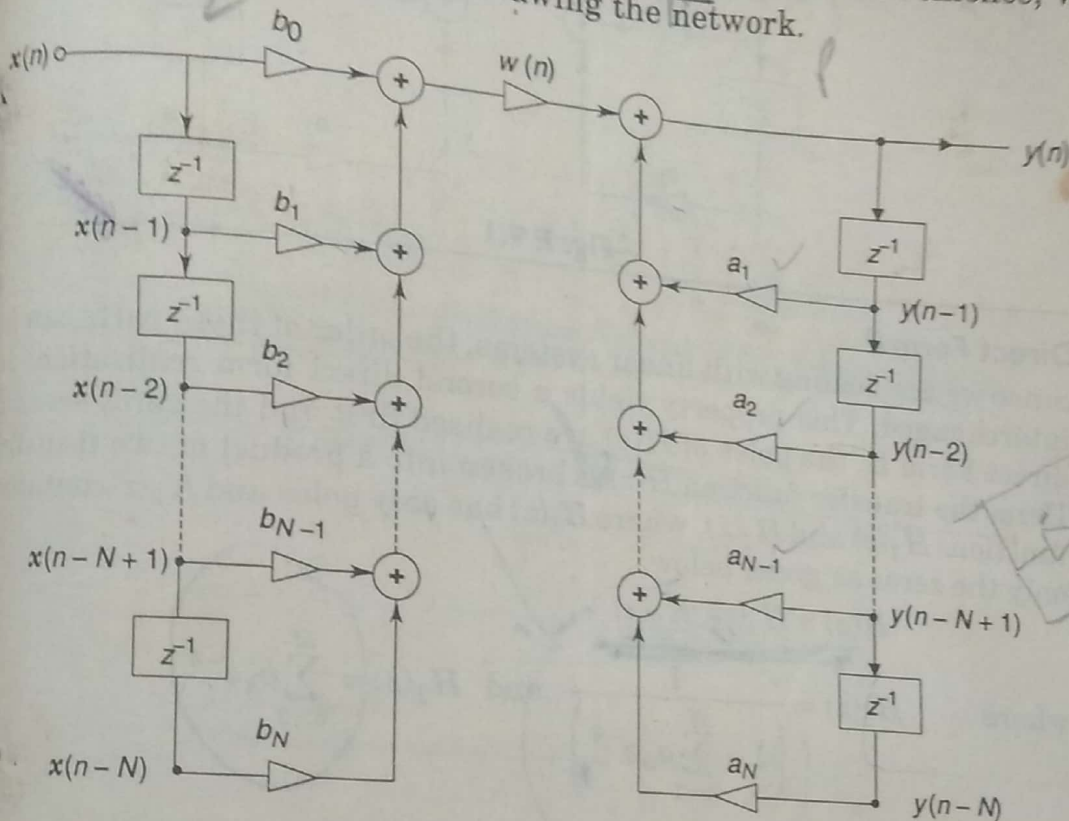


Fig. 9.4 Direct-Form I Realisation of an Nth Order Difference

Example 9.1 Develop a direct form I realisation of the difference equation

$$y(n) = b_0 x(n) + b_1 x(n-1) + b_2 x(n-2) + b_3 x(n-3) - a_1 y(n-1) - a_2 y(n-2) - a_3 y(n-3)$$

Solution Taking z -transform of the given difference equation and simplifying, we get the transfer function

$$H(z) = \frac{Y(z)}{X(z)} = \frac{b_0 + b_1 z^{-1} + b_2 z^{-2} + b_3 z^{-3}}{1 - a_1 z^{-1} - a_2 z^{-2} - a_3 z^{-3}}$$

The corresponding direct form I realisation structure is illustrated in Fig. E9.1.

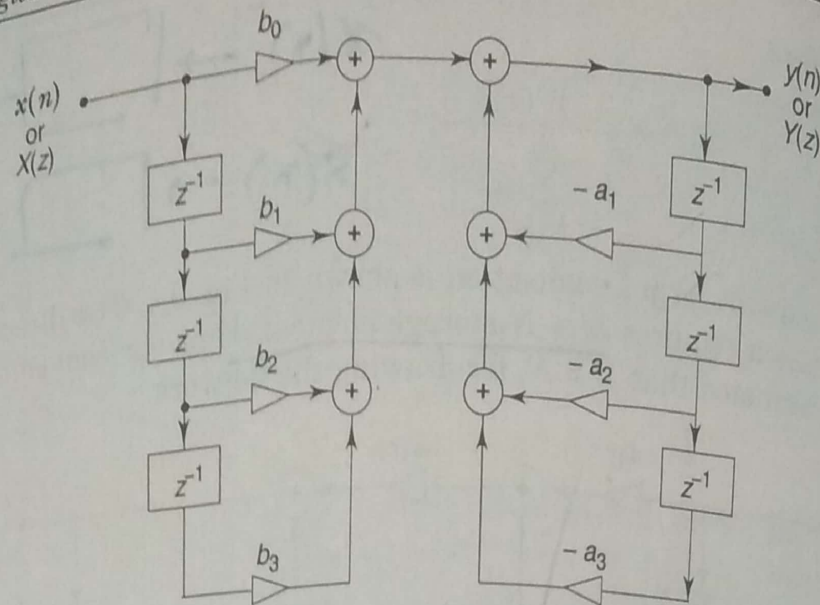


Fig. E 9.1

Direct Form II

Since we are dealing with linear systems, the order of these parts can be interchanged. This property yields a second direct form realisation. In direct Form II, the poles of $H(z)$ are realised first and the zeros second. Here, the transfer function $H(z)$ is broken into a product of two transfer functions $H_1(z)$ and $H_2(z)$, where $H_1(z)$ has only poles and $H_2(z)$ contains only the zeros as given below:

where $H(z) = H_1(z) \cdot H_2(z)$

$$H_1(z) = \frac{1}{\left(1 - \sum_{k=1}^N a_k z^{-k}\right)}$$

and $H_2(z) = \sum_{k=0}^M b_k z^{-k}$

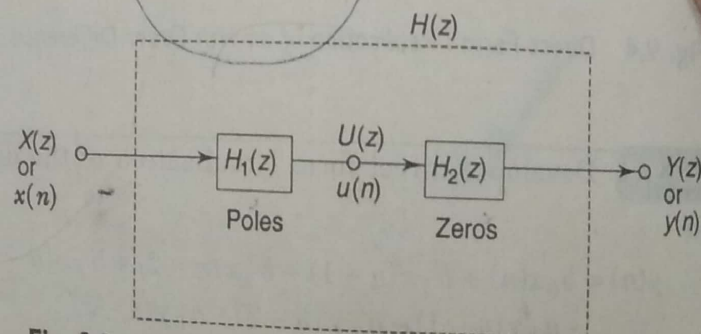


Fig. 9.5 The Decomposition for Direct Form II Realisation

The decomposition for direct Form II realisation is shown in Fig. 9.5. An intermediate sequence $u(n)$ is introduced to obtain the output of the filter.

$$u(n) = \sum_{k=1}^N a_k u(n-k) + x(n)$$

and

$$y(n) = \sum_{k=0}^M b_k u(n-k)$$

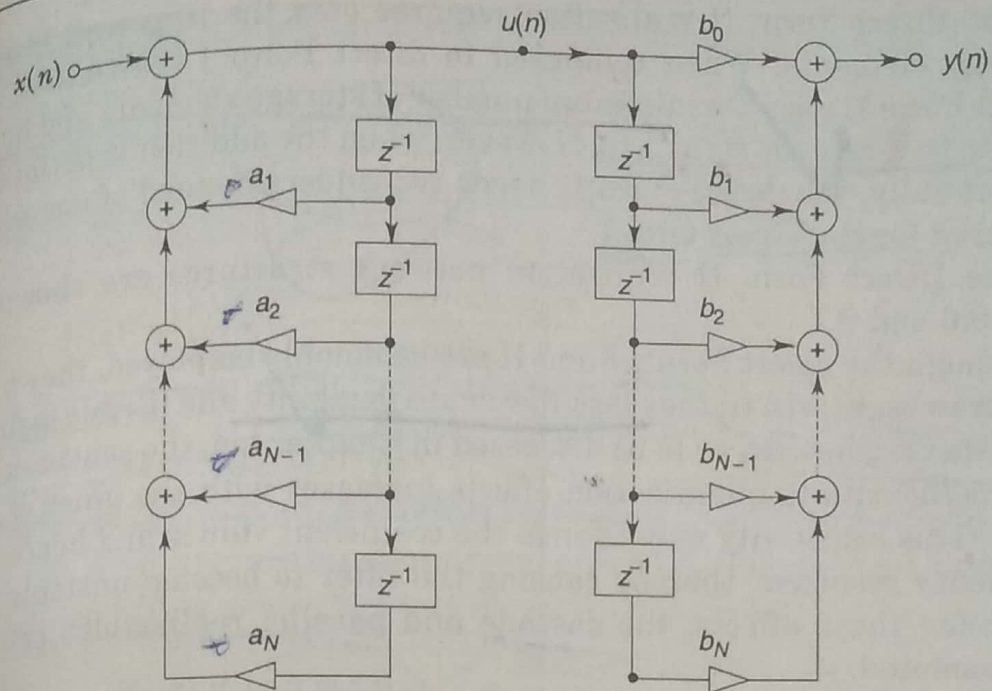


Fig. 9.6 Direct Form II—Realization Network of Fig. 9.4 with Poles and Zeros Interchanged

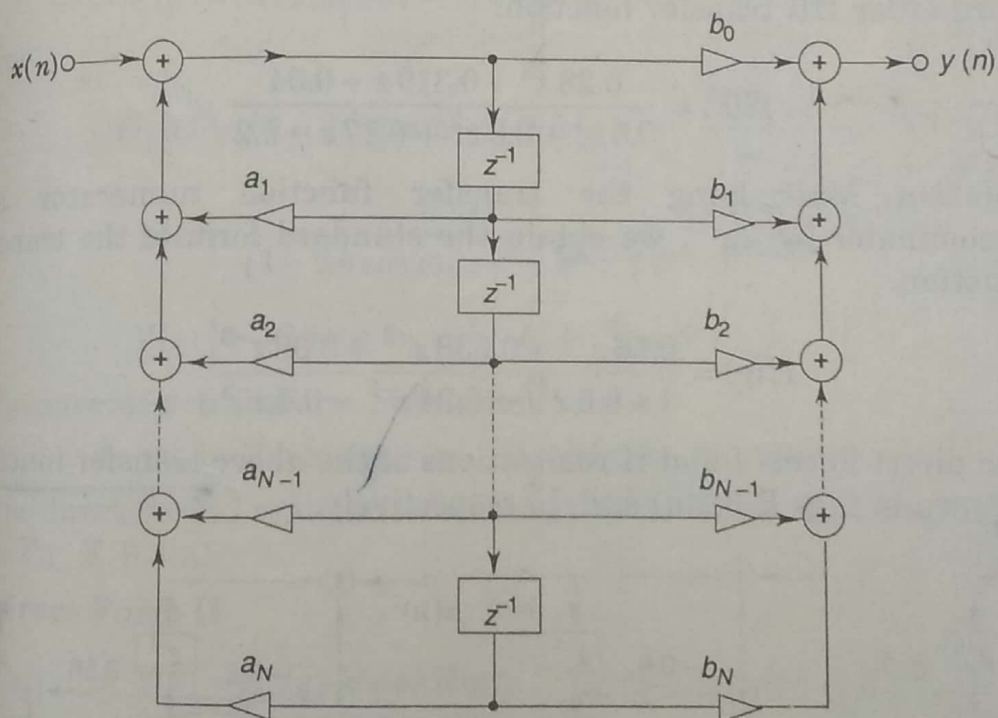


Fig. 9.7 Direct Form II—Realization Network of Fig. 9.5 with Combined Common Delays

Taking z -transform of the above equations, we get

$$U(z) = \frac{X(z)}{1 - \sum_{k=1}^M a_k z^{-k}}$$

and

$$Y(z) = U(z) \sum_{k=0}^M b_k z^{-k}$$

The direct Form II realisation requires only the larger of M or N storage elements. When compared to direct Form I realisation, the direct Form II uses the minimum number of storage elements and hence said to be a canonic structure. However, when the addition is performed sequentially, the direct Form II needs two adders instead of one adder required for the direct form I.

The Direct Form II realisation network structures are shown in Figs 9.6 and 9.7.

Though the direct Forms I and II are commonly employed, they have two drawbacks, viz. (i) they lack hardware flexibility and (ii) due to finite precision arithmetic, as to be discussed in Chapter ten, the sensitivity of the coefficients to quantisation effects increases with the order of the filter. This sensitivity may change the coefficient values and hence the frequency response, thereby causing the filter to become unstable. To overcome these effects, the cascade and parallel realisations can be implemented.

Example 9.2 Determine the direct Forms I and II realisations for a third-order IIR transfer function.

$$H(z) = \frac{0.28z^2 + 0.319z + 0.04}{0.5z^3 + 0.3z^2 + 0.17z - 0.2}$$

Solution Multiplying the transfer function numerator and denominator by $2z^{-3}$, we obtain the standard form of the transfer function.

$$H(z) = \frac{0.56z^{-1} + 0.638z^{-2} + 0.08z^{-3}}{1 + 0.6z^{-1} + 0.34z^{-2} - 0.4z^{-3}}$$

The direct Forms I and II realisations of the above transfer function are shown in Figs E 9.2(a) and (b) respectively.

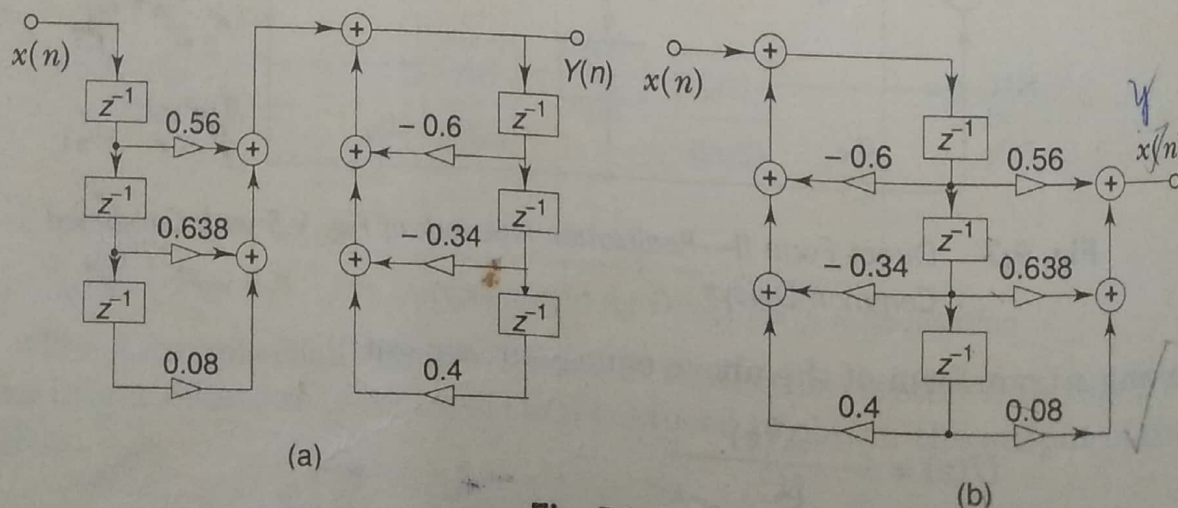


Fig. E9.2

Example 9.3

Determine direct Forms I and II for the second-order filter given by

$$y(n) = 2b \cos \omega_0 y(n-1) - b^2 y(n-2) + x(n) - b \cos \omega_0 x(n-1)$$

Solution Taking z -transform for the given function, we get

$$Y(z) = 2b \cos \omega_0 z^{-1} Y(z) - b^2 z^{-2} Y(z) + X(z) - b \cos \omega_0 z^{-1} X(z)$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{1 - b \cos \omega_0 z^{-1}}{1 - 2b \cos \omega_0 z^{-1} + b^2 z^{-2}}$$

Direct form I

$$H(z) = H_1(z) H_2(z)$$

$$\text{Therefore, } Y(z) = H_1(z) H_2(z) X(z)$$

In this form, the intermediate sequence $w(n)$ is introduced between $H_1(n)$ and $H_2(n)$

$$\text{Let } H_1(z) = 1 - b \cos \omega_0 z^{-1} = \frac{W(z)}{X(z)}$$

$$\text{Therefore, } X(z) (1 - b \cos \omega_0 z^{-1}) = W(z)$$

$$x(n) - b \cos \omega_0 x(n-1) = w(n)$$

and

$$H_2(z) = (1 - 2b \cos \omega_0 z^{-1} + b^2 z^{-2})^{-1} = \frac{Y(z)}{W(z)}$$

$$Y(z) = \frac{W(z)}{(1 - 2b \cos \omega_0 z^{-1} + b^2 z^{-2})}$$

$$Y(z) [1 - 2b \cos \omega_0 z^{-1} + b^2 z^{-2}] = W(z)$$

The inverse z -transform of this function is

$$y(n) - 2b \cos \omega_0 y(n-1) + b^2 y(n-2) = w(n)$$

The direct Form I realisation structure of the above function is shown in Fig. E 9.3(a).

Direct Form II

$$Y(z) = H_2(z) H_1(z) X(z)$$

$$H_2(z) = (1 - 2b \cos \omega_0 z^{-1} + b^2 z^{-2})^{-1} = \frac{U(z)}{X(z)}$$

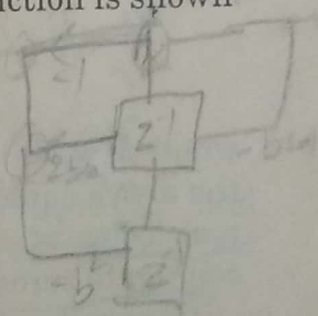
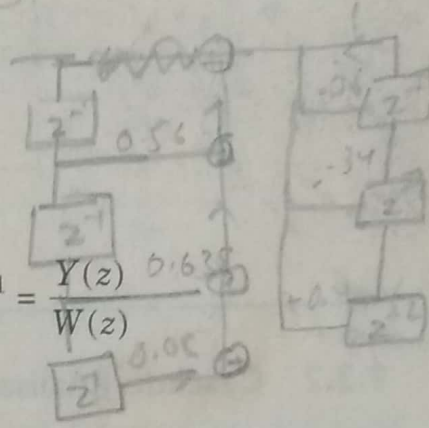
$$\text{Let } H_1(z) = 1 - b \cos \omega_0 z^{-1} = \frac{Y(z)}{U(z)}$$

$$X(z) = U(z) \{1 - 2b \cos \omega_0 z^{-1} + b^2 z^{-2}\}$$

$$\text{Hence, } x(n) = u(n) - 2b \cos \omega_0 u(n-1) + b^2 u(n-2)$$

$$Y(z) = U(z) \{1 - b \cos \omega_0 z^{-1}\}$$

$$\text{Hence, } y(n) = u(n) - b \cos \omega_0 u(n-1)$$



The Direct Form II realisation structure of the above function is shown in Fig. 9.3(b).

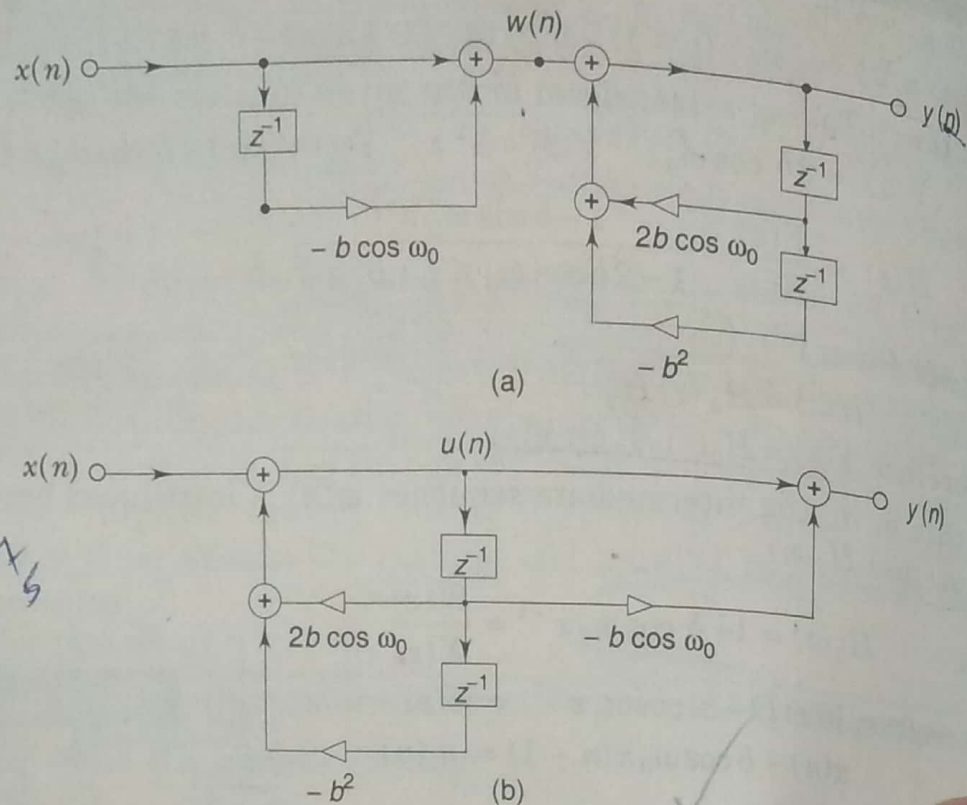


Fig. E9.3

9.3.2 Cascade Realisation of IIR Systems

In cascade realisation, the transfer function $H(z)$ is broken into a product of transfer functions $H_1(z)$, $H_2(z)$,, $H_k(z)$. Factoring the numerator and denominator polynomials of the transfer function $H(z)$, we obtain

$$H(z) = G \cdot \frac{\prod_{k=1}^{M_1} (1 - g_k z^{-1}) \prod_{k=1}^{M_2} (1 - h_k z^{-1}) (1 - h_k^* z^{-1})}{\prod_{k=1}^{N_1} (1 - c_k z^{-1}) \prod_{k=1}^{N_2} (1 - d_k z^{-1}) (1 - d_k^* z^{-1})} \quad (9.3)$$

where $M = M_1 + 2M_2$, $N = N_1 + 2N_2$ and the constant G is a gain term. In the above equation, the first-order factors represent real zeros at g_k and real poles at c_k and the second-order factors represent complex conjugate zeros at h_k and h_k^* , and complex conjugate poles at d_k and d_k^* .

Generally, the numerator and denominator polynomials of the transfer function $H(z)$ are factored into a product of first-order and second-order polynomials. Here $H(z)$ can be expressed as

$$H(z) = G \prod_{k=1}^{[(N+1)/2]} \frac{1 + \beta_{1k} z^{-1} + \beta_{2k} z^{-2}}{1 - \alpha_{1k} z^{-1} - \alpha_{2k} z^{-2}} \quad (9.4)$$

where $[(N+1)/2]$ means the largest integer contained in $(N+1)/2$. For a first-order factor, $\alpha_{2k} = \beta_{2k} = 0$. Here, it is assumed that $M \leq N$ and the real poles and the real zeros are combined in pairs. For the filter coefficients to be real, two complex conjugate zeros of the system function $H(z)$ are combined for each numerator term and two complex conjugate poles are combined for each denominator term. Two real-valued singularities can also be combined to form a second-order term as well. If the number of poles or zeros is odd, the remaining single singularity is implemented by making $\beta_{2k} = 0$ or $\alpha_{2k} = 0$ for one of the sections.

There are several cascade realisations for the factored form, based on pole-zero pairings and ordering. A possible realisation of a third-order transfer function is shown in Fig. 9.8.

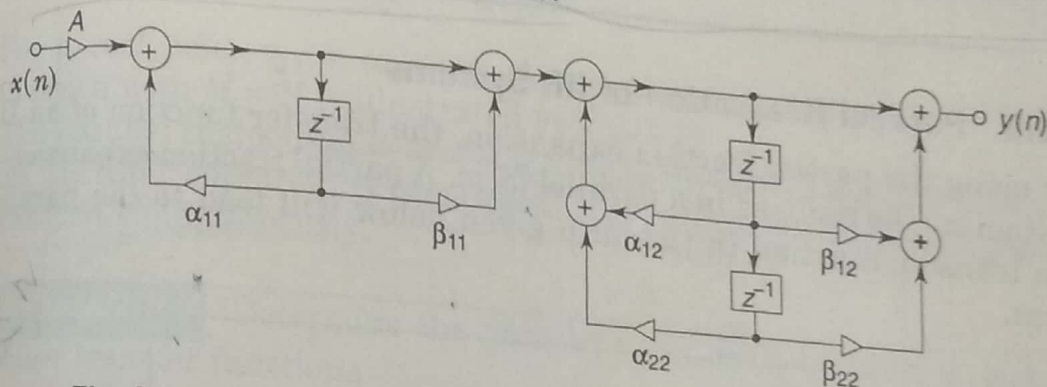


Fig. 9.8 Cascade Structure With Direct Form II Realisation of a Third-Order IIR Transfer Function

The advantage of the cascade form is that the overall transfer function of the filter can be determined, and also it has the same zeros and poles as the individual components, since the transfer function has the product of the components.

Example 9.4 Obtain a cascade realisation of the system characterised by the transfer function

$$H(z) = \frac{2(z+2)}{z(z-0.1)(z+0.5)(z+0.4)}$$

Solution Multiplying the transfer function numerator and denominator by z^{-4} , we obtain the standard form of the transfer function given by

$$\begin{aligned} H(z) &= \frac{2z^{-3}(1+2z^{-1})}{(1-0.1z^{-1})(1+0.5z^{-1})(1+0.4z^{-1})} \\ &= 2 \cdot z^{-3} \cdot \frac{(1+2z^{-1})}{(1-0.1z^{-1})} \cdot \frac{1}{(1+0.5z^{-1})} \cdot \frac{1}{(1+0.4z^{-1})} \\ &= 2 \cdot H_1(z) \cdot H_2(z) \cdot H_3(z) \cdot H_4(z) \end{aligned}$$

The cascade realisation of the system transfer function is shown in Fig. E9.4.

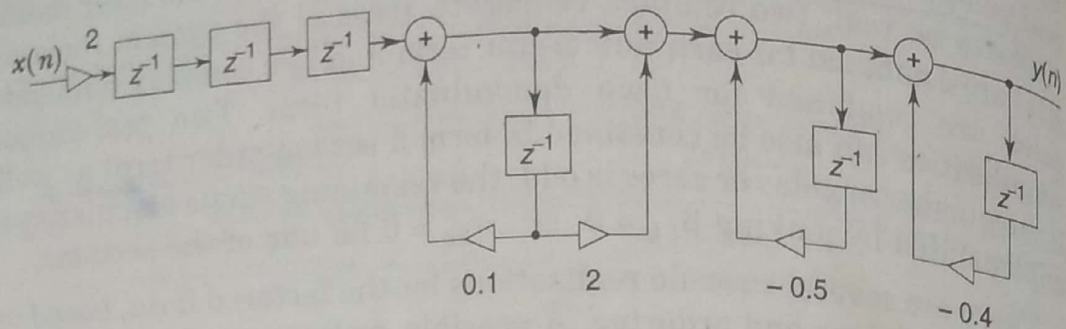


Fig. E9.4

9.3.3 Parallel Realisation of IIR Systems

By using the partial fraction expansion, the transfer function of an IIR system can be realised in a parallel form. A partial fraction expansion of the transfer function in the form given below will lead to the parallel form.

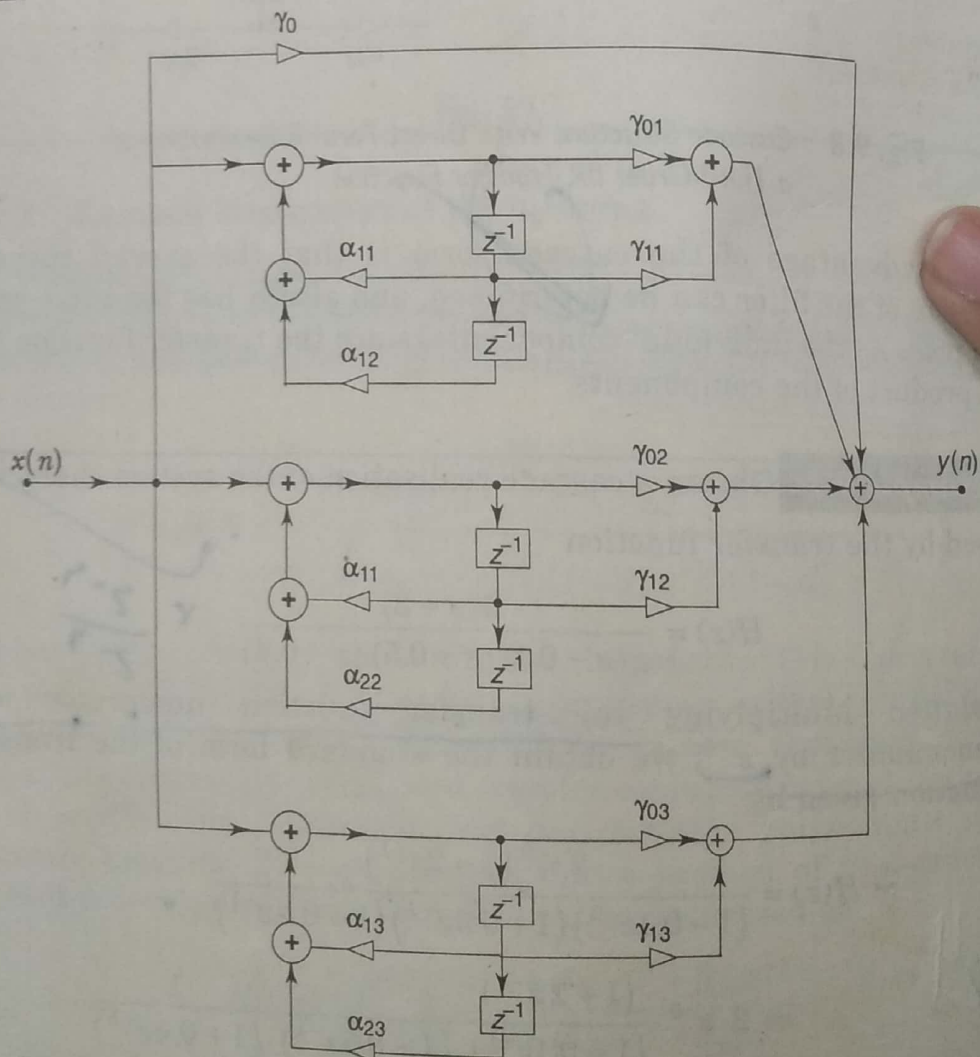


Fig. 9.9 Parallel Form Realisation Structure With the Real and Complex Poles Grouped in Pairs

$$H(z) = \sum_{k=1}^{N_1} \frac{A_k}{1 - c_k z^{-1}} + \sum_{k=1}^{N_2} \frac{B_k (1 - e_k z^{-1})}{(1 - d_k z^{-1})(1 - d_k^* z^{-1})} + \sum_{k=0}^{M-N} C_k z^{-k} \quad (9.5)$$

The quantities A_k , B_k , C_k , c_k and e_k are real when the coefficients a_k and b_k of Eq. 9.2 are real. If $M < N$, then the last term $\sum_{k=0}^{M-N} C_k z^{-k}$ will become zero.

Thus, assuming simple poles, $H(z)$ is expressed in the form

$$H(z) = \gamma_0 + \sum_{k=1}^{(N+1)/2} \frac{\gamma_{0k} + \gamma_{1k} z^{-1}}{1 - \alpha_{1k} z^{-1} - \alpha_{2k} z^{-2}} \quad (9.6)$$

For a real pole, $\gamma_{1k} = \alpha_{2k} = 0$. As an example, the parallel form realisation with $M = N$ is illustrated in Fig. 9.9.

The parallel realisation is useful for high speed filtering applications since the filter operation is performed in parallel, i.e. the processing is performed simultaneously.

Example 9.5 Determine the parallel realisation of the IIR digital filter transfer functions

$$(a) \quad H(z) = \frac{3(2z^2 + 5z + 4)}{(2z + 1)(z + 2)}$$

$$(b) \quad H(z) = \frac{3z(5z - 2)}{\left(z + \frac{1}{2}\right)(3z - 1)}$$

Solution (a) In order to find the parallel realisation, the partial fraction expansion of $H(z)/z$ is first determined, just as we did for inverse z -transforms. This gives

$$F(z) = \frac{H(z)}{z} = \frac{\frac{3}{2}(2z^2 + 5z + 4)}{z\left(z + \frac{1}{2}\right)(z + 2)} = \frac{A_1}{z} + \frac{A_2}{\left(z + \frac{1}{2}\right)} + \frac{A_3}{(z + 2)}$$

where

$$A_1 = zF(z)|_{z=0} = \frac{\frac{3}{2}(2z^2 + 5z + 4)}{\left(z + \frac{1}{2}\right)(z + 2)} \Big|_{z=0} = 6$$

$$A_2 = \left(z + \frac{1}{2}\right)F(z) \Big|_{z=-\frac{1}{2}} = \frac{\frac{3}{2}(2z^2 + 5z + 4)}{z(z + 2)} \Big|_{z=-\frac{1}{2}} = -4$$

$$A_3 = (z+2)F(z)|_{z=-2} = \frac{\frac{3}{2}(2z^2 + 5z + 4)}{z\left(z + \frac{1}{2}\right)} \Big|_{z=-2} = 1$$

Therefore, $\frac{H(z)}{z} = \frac{6}{z} - \frac{4}{z + \frac{1}{2}} + \frac{1}{(z+2)}$ ✓

Hence, $H(z) = 6 - \frac{4z}{z + \frac{1}{2}} + \frac{z}{(z+2)} = 6 - \frac{4}{\left(1 + \frac{1}{2}z^{-1}\right)} + \frac{1}{1 + 2z^{-1}}$ ✓

The parallel realisation of this transfer function is shown in Fig. E9.5(a).

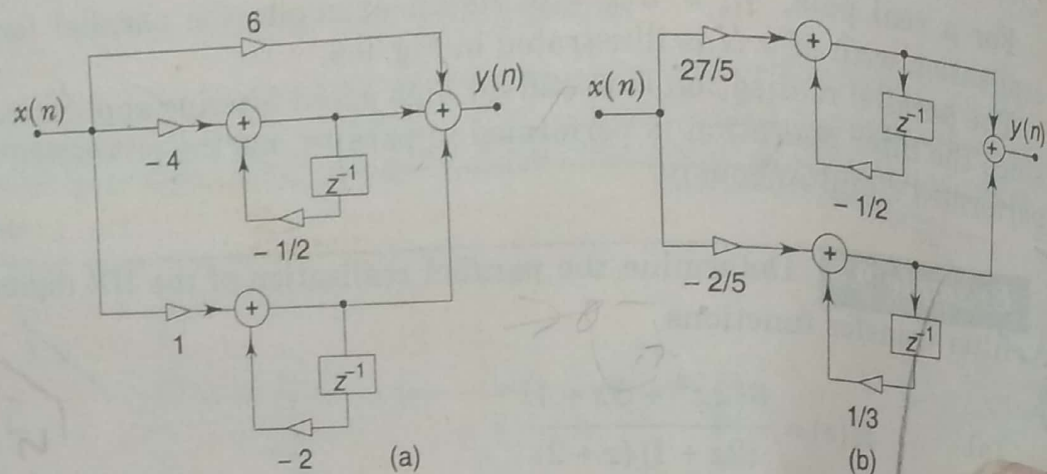


Fig. E9.5

(b) $H(z) = \frac{3z(5z-2)}{\left(z + \frac{1}{2}\right)(3z-1)} = \frac{z(5z-2)}{\left(z + \frac{1}{2}\right)\left(z - \frac{1}{3}\right)}$

$$F(z) = \frac{H(z)}{z} = \frac{5z-2}{\left(z + \frac{1}{2}\right)\left(z - \frac{1}{3}\right)} = \frac{A_1}{z + \frac{1}{2}} + \frac{A_2}{\left(z - \frac{1}{3}\right)}$$

where

$$A_1 = F(z)\left(z + \frac{1}{2}\right) \Big|_{z=-\frac{1}{2}} = \frac{5z-2}{\left(z - \frac{1}{3}\right)} \Big|_{z=-\frac{1}{2}} = \frac{27}{5}$$

$$A_2 = F(z)\left(z - \frac{1}{3}\right) \Big|_{z=\frac{1}{3}} = \frac{5z-2}{\left(z + \frac{1}{2}\right)} \Big|_{z=\frac{1}{3}} = -\frac{2}{5}$$

Therefore, $H(z) = \frac{27}{5} \frac{z}{\left(z + \frac{1}{2}\right)} - \frac{2}{5} \frac{z}{\left(z - \frac{1}{3}\right)}$

$$= \frac{27}{5} \frac{1}{\left(1 + \frac{1}{2}z^{-1}\right)} - \frac{2}{5} \frac{1}{\left(1 - \frac{1}{3}z^{-1}\right)}$$

The parallel realisation of this transfer function is shown in Fig. E9.5(b).

9.3.4 Transposed Forms

If two digital filter structures have the same transfer function, then they are called **equivalent structures**. A simple way to generate an equivalent structure from a given realisation structure is via the **transpose operation**. The transposed form is obtained by (i) reversing the paths, (ii) replacing pick-off nodes by adders, and vice-versa, and (iii) interchanging the input and output nodes. For a single input-output system, the transposed structure has the same transfer function as the original realisation structure.

While using infinite precision arithmetic, any realisation structure will behave identically to any other equivalent structure. But, due to the finite word length limitations, the behaviour of a particular realisation structure is totally different from its equivalent structures. (Therefore, it is important to select a structure which has the minimum quantization effects from the finite word length implementation))

Figure 9.10 shows the transposed direct form I realisation network of Fig 9.4. Figure 9.11 also illustrates the direct form I network of Fig. 9.10 which is redrawn to have the input on the left and output on the right conventionally. Figure 9.12 shows the transposed direct form II structure.

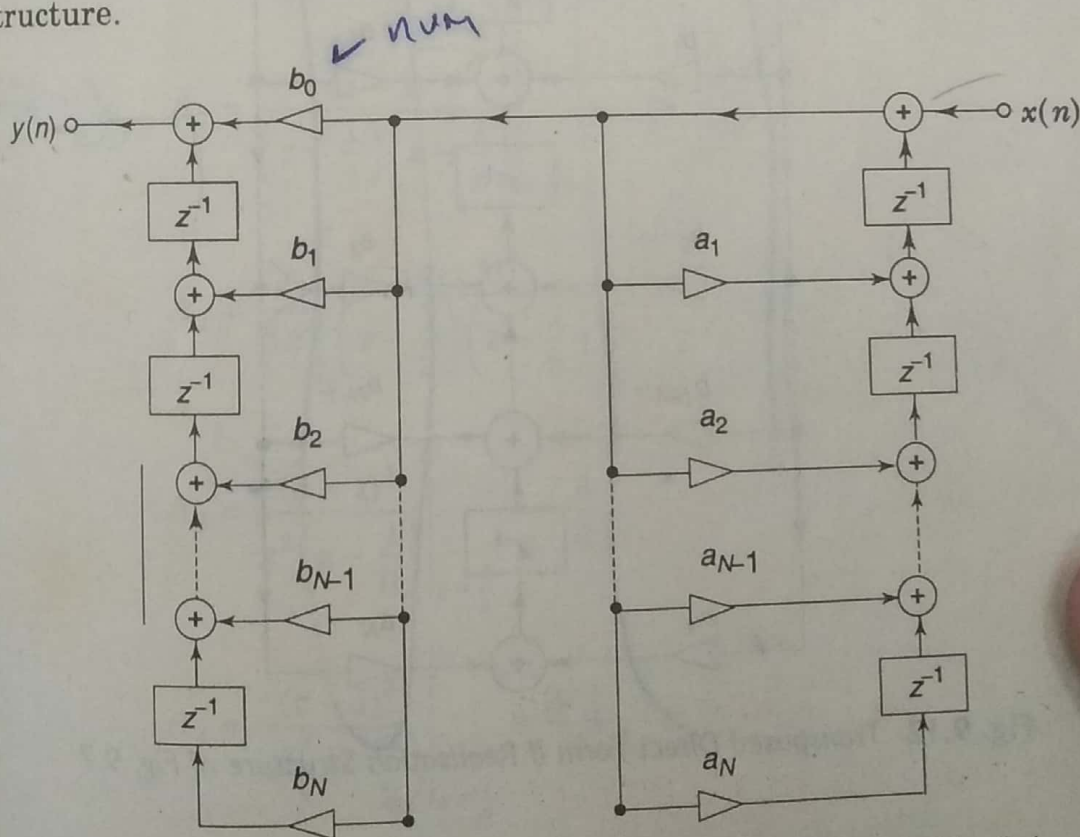


Fig. 9.10 Transposed Direct Form I Realisation Structure of Fig. 9.4

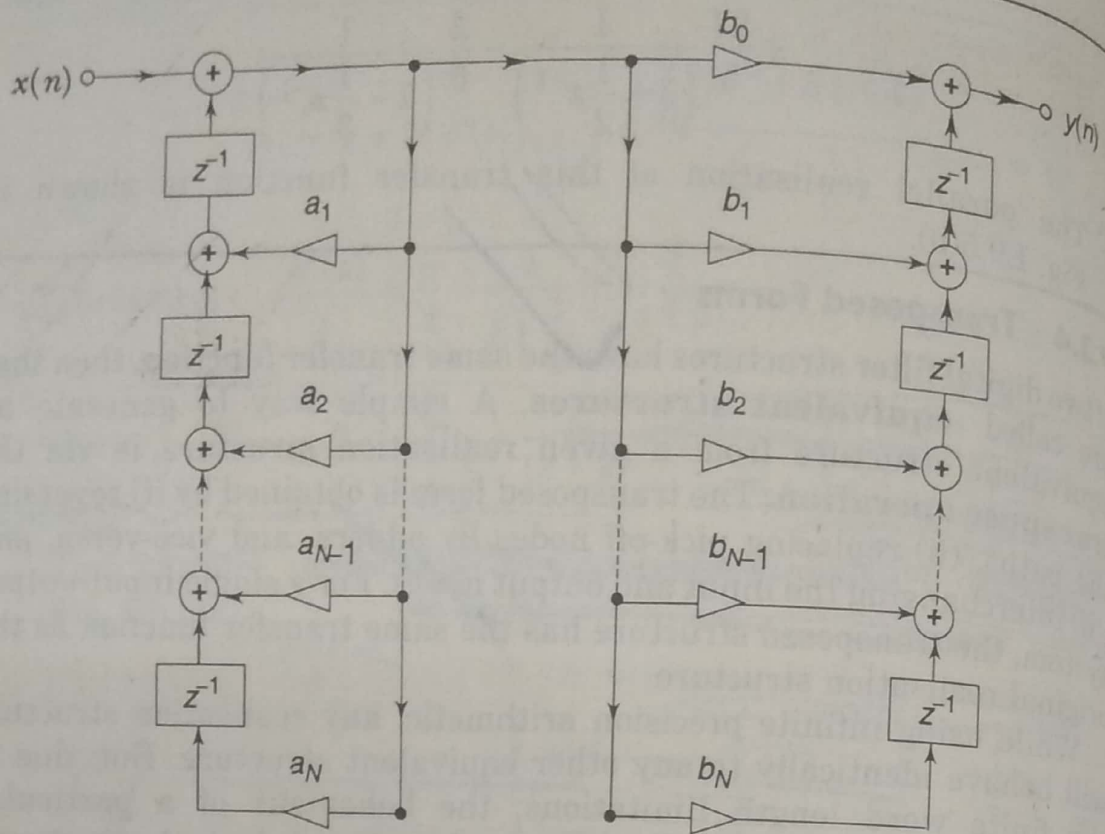


Fig. 9.11 Transposed Direct Form I Realisation Structure

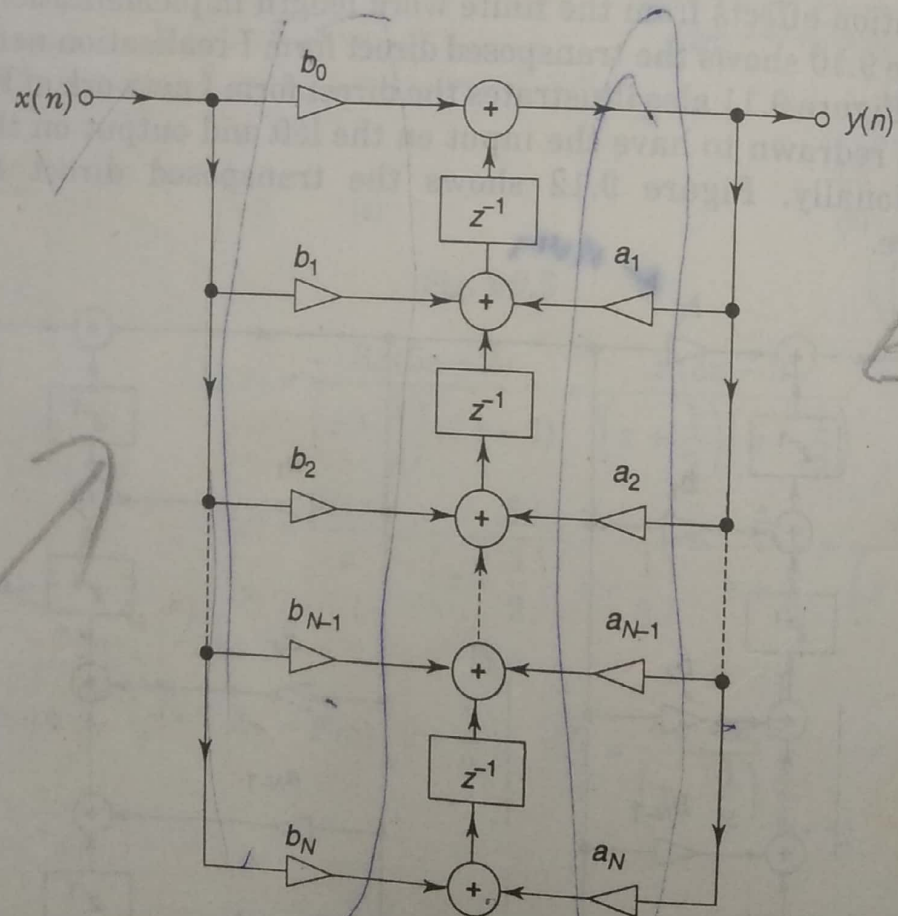


Fig. 9.12 Transposed Direct Form II Realisation Structure of Fig. 9.7

Example 9.6

Draw the structures of cascade and parallel realisations of

$$H(z) = \frac{(1 - z^{-1})^3}{\left(1 - \frac{1}{2}z^{-1}\right)\left(1 - \frac{1}{8}z^{-1}\right)}$$

Solution

Cascade realisation The given transfer function $H(z)$ can be written in the form

$$H(z) = \frac{(1 - z^{-1})}{\left(1 - \frac{1}{2}z^{-1}\right)} \cdot \frac{(1 - z^{-1})}{\left(1 - \frac{1}{8}z^{-1}\right)} \cdot (1 - z^{-1}) = H_1(z) \cdot H_2(z) \cdot H_3(z)$$

This function is realised in cascade structure as shown in Fig. E9.6(a).

Parallel realisation The transfer function $H(z)$ can be written as

$$H(z) = \frac{(1 - z^{-1})^3}{\left(1 - \frac{1}{2}z^{-1}\right)\left(1 - \frac{1}{8}z^{-1}\right)} = \frac{(z - 1)^3}{z\left(z - \frac{1}{2}\right)\left(z - \frac{1}{8}\right)}$$

For finding the parallel realisation, the partial fraction expansion of $H(z)/z$ is determined as

$$F(z) = \frac{H(z)}{z} = \frac{(z - 1)^3}{z^2\left(z - \frac{1}{2}\right)\left(z - \frac{1}{8}\right)} = \frac{A_1}{z^2} + \frac{A_2}{z} + \frac{A_3}{z - \frac{1}{2}} + \frac{A_4}{z - \frac{1}{8}}$$

where

$$A_1 = \frac{(z - 1)^3}{\left(z - \frac{1}{2}\right)\left(z - \frac{1}{8}\right)} \Big|_{z=0} = -16$$

$$A_2 = \frac{d}{dz} \left[\frac{(z - 1)^3}{\left(z - \frac{1}{2}\right)\left(z - \frac{1}{8}\right)} \right] \Big|_{z=0} = -112$$

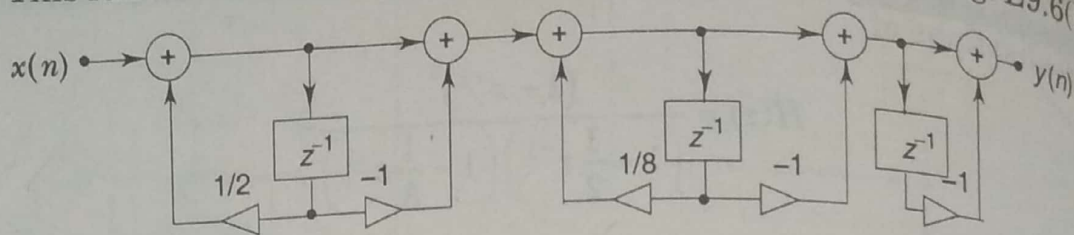
$$A_3 = \frac{(z - 1)^3}{z^2\left(z - \frac{1}{8}\right)} \Big|_{z=\frac{1}{2}} = -\frac{4}{3}$$

$$A_4 = \frac{(z - 1)^3}{z^2\left(z - \frac{1}{2}\right)} \Big|_{z=\frac{1}{8}} = \frac{343}{3}$$

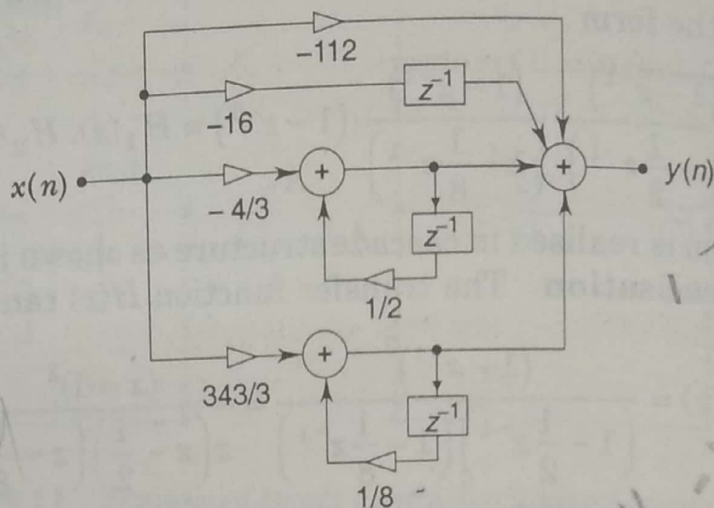
Thus

$$H(z) = -112 - 16z^{-1} - \frac{4}{3} \frac{1}{\left(1 - \frac{1}{2}z^{-1}\right)} + \frac{343}{3} \frac{1}{\left(1 - \frac{1}{8}z^{-1}\right)}$$

This function is realised in parallel structure as shown in Fig. E9.6(b).



(a)



(b)

Fig. E9.6

Example 9.7 Draw the cascade and parallel realisation structures for the system described by the system function

$$H(z) = \frac{5 \left(1 - \frac{1}{4}z^{-1}\right) \left(1 - \frac{2}{3}z^{-1}\right) (1 + 2z^{-1})}{\left(1 - \frac{3}{4}z^{-1}\right) \left(1 - \frac{1}{8}z^{-1}\right) \left[1 - \left(\frac{1}{2} + j\frac{1}{2}\right)z^{-1}\right] \left[1 - \left(\frac{1}{2} - j\frac{1}{2}\right)z^{-1}\right]}$$

Solution

Cascade realisation This realisation can be easily obtained by grouping poles and zeros of the system function in several possible ways; one possible pairing of poles and zeros is

$$H_1(z) = \frac{1 - \frac{2}{3}z^{-1}}{1 - \frac{7}{8}z^{-1} + \frac{3}{32}z^{-2}}, \quad H_2(z) = \frac{1 + \frac{7}{4}z^{-1} - \frac{1}{2}z^{-2}}{1 - z^{-1} + \frac{1}{2}z^{-2}}$$

Hence, $H(z) = 5H_1(z)H_2(z)$

The difference equations for $H_1(z)$ and $H_2(z)$ are

$$y_1(n) = \frac{7}{8}y_1(n-1) - \frac{3}{32}y_1(n-2) + x_1(n) - \frac{2}{3}x_1(n-1)$$

$$y_2(n) = y_2(n-1) - \frac{1}{2}y_2(n-2) + x_2(n) + \frac{7}{4}x_2(n-1) - \frac{1}{2}x_2(n-2)$$

These functions are realised in cascade form as shown in Fig. E9.7(a).

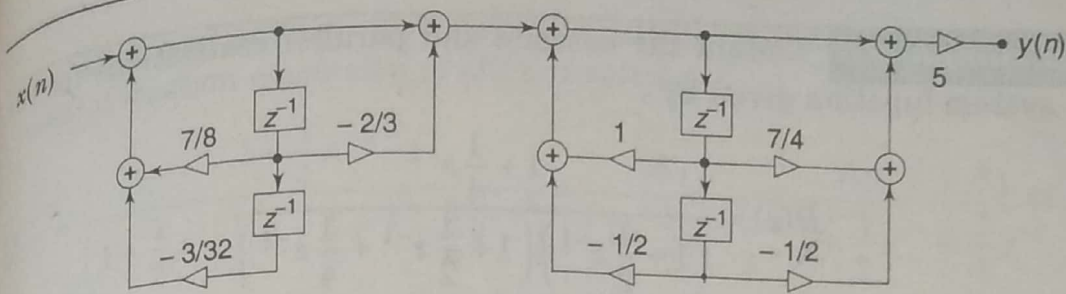


Fig. E9.7(a)

Parallel realisation To obtain the parallel form realisation, $H(z)$ must be expanded in partial fractions. Thus we have

$$H(z) = \frac{A_1}{1 - \frac{3}{4}z^{-1}} + \frac{A_2}{1 - \frac{1}{8}z^{-1}} + \frac{A_3}{1 - \left(\frac{1}{2} + j\frac{1}{2}\right)z^{-1}} + \frac{A_3^*}{1 - \left(\frac{1}{2} - j\frac{1}{2}\right)z^{-1}}$$

Upon solving, we find $A_1 = 2.933$, $A_2 = -2.947$,

$$A_3 = 2.507 - j10.45 \text{ and } A_3^* = 2.507 + j10.45$$

$$\begin{aligned} \text{Therefore, } H(z) &= \frac{2.933}{\left(1 - \frac{3}{4}z^{-1}\right)} - \frac{2.947}{\left(1 - \frac{1}{8}z^{-1}\right)} \\ &\quad + \frac{2.507 - j10.45}{1 - \frac{1}{2}(1 + j)z^{-1}} + \frac{2.507 + j10.45}{1 - \frac{1}{2}(1 - j)z^{-1}} \\ &= \frac{-0.02 + 1.846z^{-1}}{1 - \frac{7}{8}z^{-1} + \frac{3}{32}z^{-2}} + \frac{5.02 + 7.743z^{-1}}{1 - z^{-1} + \frac{1}{2}z^{-2}} \end{aligned}$$

The parallel form realisation is illustrated in Fig. E9.7(b).

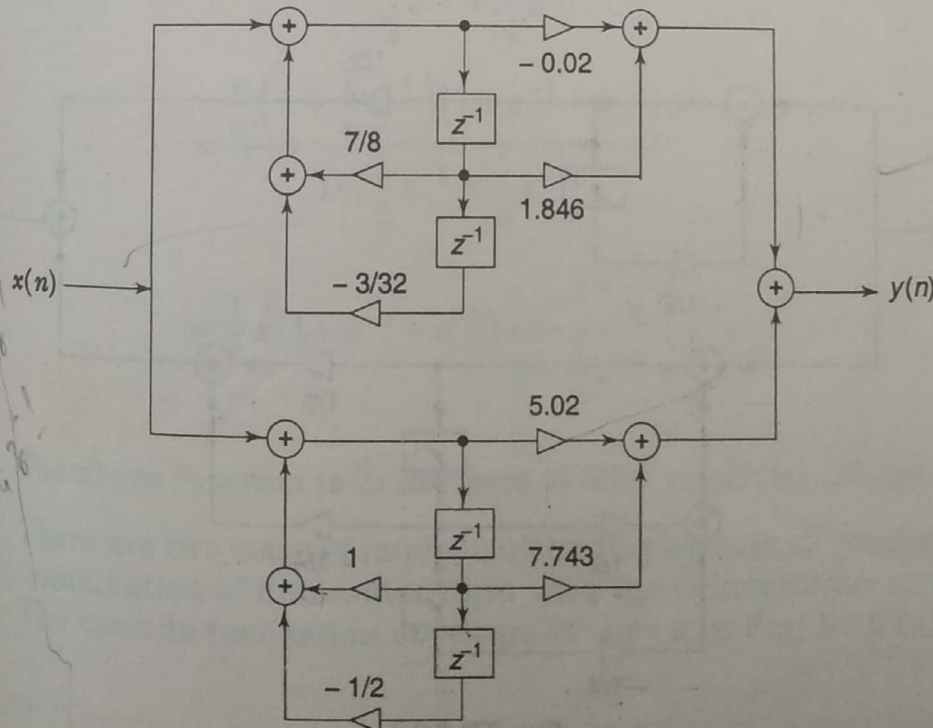


Fig. E9.7(b)

Example 9.8 Obtain the cascade and parallel realisations for the system function given by

$$H(z) = \frac{1 + \frac{1}{4}z^{-1}}{\left(1 + \frac{1}{2}z^{-1}\right)\left(1 + \frac{1}{2}z^{-1} + \frac{1}{4}z^{-2}\right)}$$

Solution

Cascade Realisation To obtain the cascade realisation, the transfer function is broken into a product of two functions as

$$H(z) = H_1(z) H_2(z)$$

where $H_1(z) = \frac{1 + \frac{1}{4}z^{-1}}{1 + \frac{1}{2}z^{-1}}$ and $H_2(z) = \frac{1}{1 + \frac{1}{2}z^{-1} + \frac{1}{4}z^{-2}}$

The cascade realisation structure for this system function is shown in Fig. E9.8(a).

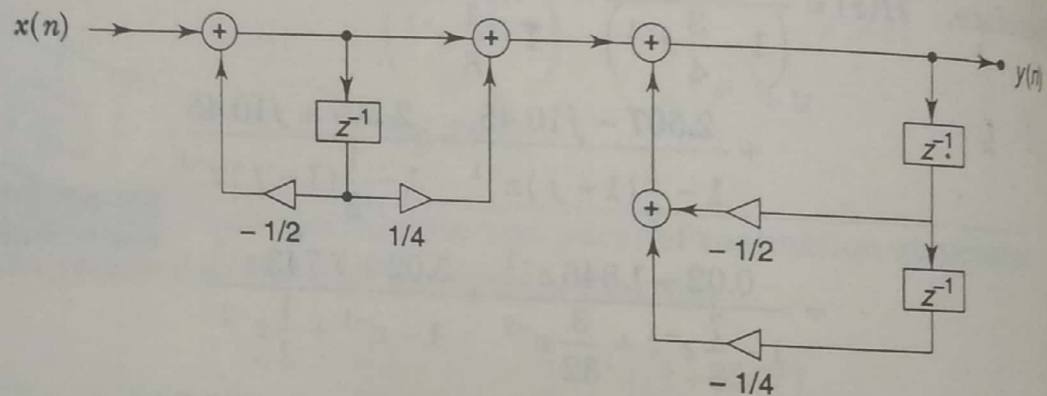


Fig. E9.8(a)

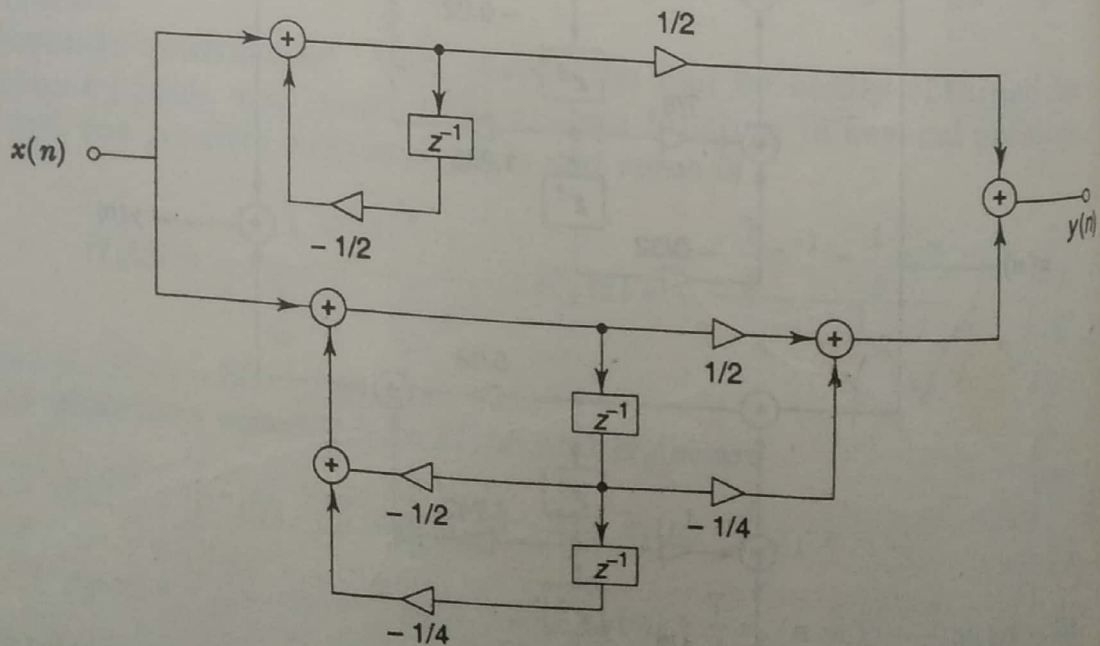


Fig. E9.8(b)

Parallel realisation To obtain the parallel realisation, we make a partial fraction expansion of $H(z)$, resulting in

$$H(z) = \frac{1 + \frac{1}{4}z^{-1}}{\left(1 + \frac{1}{2}z^{-1}\right)\left(1 + \frac{1}{2}z^{-1} + \frac{1}{4}z^{-2}\right)} = \frac{A_1}{\left(1 + \frac{1}{2}z^{-1}\right)} + \frac{A_2z^{-1} + A_3}{1 + \frac{1}{2}z^{-1} + \frac{1}{4}z^{-2}}$$

Multiplying out and equating the coefficients of negative powers of z and then solving, we obtain, $A_1 = \frac{1}{2}$, $A_2 = -\frac{1}{4}$ and $A_3 = \frac{1}{2}$

$$\text{Therefore, } H(z) = \frac{\frac{1}{2}}{1 + \frac{1}{2}z^{-1}} + \frac{\frac{1}{2} - \frac{1}{4}z^{-1}}{1 + \frac{1}{2}z^{-1} + \frac{1}{4}z^{-2}}$$

The parallel realisation structure is shown in Fig. E9.8(b).

Example 9.9 Develop cascade and parallel realisation structures for

$$H(z) = \frac{\frac{z}{6} + \frac{5}{24} + \frac{5}{24}z^{-1} + \frac{1}{24}z^{-2}}{1 - \frac{1}{2}z^{-1} + \frac{1}{4}z^{-2}}$$

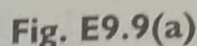
Solution

Cascade realisation The transfer function can be rearranged as

$$\begin{aligned} H(z) &= \frac{\frac{z}{6} \left(1 + \frac{5}{4}z^{-1} + \frac{5}{4}z^{-2} + \frac{1}{4}z^{-3}\right)}{1 - \frac{1}{2}z^{-1} + \frac{1}{4}z^{-2}} \\ &= \frac{\frac{z}{6} \left(1 + \frac{1}{4}z^{-1}\right) (1 + z^{-1} + z^{-2})}{1 - \frac{1}{2}z^{-1} + \frac{1}{4}z^{-2}} \\ &= \frac{1}{6} z (1 + z^{-1} + z^{-2}) \frac{1 + \frac{1}{4}z^{-1}}{1 - \frac{1}{2}z^{-1} + \frac{1}{4}z^{-2}} \end{aligned}$$

The above function is in the form of $H(z) = \frac{1}{6} H_1(z) \cdot H_2(z) \cdot H_3(z)$.

There are two ways of implementing this system, depending upon the combination of numerator term with the denominator term. The cascade realisation structure is shown in Fig. E9.9 (a).



Parallel Realisation

$$H(z) = \frac{\frac{z}{6} + \frac{5}{24} + \frac{5}{24}z^{-1} + \frac{1}{24}z^{-2}}{1 - \frac{1}{2}z^{-1} + \frac{1}{4}z^{-2}} = \frac{\frac{z}{6} \left(1 + \frac{5}{4}z^{-1} + \frac{5}{4}z^{-2} + \frac{1}{4}z^{-3} \right)}{1 - \frac{1}{2}z^{-1} + \frac{1}{4}z^{-2}}$$

Dividing the denominator by the numerator, we obtain

$$\begin{array}{r} z^{-1} + 7 \\ \hline \frac{1}{4}z^{-2} - \frac{1}{2}z^{-1} + 1 \quad \left| \begin{array}{l} \frac{1}{4}z^{-3} + \frac{5}{4}z^{-2} + \frac{5}{4}z^{-1} + 1 \\ \frac{1}{4}z^{-3} - \frac{1}{2}z^{-2} + z^{-1} \end{array} \right. \\ \hline \frac{7}{4}z^{-2} + \frac{1}{4}z^{-1} + 1 \\ \frac{7}{4}z^{-2} + \frac{7}{2}z^{-1} + 7 \\ \hline \frac{15}{4}z^{-1} - 6 \end{array}$$

Therefore ,
$$H(z) = \frac{z}{6} \left[7 + z^{-1} + \frac{-6 + \frac{15}{4}z^{-1}}{1 - \frac{1}{2}z^{-1} + \frac{1}{4}z^{-2}} \right]$$

$$= z \left[\frac{7}{6} + \frac{1}{6}z^{-1} + \frac{-1 + \frac{5}{8}z^{-1}}{1 - \frac{1}{2}z^{-1} + \frac{1}{4}z^{-2}} \right]$$

$$H(z) = z[H_1(z) + H_2(z)]$$

where $H_1(z) = \frac{7}{6} + \frac{1}{6}z^{-1}$ and $H_2(z) = \frac{-1 + \frac{5}{8}z^{-1}}{1 - \frac{1}{2}z^{-1} + \frac{1}{4}z^{-2}}$

The parallel realisation structure is shown in Fig. E9.9(b).

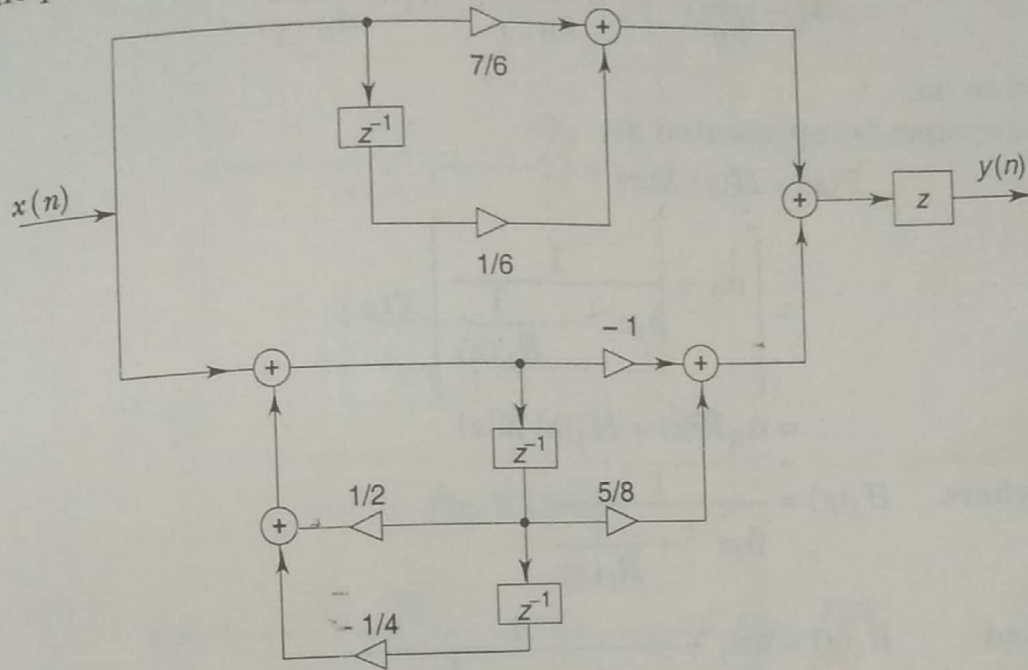


Fig. E9.9(b)

9.3.5 Ladder Structures

Filters realised using ladder structures have desirable coefficient sensitivity properties. Small changes in the filter parameters have little effect on its performance.

Consider a filter represented by the transfer function

$$H(z) = \frac{a_N z^{-N} + a_{N-1} z^{-N+1} + \dots + a_1 z^{-1} + a_0}{b_N z^{-N} + b_{N-1} z^{-N+1} + \dots + b_1 z^{-1} + b_0}$$

$$= \alpha_0 + \frac{1}{\beta_1 z^{-1} + \frac{1}{\alpha_1 + \frac{1}{\beta_2 z^{-1} + \dots + \frac{1}{\alpha_N}}}}$$

To calculate the values of α_i s and β_i s, form the Routh array.

z^{-N}	a_N	a_{N-1}	a_{N-2} ...	$a_1 a_0$
z^{-N}	b_N	b_{N-1}	b_{N-2} ...	$b_1 b_0$
z^{-N+1}	c_{N-1}	c_{N-2}	c_{N-3} ...	c_0
z^{-N+1}	d_{N-1}	d_{N-2}	d_{N-3} ...	d_0
z^{-N+2}	e_{N-2}	e_{N-3}	e_{N-4} ...	
z^{-N+2}	f_{N-2}	f_{N-3}	f_{N-4} ...	
1	g_0			
1	h_0			

The ladder structure parameters are given by,

$$\alpha_0 = \frac{a_N}{b_N}, \quad \beta_1 = \frac{b_N}{c_{N-1}}, \quad \alpha_1 = \frac{c_{N-1}}{d_{N-1}}$$

and so on.

The system is represented as,

$$Y(z) = H(z) X(z)$$

$$= \left[\alpha_0 + \frac{1}{\beta_1 z^{-1} + \frac{1}{R_1(z)}} \right] X(z)$$

$$= \alpha_0 X(z) + H_1(z) X(z)$$

where $H_1(z) = \frac{1}{\beta_1 z^{-1} + \frac{1}{R_1(z)}}$

and $R_1(z) = \frac{1}{\alpha_1 + \frac{1}{\beta_2 z^{-1} + \frac{1}{\alpha_2 + \frac{1}{\ddots + \frac{1}{\alpha_N}}}}}$

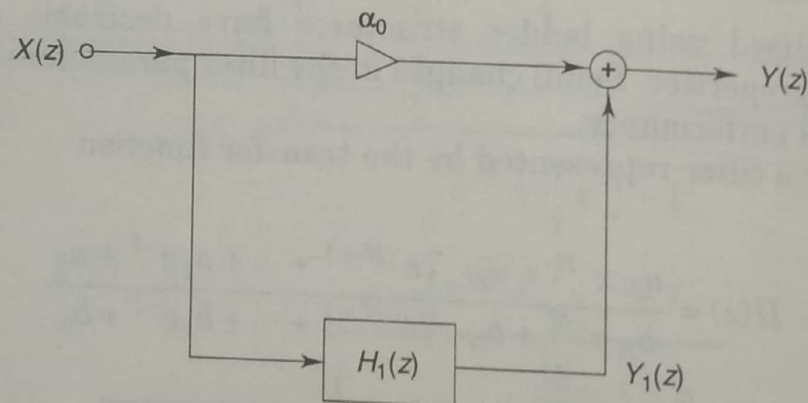


Fig. 9.13(a)

$$H_1(z) = \frac{1}{\beta_1 z^{-1} + \frac{1}{R_1(z)}} = \frac{Y_1(z)}{X(z)}$$

$$Y_1(z) \left\{ \beta_1 z^{-1} + \frac{1}{R_1(z)} \right\} = X(z)$$

$$Y_1(z) = R_1(z) \{X(z) - \beta_1 z^{-1} Y_1(z)\}$$

$$= R_1(z) X_1(z)$$

Combining Figs 9.13 (a) and (b), we obtain the structure as shown in Fig. 9.13 (c).

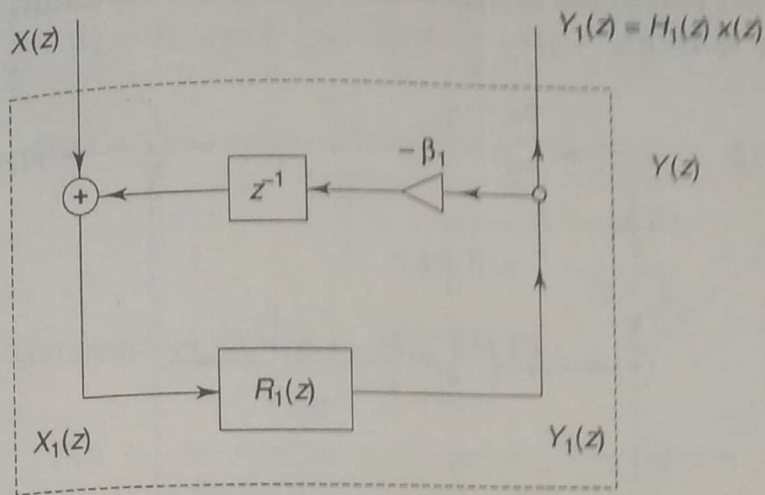


Fig. 9.13(b)

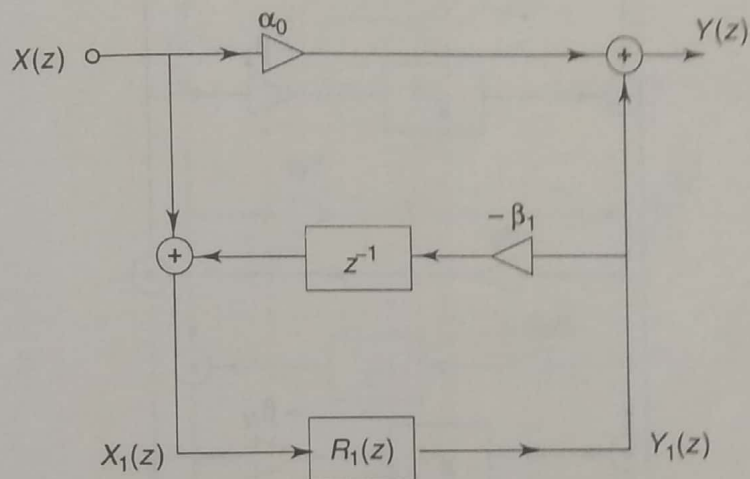


Fig. 9.13(c)

$$R_1(z) = \frac{Y_1(z)}{X_1(z)} = \alpha_1 + \frac{1}{\beta_2 z^{-1} + \frac{1}{R_2(z)}}$$

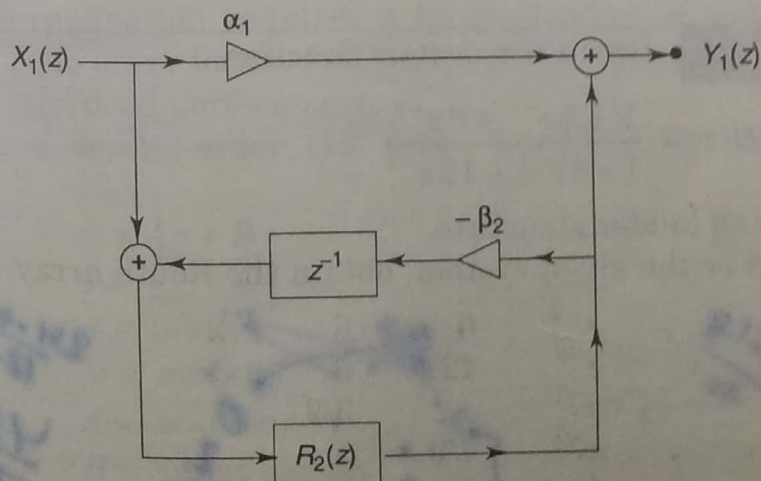


Fig. 9.13(d)

With continued fractional realisation of $H(z)$, the resultant structure is shown in Fig. 9.13(e).

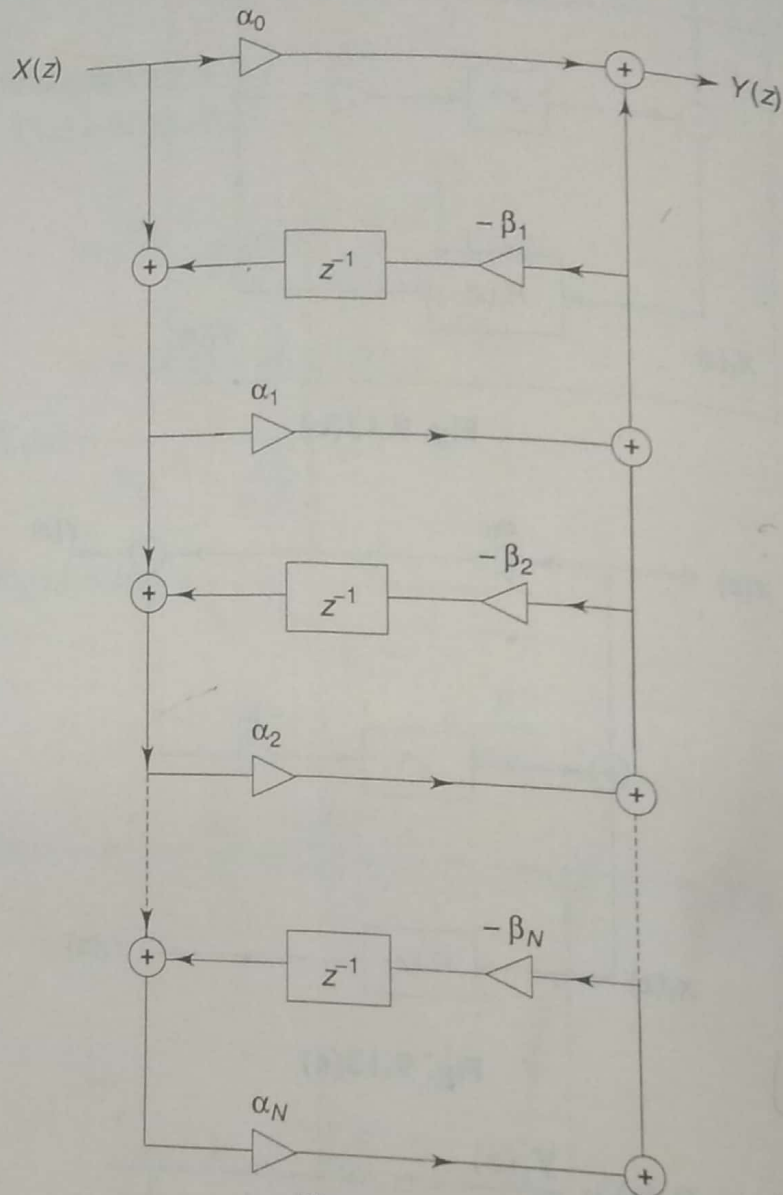


Fig. 9.13(e)

Example 9.10 Given the system function

$$H(z) = \frac{2 + 8z^{-1} + 6z^{-2}}{1 + 8z^{-1} + 12z^{-2}}$$

Realise using ladder structure.

Solution For the given system, obtain the Routh array

$$\frac{12(8) - 6(8)}{12} = \frac{48 - 48}{12} = 0$$

$$\frac{24 - 6}{12} = \frac{18}{12} = \frac{3}{2}$$

$$\frac{48 - 6}{12} = \frac{42}{12} = \frac{7}{2}$$

z^{-2}	6	8	2
z^{-2}	12	8	1
z^{-1}	4	$3/2$	0
z^{-1}	$7/2$	1	0
1	$5/14$	0	0
1	1	0	0

The ladder structure parameters are

$$\alpha_0 = \frac{1}{2}, \quad \beta_1 = 3, \quad \alpha_1 = \frac{8}{7}, \quad \beta_2 = \frac{49}{5}, \quad \alpha_2 = \frac{5}{14}$$

$$H(z) = \frac{1}{2} + \frac{1}{3z^{-1} + \frac{8}{7} + \frac{1}{(49/5)z^{-1} + \frac{5}{14}}}$$

The ladder structure is shown in Fig. E9.10.

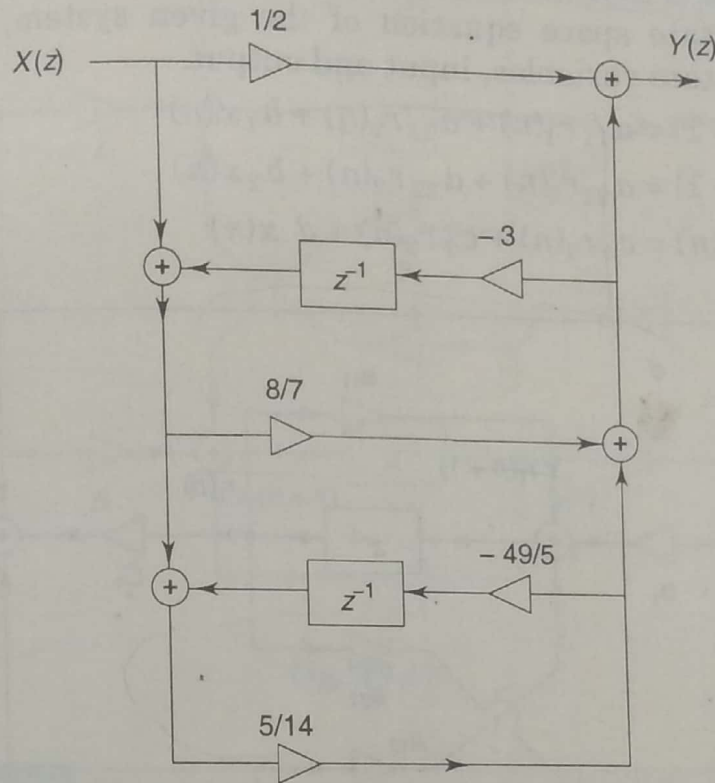


Fig. E9.10

9.3.6 State Space Structure

State space realisation requires a large number of computations as compared to the other forms. But under finite word length constraints, it provides improved performance.

Consider a second-order IIR filter described by its state space equations,

$$\dot{r} = Ar + Bx$$

$$y = Cr + Dx$$

where

x = input

y = output

A = state matrix

B = input matrix

C = output matrix

D = direct transmission matrix

r = state variables

$$\begin{bmatrix} r_1(n+1) \\ r_2(n+2) \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} r_1(n) \\ r_2(n) \end{bmatrix} + \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} x(n)$$

$$y(n) = [c_1 \ c_2] \begin{bmatrix} r_1(n) \\ r_2(n) \end{bmatrix} + d x(n)$$

The state space realisation of an N th order IIR filter requires $(N+1)^2$ multipliers and $N(N+1)$ two input adders.

From the state space equation of the given system, express the intermediate state variables, input and output.

$$r_1(n+1) = a_{11}r_1(n) + a_{12}r_2(n) + b_1x(n)$$

$$r_2(n+1) = a_{21}r_1(n) + a_{22}r_2(n) + b_2x(n)$$

$$y(n) = c_1r_1(n) + c_2r_2(n) + d x(n)$$

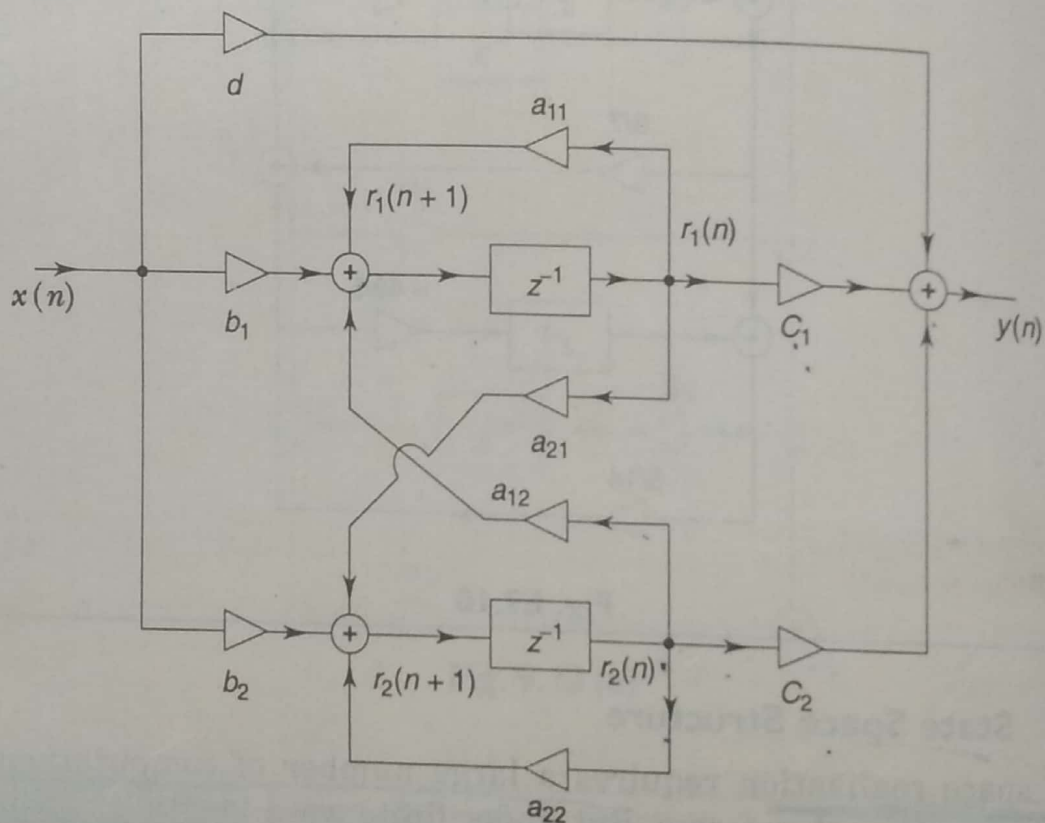


Fig. 9.14

Example 9.11

Realise a system defined by the following state space equations

$$\begin{bmatrix} r_1(n+1) \\ r_2(n+2) \end{bmatrix} = \begin{bmatrix} -5 & -1 \\ 3 & -1 \end{bmatrix} \begin{bmatrix} r_1(n) \\ r_2(n) \end{bmatrix} + \begin{bmatrix} 4 \\ 6 \end{bmatrix} x(n)$$

$$y(n) = \begin{bmatrix} 2 & 3 \end{bmatrix} \begin{bmatrix} r_1(n) \\ r_2(n) \end{bmatrix}$$

Solution Expand the state space equations and obtain the relation between the state variables, input and output.

$$r_1(n+1) = -5r_1(n) - r_2(n) + 4x(n)$$

$$r_2(n+1) = 3r_1(n) - r_2(n) + 6x(n)$$

$$y(n) = 2r_1(n) + 3r_2(n)$$

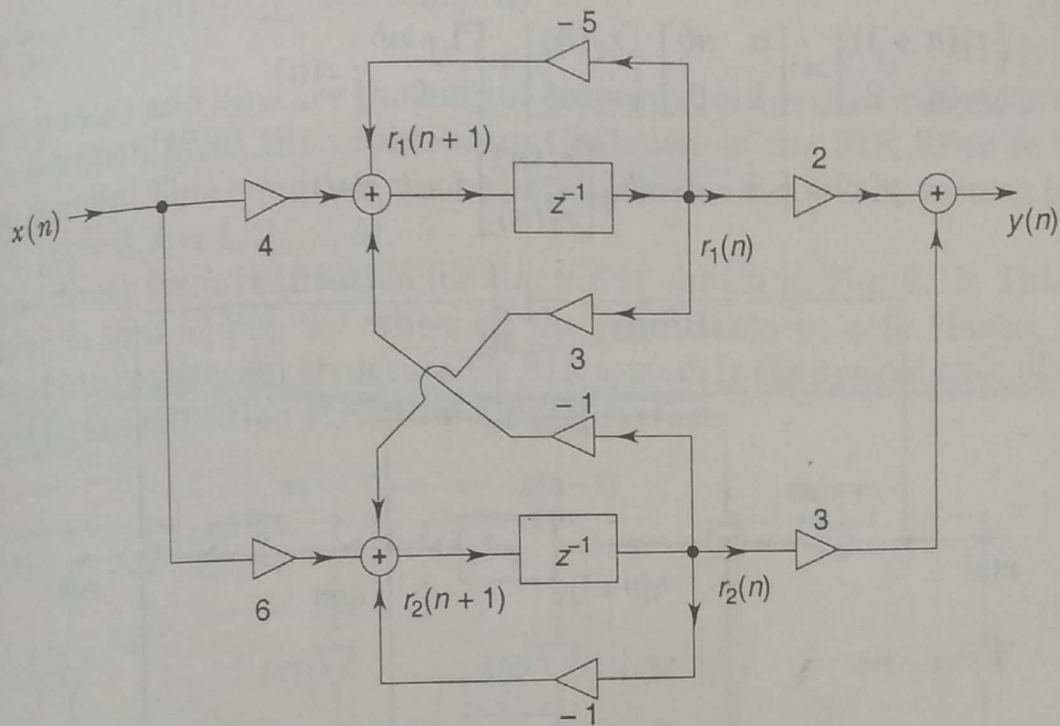


Fig. E9.11

Example 9.12 Develop the state space representation of the digital filter structure shown in Fig. E9.12 and draw the block diagram representation of this state space structure.

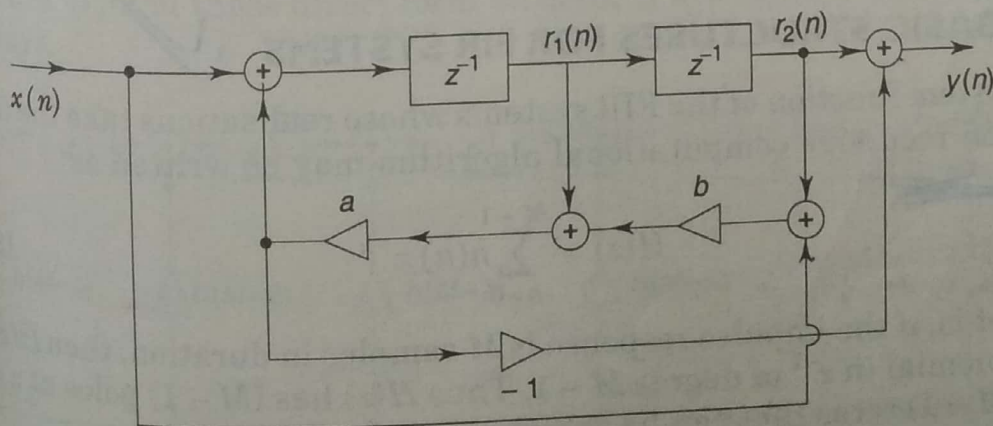


Fig. E9.12(a)

$$y(n) = 2r_1(n) + 3r_2(n)$$

$$r_2(n+1) = r_1(n)$$

$$\begin{aligned} r_1(n+1) &= x(n) + \{(r_2(n) + x(n))b + r_1(n)\}a \\ &= x(n)(1+ab) + ab r_2(n) + a r_1(n) \end{aligned}$$

$$\begin{aligned} y(n) &= r_2(n) - a \{(r_2(n) + x(n))b + r_1(n)\} \\ &= r_2(n)(1-ab) - ab x(n) - a r_1(n) \end{aligned}$$

The state space equation is given by

$$\begin{bmatrix} r_1(n+1) \\ r_2(n+1) \end{bmatrix} = \begin{bmatrix} a & ab \\ 1 & 0 \end{bmatrix} \begin{bmatrix} r_1(n) \\ r_2(n) \end{bmatrix} + \begin{bmatrix} 1+ab \\ 0 \end{bmatrix} x(n)$$

$$y(n) = [-a \quad 1-ab] \begin{bmatrix} r_1(n) \\ r_2(n) \end{bmatrix} - ab x(n)$$

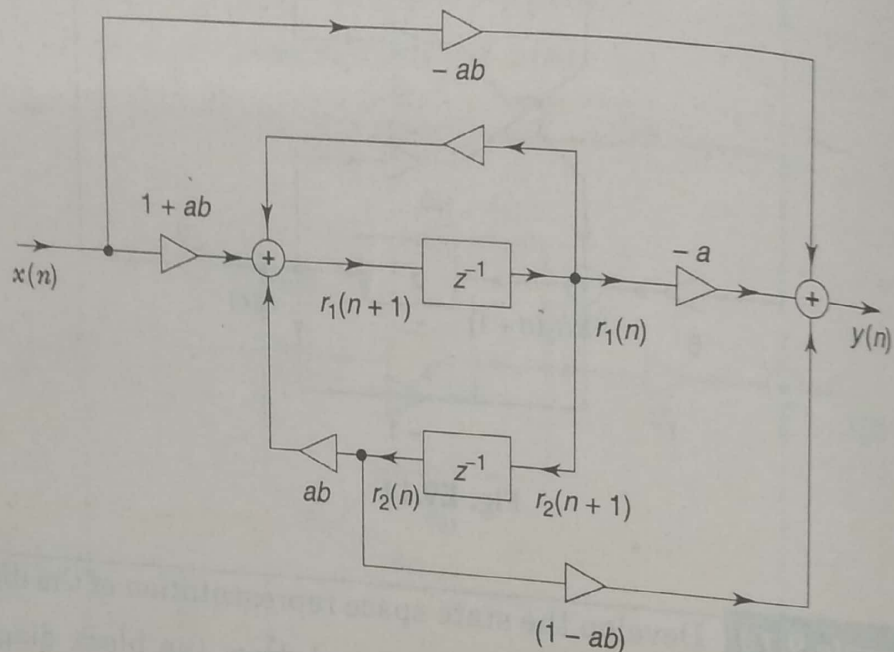


Fig. E9.12(b)

9.4 BASIC STRUCTURES FOR FIR SYSTEMS

The system function of the FIR systems whose realisations take the form of a non-recursive computational algorithm may be written as

$$H(z) = \sum_{n=0}^{M-1} h(n) z^{-n} \quad (9.7)$$

That is, if the impulse response is M samples in duration, then $H(z)$ is a polynomial in z^{-1} of degree $M-1$. Thus $H(z)$ has $(M-1)$ poles at $z=0$ and $(M-1)$ zeros that can be anywhere in the finite z -plane.

FIR filters are often preferred in many applications, since they provide an exact linear phase over the whole frequency range and they

are always BIBO stable independent of the filter coefficients. Two simple realisation methods for FIR filters, viz. direct form and cascade form are outlined in this section.

9.4.1 Direct Form Realisation of FIR Systems

The convolution sum relationship gives the system response as

$$y(n) = \sum_{k=0}^{M-1} h(k) x(n-k) \quad (9.8)$$

where $y(n)$ and $x(n)$ are the output and input sequences, respectively. This equation gives the input-output relation of the FIR filter in the time-domain. This equation can be obtained from Eq. 9.1 by setting $b_k = h(k)$, $a_k = 0$, $k = 1, 2, \dots, M$.

The direct form realisation for Eq. 9.8 is shown in Fig. 9.15. This is similar to that of Fig. 9.7 when all the coefficients $a_k = 0$. Hence, the direct form realisation structure for FIR system is the special case of the direct form realisation structure for IIR system.

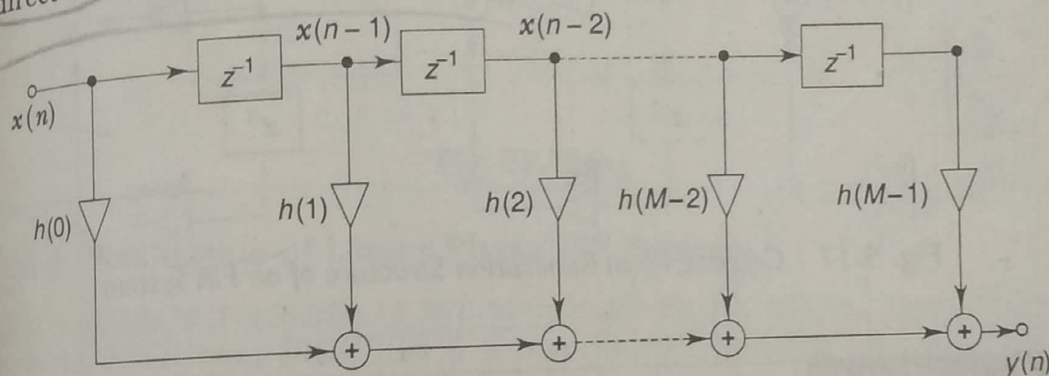


Fig. 9.15 Direct Form Realisation Structure of an FIR System

The transposed structure shown in Fig. 9.16 is the second direct form structure. Both of these direct form structures are canonic with respect to delays.

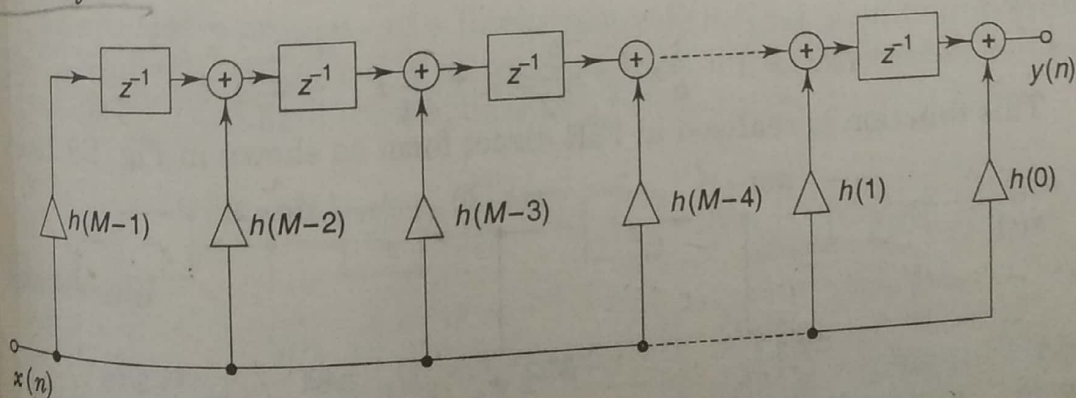


Fig. 9.16 Transposed Direct Form Realisation Structure of an FIR System of Fig. 9.15

9.4.2 Cascade form Realisation of FIR Systems

As an alternative to the Direct-Form, we factor the FIR transfer function $H(z)$ as a product of second-order factors given by

$$H(z) = \prod_{k=1}^{M/2} (\beta_{0k} + \beta_{1k}z^{-1} + \beta_{2k}z^{-2}) \quad (9.9)$$

where if M is even, $\beta_{2k} = 0$ and $H(z)$ has an odd number of real roots. A realisation of Eq. 9.9 is shown in Fig. 9.17 for a cascade of second-order sections. This cascade form is canonic with respect to delay, and also employs $(M-1)$ two input adders and M multipliers for an $(M-1)$ th order FIR transfer function.

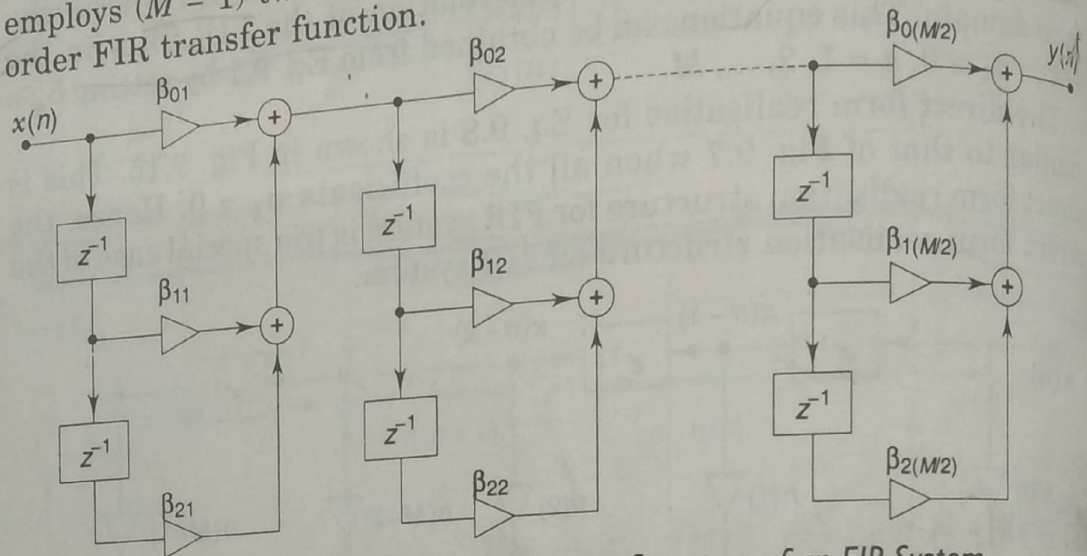


Fig. 9.17 Cascade Form Realisation Structure of an FIR System

Example 9.13 Obtain direct form and cascade form realisations for the transfer function of an FIR system given by

$$H(z) = \left(1 - \frac{1}{4}z^{-1} + \frac{3}{8}z^{-2}\right) \left(1 - \frac{1}{8}z^{-1} - \frac{1}{2}z^{-2}\right)$$

Solution

Direct form realisation Expanding the transfer function $H(z)$, we have

$$H(z) = 1 - \frac{3}{8}z^{-1} - \frac{3}{32}z^{-2} + \frac{5}{64}z^{-3} - \frac{3}{16}z^{-4}$$

This function is realised in FIR direct form as shown in Fig. E9.13(a)

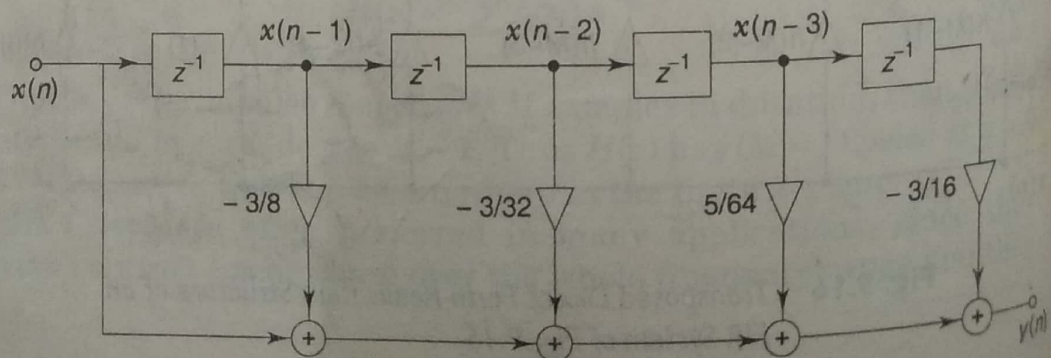


Fig. E9.13(a)

Cascade realisation

$$H(z) = \left(1 - \frac{1}{4}z^{-1} + \frac{3}{8}z^{-2}\right) \left(1 - \frac{1}{8}z^{-1} - \frac{1}{2}z^{-2}\right)$$

$$= H_1(z) \cdot H_2(z)$$

This function is realised in FIR cascade form as shown in Fig. E9.13(b)

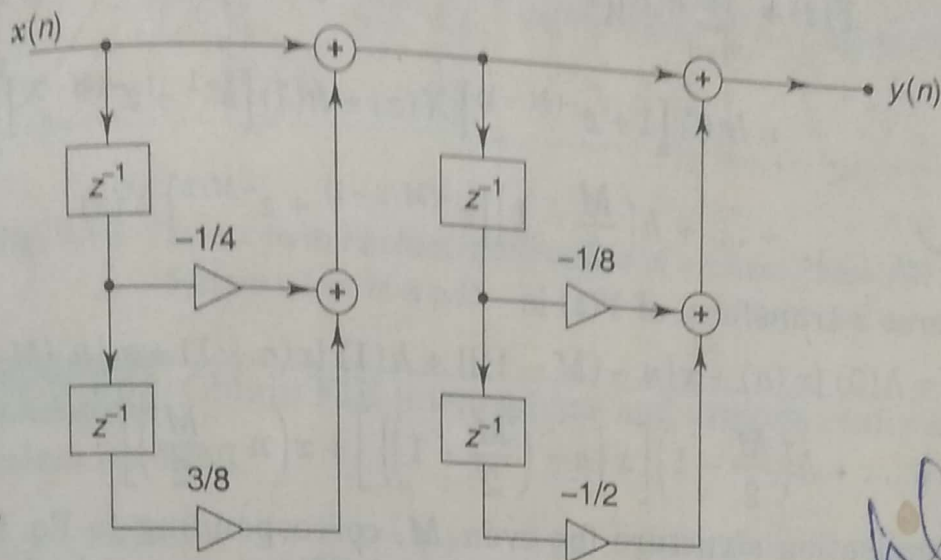


Fig. E9.13(b)

9.4.3 Realisation of Linear Phase FIR Systems

If the impulse response is symmetric about its origin, linear phase results. If all zero filter has a linear phase response, a special non-recursive structure that reduces the number of multiplications by approximately one-half can be implemented. The impulse response for a causal filter begins at zero and ends at $M - 1$.

As discussed in Section 7.2, a linear phase FIR filter of length M is characterised by

$$h(n) = h(M - 1 - n) \quad (9.10)$$

The symmetry property of a linear phase FIR filter is used to reduce the multipliers required in these realisations. Using this condition, the z -transform of the impulse response can be expressed as

$$H(z) = Z[h(n)] = \sum_{n=0}^{M-1} h(n) z^{-n} \quad (9.11)$$

For M even

$$H(z) = \sum_{n=0}^{M/2-1} h(n) [z^{-n} + z^{-(M-1-n)}] \quad (9.12)$$

For M odd

$$H(z) = h\left(\frac{M-1}{2}\right)z^{-(M-1)/2} + \sum_{n=0}^{(M-3)/2} h(n)[z^{-n} + z^{-(M-1-n)}] \quad (9.13)$$

We know that the output transform $Y(z) = H(z)X(z)$. When M is even,

$$\begin{aligned} Y(z) &= \sum_{n=0}^{\frac{M}{2}-1} h(n)[z^{-n} + z^{-(M-1-n)}]X(z) \\ &= h(0)[1 + z^{-(M-1)}]X(z) + h(1)[z^{-1} + z^{-(M-2)}]X(z) \\ &\quad + \dots + h\left(\frac{M}{2}-1\right)[z^{-(M/2-1)} + z^{-M/2}]X(z) \end{aligned} \quad (9.14)$$

The inverse z -transform of $Y(z)$ is

$$\begin{aligned} y(n) &= h(0)[x(n) + x\{n-(M-1)\}] + h(1)[x(n-1) + x\{n-(M-2)\}] \\ &\quad + \dots + h\left(\frac{M}{2}-1\right)[x\{n-(\frac{M}{2}-1)\} + x\{n-\frac{M}{2}\}] \end{aligned} \quad (9.15)$$

The realisation structure for even M , corresponding to Eq. 9.15, is shown in Fig. 9.18. For this case, $M/2$ multipliers are required.

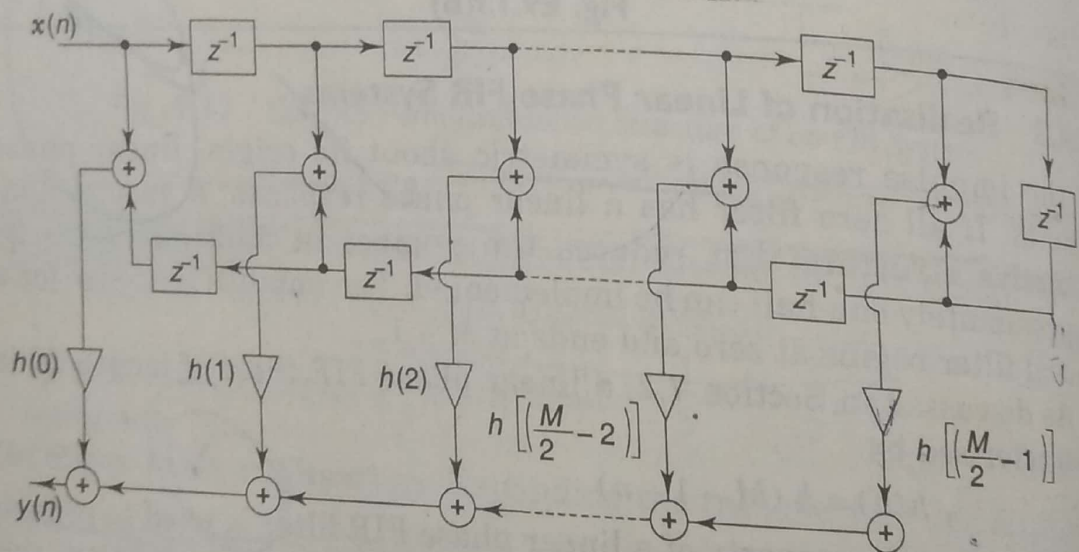


Fig. 9.18 Direct form Realisation Structure of a Linear Phase FIR System when M is Even

The same procedure can be applied to Eq. 9.13 for M odd, resulting in the realisation shown in Fig. 9.19. Here, $\frac{M-3}{2} + 1 + 1$ or $\frac{(M+1)}{2}$ multipliers are required.

The linear phase FIR system realisations require approximately only half the number of multipliers than direct form and cascade form realisations.

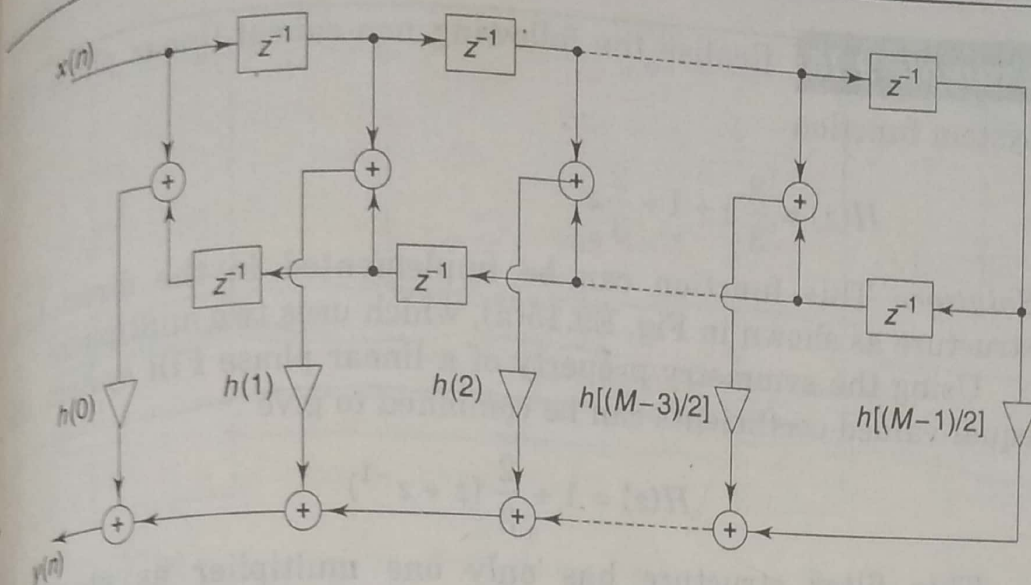


Fig. 9.19 Direct form Realisation Structure of a Linear Phase FIR System when M is odd

Example 9.14 Obtain FIR linear-phase and cascade realisations of the system function.

$$H(z) = \left(1 + \frac{1}{2}z^{-1} + z^{-2}\right) \left(1 + \frac{1}{4}z^{-1} + z^{-2}\right)$$

Solution

Linear phase realisation Expanding $H(z)$, we have

$$H(z) = 1 + \frac{3}{4}z^{-1} + \frac{17}{8}z^{-2} + \frac{3}{4}z^{-3} + z^{-4}$$

Here $M = 5$ and the realisation is shown in Fig. E9.14(a).

Cascade realisation

$$H(z) = \left(1 + \frac{1}{2}z^{-1} + z^{-2}\right) \left(1 + \frac{1}{4}z^{-1} + z^{-2}\right)$$

$H(z)$ has a product of two sections which have the linear phase symmetry property. The corresponding cascade realisation is shown in Fig. E9.14(b).

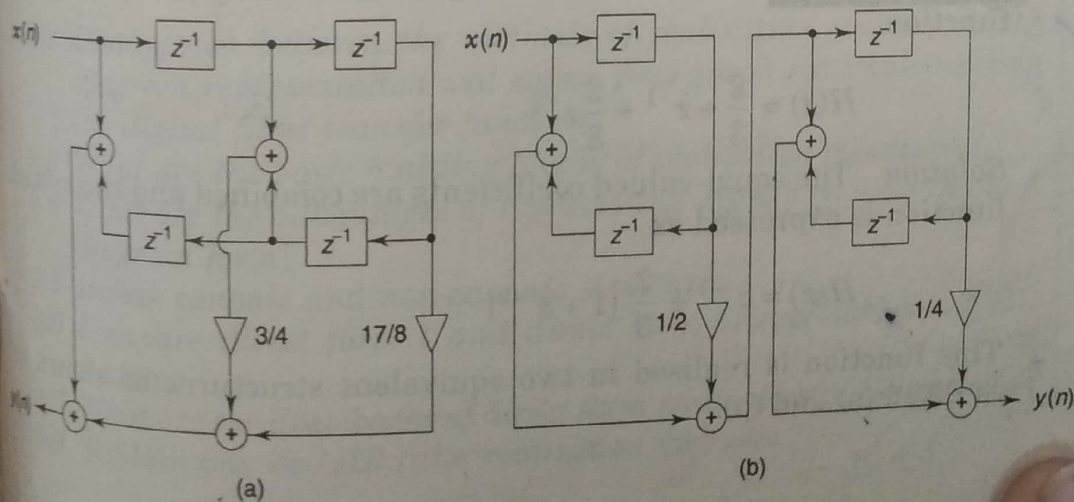


Fig. E9.14

Example 9.15 Realise the following non-causal linear phase FIR system function

$$H(z) = \frac{2}{3}z + 1 + \frac{2}{3}z^{-1}$$

Solution This function can be implemented in the direct form structure as shown in Fig. E9.15(a), which uses two multipliers. Using the symmetry property of a linear phase FIR system, the equal-valued coefficients can be combined to give

$$H(z) = 1 + \frac{2}{3}(z + z^{-1})$$

This filter structure has only one multiplier as shown in Fig. E9.15(b). Both the structures shown in Figs E9.15(a) and (b) are equivalent to a non-causal linear phase FIR system.

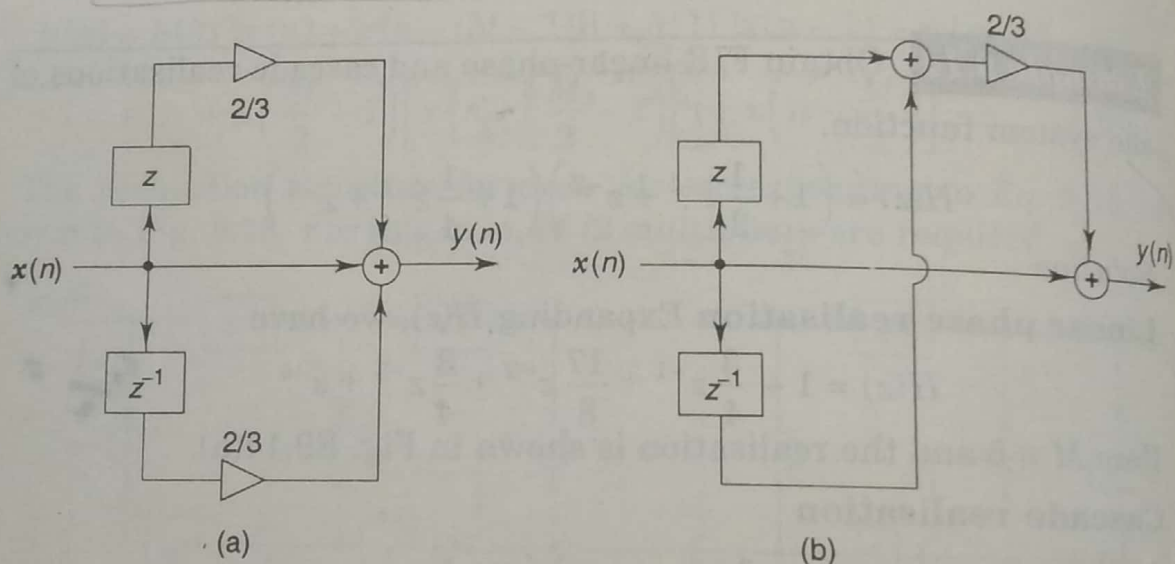


Fig. E9.15

Example 9.16 Realise the following causal linear phase FIR system function

$$H(z) = \frac{2}{3} + z^{-1} + \frac{2}{3}z^{-2}$$

Solution The equal-valued coefficients are combined and the system function is expressed as

$$H(z) = z^{-1} + \frac{2}{3}[1 + z^{-2}]$$

This function is realised in two equivalent structures as shown in Figs E9.16 (a) and (b).

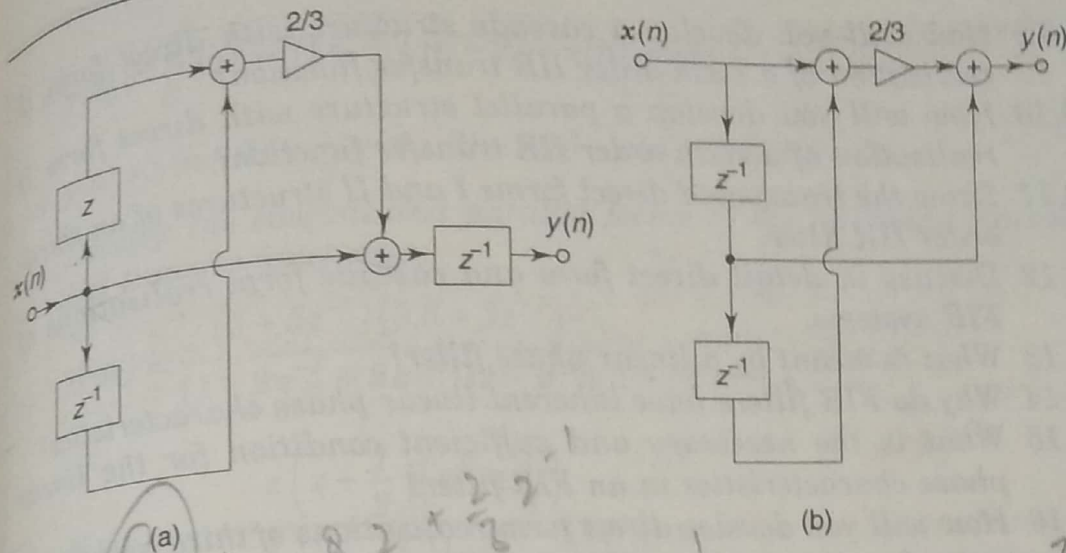


Fig. E9.16

REVIEW QUESTIONS

9.1 How will you obtain the transfer function

$$H(z) = \frac{Y(z)}{X(z)} = \frac{\sum_{k=0}^M b_k z^{-k}}{1 - \sum_{k=1}^N a_k z^{-k}}$$

from the linear difference equation

$$y(n) = \sum_{k=1}^N a_k y(n-k) + \sum_{k=0}^M b_k x(n-k)?$$

- 9.2 Distinguish between the methods of realisation namely, block diagram representation and signal flow graph for implementing the digital filter transfer function.
- 9.3 What are the basic building blocks of realisation structures?
- 9.4 What are the advantages of representing digital systems in block diagram form?
- 9.5 Define canonic and non-canonic structures.
- 9.6 Compare direct form I and direct form II realisations of IIR systems.
- 9.7 What are the drawbacks of direct form realisation of IIR systems?
- 9.8 Explain any two IIR filter realisation methods.